

Rink Reports

Operations & Training Manual

The complete operations manual: how the app is organized, who sees what, and one chapter per module — twelve report and scheduling modules plus the Admin Control Center that configures them — plus a cross-module glossary.

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RinkReports — Operations & Training Manual

RinkReports

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RinkReports is a progressive web app (PWA) for ice-rink facility operations. It replaces the clipboards and binders at the rink desk with structured, mobile-first digital flows: daily checklists, refrigeration and air-quality readings, ice operations logs, ice-depth measurements, incident and accident reports, and employee scheduling — all configured per facility from a single Admin Control Center.

You only see data for your own facility — this is automatic. There is no facility switcher for staff or facility admins; the app keeps each rink's records separate on its own. (The one exception is the platform-wide Super Admin console.)

Table of contents

Onboarding guides (start here)

- Admin onboarding — stand up a new facility
- Supervisor onboarding — day-to-day oversight
- Staff onboarding — here's how to do your job

Module chapters

1. Admin Control Center
2. Daily Reports
3. Refrigeration Logs
4. Incident Reporting
5. Accident Reports
6. Ice Operations
7. Air Quality
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9. Dasher Boards
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- How RinkReports is organized
- Module summaries
- Glossary of cross-module terms
- Open questions / [△](#) VERIFY items

How RinkReports is organized

The thirteen modules

RinkReports is built from twelve toggleable report and scheduling modules, plus the Admin Control Center that configures all of them (always on, never toggleable). Each toggleable module appears in the staff menu only when the facility has it enabled and the person has permission.

1. **Admin Control Center** — the back office: facility settings, employees, roles, permissions, lists, exports, retention, and the audit log.
2. **Daily Reports** — digital shift checklists, one per work area, ticked off and submitted each shift.
3. **Refrigeration Logs** — structured readings for the refrigeration plant, checked live against normal ranges.
4. **Incident Reporting** — records of something that went wrong (slip, collision, medical), reviewed and tracked by managers.
5. **Accident Reports** — records of a personal injury (who was hurt, how, where, witnesses, and any workers'-comp claim), reviewed by managers with no separate status lifecycle.
6. **Ice Operations** — the maintenance logbook: ice makes, edging, blade changes, and circle-check inspections.
7. **Air Quality** — CO / NO₂ / CO₂ readings checked live against the facility's compliance thresholds.
8. **Ice Depth** — point-by-point ice thickness measured against a tappable rink diagram.
9. **Dasher Boards** — the rink-perimeter condition map: tap a board, glass panel, or door to report a problem, plus scheduled inspection walks.
10. **Facility Paperwork** — a read-only library of admin-uploaded documents (policies, handbooks, safety manuals) staff can browse and download.
11. **Communications** — the facility's alert and messaging hub: routing rules fan submissions out to people, plus direct staff messaging.
12. **Employee Scheduling** — shift planning (admin grid) plus staff self-service (availability, time off, swaps, open shifts).
13. **Rink Scheduling** — ice-time booking and billing: the admin booking calendar, rate cards, season contracts, invoices, and the front-desk/customer-facing views. Distinct from Employee Scheduling, which schedules staff shifts, not ice time.

There's also a 13th toggle, **Insights**, a cross-module analytics view with no admin console or training chapter of its own — see Admin Control Center §Modules.

Role tiers

This manual uses a five-tier vocabulary, highest to lowest:

Doc tier	One-line description
super_admin	Platform owner. Cross-facility access; the only tier that sees the Super Admin console, creates facilities, and hard-deletes records.
org_admin	A multi-facility / organizational administrator sitting between super_admin and a single facility admin.
facility_manager	The person who runs the Admin Control Center for one facility — configures modules, employees, and settings.

Doc tier	One-line description
supervisor	A mid-tier overseer with some elevated approval/review rights (e.g. scheduling decisions, submission review).
staff	Front-line workers who submit reports and use self-service scheduling.

⚠ **VERIFY — role mapping.** The live application does **not** use those five fixed tiers. The real model is **super_admin / admin / manager / staff**, plus optional **per-facility custom roles** (e.g. **driver**). Access is resolved **per module and per action** (View / Submit / Edit / Admin) through a permissions system — roles only seed default permissions, and an admin can override any individual person's access. The closest mapping: | Doc tier (this manual) | Live app role | Notes | ---|---|---| | super_admin | super_admin (platform flag) | Cross-facility; sees the Super Admin console. | | org_admin | (no exact equivalent) | ⚠ **VERIFY** — the app has **no separate org/multi-facility admin tier**. Treat as super_admin for now. | | facility_manager | admin (facility administrator) | Runs the Admin Center for one facility. | | supervisor | manager (or a custom role) | Some elevated rights — **but permission-driven, not title-driven**. | | staff | staff (or a custom role) | Submits reports; self-service scheduling. | Because access is permission-driven, two people with the same title can have different access if an admin has customized it. Read every "tier X can do Y" statement as "a user with the matching permission can do Y."

You only see your own facility's data

This is automatic and not configurable by staff or facility admins. There is no facility switcher, no way to view another rink's records, and nothing to set up — the separation is built in. Only the platform-wide Super Admin console works across facilities.

Module summaries

Each summary links to the full chapter. The chapters hold the screen-by-screen detail, field references, and per-module troubleshooting.

Admin Control Center — The configuration hub for one rink. Here a facility_manager sets the facility's information, turns report modules on or off, adds and manages employees, assigns roles and per-person permissions, manages shared lists (departments, facility spaces, dropdowns), and controls system settings for exports, data retention, and the audit log. Nothing a staff member sees is hardcoded — it all flows from here. Facility creation and granting Admin-module Admin access are super_admin-only.

Daily Reports — The digital shift-checklist clipboard. The facility is divided into **work areas**; each area has one or more checklist **templates** ("shifts"). Staff pick an area and shift, tick off tasks, leave an optional note, and submit. Every submission is an independent, append-only record auto-tagged with the facility's local date; reports auto-delete after 14 days. Submit access is granted area-by-area.

Refrigeration Logs — Structured readings for the refrigeration plant (compressors, pumps, condensers, brine, machine hours, alarms). Admins build sections, equipment, and reading fields, and set thresholds (normal ranges). Staff enter a round of readings, see "Normal: min – max" hints live, and must write a corrective-action note for any critically out-of-range value. A °F/°C toggle changes display only. Append-only.

Incident Reporting — Records of something that went wrong: when/where, what happened, severity, whether an ambulance was called, who was involved, and up to three witnesses. Uniquely, the reporter

can **edit their own report for 24 hours**, after which it locks. No photo/file upload. Admins review in History, change status, and add follow-up notes.

Accident Reports — Records of a personal injury: who was hurt, an interactive body diagram for where, severity, witnesses, and an optional workers'-comp acknowledgement. Shares Incident Reporting's 24-hour reporter-edit window, but — unlike Incident Reporting — has **no status lifecycle**; admins track follow-up purely through append-only notes. Certain injury-severity values can be flagged to automatically raise a Communications alert.

Ice Operations — The maintenance logbook covering four built-in operation types: **Ice Make**, **Edging**, **Blade Change**, and **Circle Check** (a pass/fail inspection where failed items require a note and raise an alert). A facility chooses which types to show; equipment and circle-check items are admin-configured. Append-only.

Air Quality — CO / NO₂ / CO₂ (and more) readings, checked live against the facility's compliance profile with colored range badges (Within range / Corrective / Notification / Evacuation). Thresholds derive from a regulatory jurisdiction and can be **tightened but never loosened**. Over-threshold readings are exceedances and require a corrective-action note. Append-only.

Ice Depth — A two-phase (measure → review) flow that records ice thickness point-by-point against a tappable USA-Hockey rink diagram, tuned for Bluetooth calipers. Each point colors live by severity. Measures **depth, not temperature** (unit is inches or mm — no °F/°C toggle). Submitted sessions are immutable, with PDF / print / send options.

Dasher Boards — The rink-perimeter condition map. Admins build the physical board/glass/door sequence around each rink and assign it permanent labels that never renumber, even as the layout changes. Staff tap any segment during a walk to report a problem or complete a scheduled checklist inspection; every walk is a permanent, append-only record with pass/fail tallies.

Facility Paperwork — A read-only document library. Admins bulk-upload the facility's own policies, handbooks, and safety manuals into fixed categories; staff browse and download, with no submission workflow of their own. Distinct from the app's built-in Training & Documentation, which is Rink Reports' own manuals, not the facility's.

Communications — The facility's alert and messaging hub. Routing rules automatically fan a submission from another module (an out-of-range reading, a flagged accident, a failed circle check) out to the right people, on the timing an admin configures; staff also compose direct messages and broadcasts. Acknowledgement is tracked per alert.

Employee Scheduling — Two sides: the **admin grid** (drag-build shifts, assign people, publish via a two-person request/approve flow) and the **staff app** (my shifts, availability, time off, claim open shifts, request swaps). Runs cert/hour-cap/overtime rule checks. **No payroll, timekeeping, or clock-in/out** — it is assignment only.

Rink Scheduling — Ice-time booking and billing: not to be confused with Employee Scheduling. A rate-card engine prices bookings by prime/non-prime time windows and per-type overrides; a drag-and-drop admin calendar creates and manages bookings, recurring series, and resurface plans; a front-desk view lets staff book and check in customers; and season contracts, invoices, and a public booking-request link round out the billing side. Unlike Employee Scheduling, there is **no offline write path at all** — every write re-validates live.

Glossary of cross-module terms

- **Area / work area** — A zone of the facility. In Daily Reports it's the checklist scope (and the "tabs" staff see); submit access is granted area-by-area. In scheduling, **job areas** are positions people are scheduled into.
- **Template / checklist** — A named, ordered set of tickable items. In Daily Reports a template is a "shift" within an area; in Ice Operations a circle-check template; in scheduling a reusable shift pattern.
- **Threshold** — An admin-defined normal range (and severity) for a numeric reading. Drives the "Normal: min – max" hint (Refrigeration), the live range badges (Air Quality), and severity colors (Ice Depth).
- **Exceedance** — A reading that crosses a threshold. Flagged on the report, counted on the confirmation screen, and (if alerts are on) raised to managers. In Air Quality an exceedance forces a corrective-action note.
- **Corrective-action note** — A required free-text note describing what was done about an out-of-range / failed reading. Required for critical refrigeration breaches, air-quality exceedances, and failed circle-check items; blocks submit until written.
- **Draft / published** — Scheduling states. Drafts are admin-only and freely editable; publishing makes shifts visible to staff and opens unassigned ones for claiming.
- **Publish-lock** — Once a shift is published it can't be freely edited or deleted; changes go through a controlled, audited path that re-checks rules and notifies affected staff.
- **Soft cancel** — Deleting a published shift marks it **cancelled** (kept for the record) rather than erasing it. Draft shifts are deleted outright.
- **Append-only** — Each submit creates a new, permanent record that staff cannot edit. To correct, you submit again (incidents' 24-hour window is the exception). Admins add follow-up notes rather than editing.
- **Follow-up note** — An append-only note an admin adds to a submitted report from the History detail panel. Cannot be edited or deleted; the original report stays intact.
- **Offline queue / "Save on this device"** — When you submit offline, the report is saved on your device and syncs automatically when you reconnect, re-running the same checks server-side. Each queued item has a unique id so a retry can't create a duplicate.
- **Facility scoping** — The automatic rule that you only ever see and act on your own facility's data. Not configurable; built in.
- **Retention / purge** — Per-module rules (Admin → Data Retention) that permanently delete records older than a set number of days, optionally each night. Destructive; gated behind a warning. (Daily Reports auto-purges after 14 days.)
- **Seed defaults** — A one-click button on many setup screens that creates a standard starter configuration (sections, reading types, roles, etc.), safe to skip and edit afterward.

Open questions / ⚠ VERIFY items

This appendix collects every ⚠ VERIFY flag from the discovery manifest and all thirteen module chapters. Each should be validated against a real configured facility before publishing hard claims.

Roles, tiers & permissions (Manifest; Admin Control Center; every module's "Who can use it")

1. **No org_admin tier in the live app.** The five-tier vocabulary (super_admin / org_admin / facility_manager / supervisor / staff) does not match the live model (super_admin / admin / manager / staff + custom roles). There is **no org_admin equivalent** — treat any org_admin need as super_admin. (Manifest §Role Tiers, §Gaps 1; flagged in every module's role table.)
2. **Access is permission-driven, not title-driven.** "Supervisor" / manager abilities are whatever permission rows an admin enables, not a fixed tier. Two people with the same title can differ. Validate every "tier X can do Y" claim against a configured facility. (Manifest §2, §Gaps 1; Admin Control Center §2; Refrigeration §2; Incidents §2; Ice Operations §2; Air Quality §2; Ice Depth §2; Scheduling §2.)
3. **Per-module delete rights.** "Delete = super_admin only" holds for facilities and the general pattern, but per-module delete (an admin deleting an area/template/session) should be confirmed module-by-module. (Manifest §2, §Gaps 6.)
4. **Only super_admin can grant Admin-module Admin access and edit facilities.** Confirmed in the Admin chapter as the intended guard; carry it forward as the rule for who can hand out Admin Center access. (Admin Control Center §2, §5 Permissions, §9.)
5. **Roles screen navigation.** /admin/roles is a real screen but has **no sidebar entry** — it is reached only from the Admin Dashboard's setup checklist ("Canonical roles seeded" / "Role permission defaults") or by typing the address. Confirm this is the intended navigation. (Manifest §1, §Gaps 5; Admin Control Center §3, §9.)

Daily Reports

6. **Area cap is 30, not 20.** Sales material mentions "up to 20 tabs"; the database enforces **30 active areas**. Confirm the exact number and whether "tabs" means areas or templates-within-an-area. (Manifest §4, §Gaps 2; Daily Reports §4.)
7. **No end-of-day lock.** The package described an end-of-day lock/single rolling report. The code uses an **append-only** model (one immutable record per submit, facility-date tagged, **auto-deleted after 14 days**) with **no per-day lock**. Describe it that way. (Daily Reports §8.)

Refrigeration

8. **Supervisor / org_admin rows** in the access table are mapped to the live model and should be confirmed per facility (covered by items 1–2). (Refrigeration §2.)

Incident Reporting

9. **24-hour edit window — confirmed.** The reporter can edit their own incident for 24 hours, then it locks read-only; confirmed in code (edit_window_ends_at = now() + 24h). (Incidents §8.)
10. **Incident Type not collected from staff.** Type appears as a column/filter and on the admin detail, but the current **staff** submission form does **not** present an incident-type picker. Confirm whether your facility expects staff to pick a type before training on it as a staff field. (Incidents §4, §7.)

Ice Operations

11. **Temperature unit setting removed.** Confirmed unused on live forms (no staff field collects a temperature), so the Fahrenheit/Celsius selector was removed from Ice Operations Settings; legacy History readings still display in the previously saved unit. (Ice Operations §4.)
12. **Supervisor config rights** default — confirm whether a facility grants supervisors Ice Operations config by default (covered by items 1–2). (Ice Operations §2.)

Air Quality

13. **Thresholds are tighten-only.** Facility overrides may only **lower** (tighten) a regulatory ceiling, never raise it; the app rejects looser values and clamps internally. Confirmed in code, but call it out clearly in training. (Air Quality §4, §9.)
14. **Supervisor / org_admin rows** mapped to the live model; confirm per facility (items 1-2). (Air Quality §2.)

Ice Depth

15. **Unit is inches / mm — not cm.** The training brief said "in/cm," but the app's unit options are **inches** and **mm**. (Ice Depth §4.)
16. **Offline submit when no service worker is active.** If the device can't queue offline (no service worker), the app blocks the submit and tells the user to reconnect/reload, keeping typed readings on screen. Confirm this fallback behavior. (Ice Depth §8.)
17. **Super_admin-only session delete; supervisor rights** per facility (items 1-2). (Ice Depth §2.)

Offline / PWA (Manifest §7)

18. **Offline tech is service worker + IndexedDB — not Dexie.** The package mentioned "Dexie"; no Dexie library is used. Describe it generically ("saved on your device, syncs when you reconnect") and never name Dexie. (Manifest §7, §Gaps 3.)
19. **Scheduling offline scope.** Only **availability** and **time-off** queue offline. **Swap requests, swap acceptance, open-shift claims, and the entire admin grid are online-only** by design. Confirmed in code, but the staff-side discovery once suggested swap requests might queue — treat swap/claim as online-only. (Manifest §7, §Gaps 4; Scheduling §8.)

Other

20. **Dashboard & Account pages** were noted as routes but their on-screen content was not fully inventoried; capture if they need training coverage. (Manifest §Gaps 8.)

This manual links to, and does not duplicate, the module chapters in `modules/`. Where a chapter and this manual disagree, the chapter (closer to the code) wins — and the discrepancy belongs in the [△ VERIFY](#) list above.

Admin Control Center

1. What this module is for

The Admin Control Center is the configuration hub for one ice rink in RinkReports. It is where the person who runs a facility sets up everything the staff app depends on: the facility's basic information, which report modules are turned on, the people who work there, the roles and permissions that decide who can do what, the shared lists that feed dropdowns across the app, and the system-wide settings for exports, data retention, and the audit trail.

Think of it as the back office. Staff members spend their time in the report flows (Daily Reports, Refrigeration, Incidents, and so on). Admins spend their time here, deciding how those flows behave. Nothing a staff member sees on their device — the modules in their menu, the areas they can submit to, the dropdown options they pick from — is hardcoded. It is all driven by what an admin configures in this Control Center.

You only see data for your own facility — this is automatic. Everything you create, edit, or view here applies to your rink and no one else's. (The one exception is the Super Admin console, described in section 5, which is reserved for the platform owner who manages every facility at once.)

2. Who can use it

RinkReports does not use a rigid ladder of five fixed job titles. Access is **permission-driven**: each person's abilities are resolved per module and per action (View / Submit / Edit / Admin), and an admin can customize any individual's access. The table below maps the documentation's five-tier vocabulary onto how the live app actually behaves. Treat the "What they can do" column as the default for that tier — an admin can widen or narrow it for any single person.

Role tier	What they can see	What they can do
super_admin	Everything, across every facility. The only tier that sees the Super Admin console and the cross-facility facility list.	Create, edit, and activate/deactivate facilities; create and edit any facility's settings; promote or demote other super admins; reset any user's password; hard-delete employees; assign a person to a different facility; everything an org_admin/facility_manager can do.
org_admin	⚠ VERIFY — the live app has no separate org/multi-facility admin tier. There is no role between super_admin and the single-facility admin. For now, treat any "org_admin" need as super_admin.	(No distinct equivalent exists in the code.)

Role tier	What they can see	What they can do
facility_manager (live role: admin)	The full Admin Control Center for their own facility. Their own facility's settings appear read-only — they cannot edit facility info.	Turn modules on/off; add/edit/deactivate/reactivate employees; assign roles at or below their own rank; set per-person permission overrides; create and edit custom roles; manage departments, spaces, lists, exports, retention, and read the audit log. Cannot create/edit facilities, hard-delete employees, or grant Admin Center access.
supervisor (live role: manager, or a custom role)	Depends entirely on the permissions an admin has granted. By default a manager has no Admin Center access unless given the admin action on the admin module.	⚠ VERIFY — supervisor abilities are not a fixed tier; they are whatever permission rows an admin enables. Two people with the title "manager" can have different access.
staff (live role: staff, or a custom role)	No access to the Admin Control Center. Attempting to open /admin shows a "Forbidden" message.	Nothing here. Staff work only in the report flows and self-service scheduling.

How the gate actually works (plain version): the Admin Control Center opens for you if you are a super admin, **or** you have the Admin action enabled on the Admin module for your facility, **or** you are an active employee whose role is Administrator or Super Admin. A deactivated account is denied everywhere, immediately.

3. How to get there

From the staff app, admins see an extra **Admin Center** link in the left sidebar (staff do not). Selecting it opens the Admin Control Center, which has its own grouped sidebar.

The Admin sidebar is organized into three groups. The Setup and System groups are the Admin Control Center itself; the Module Admin group links out to each report module's own configuration (covered in those modules' own chapters).

- **Dashboard** — /admin (setup checklist + activity overview)
- **Setup**
 - Facility
 - Modules
 - People
 - Departments
 - Facility Spaces
 - Permissions
- **Module Admin** (cross-linked, documented in each module's chapter)
 - Daily Reports Admin, Ice Depth Admin, Ice Operations Admin, Incident Reports Admin, Accident Reports Admin, Refrigeration Admin, Air Quality Admin, Scheduling Admin, Communications Admin, Facility Paperwork
- **System**
 - Lists
 - PDF/Export Settings

- Data Retention
- Audit Log
- Super Admin (only functions for super_admin; others are bounced to a Forbidden screen)

Roles is a real screen but it is **not in the sidebar**. You reach it from the Admin Dashboard's setup checklist — the "Canonical roles seeded" and "Role permission defaults" checklist items both have a button that opens it. (You can also type the address directly.) The Roles screen links back out to Permissions and People, but People and Permissions do not link to Roles. ⚠ VERIFY this is the intended navigation; the screen has no sidebar entry.

4. Setup & configuration (admins)

This section is the quick map of every admin-configurable option in the Control Center, where it lives, the valid values, and the effect. Screen-by-screen detail follows in section 5.

Setting	Where	Valid values	Effect
Module on/off	Modules	On / Off per module	Shows or hides the module in staff navigation only . Does not change anyone's permissions.
Facility info (name, slug, timezone, address, contact, active)	Facility	Text/dropdown; slug must be lowercase letters/numbers/hyphens and unique	Identifies the facility. Editable by super_admin only ; a regular admin sees it read-only.
Employee record	People	See field reference (section 7)	Creates the person, their role, job areas, and optional login.
Role assignment	People (employee form)	Any role at or below your own rank	Seeds that role's default permissions onto the person; manual overrides are kept.
Per-person permission override	Permissions, or the employee detail "Module Access" tab	View / Submit / Edit / Admin per module	Overrides the role default for that one person. Clearing falls back to the role default.
Custom roles & role defaults	Roles	Display name, key, hierarchy level, per-module action matrix	Defines a reusable permission template; changing it re-applies to that role's staff (overrides preserved).
Departments	Departments	Name, slug, color, sort order, active	Feeds the schedule department filter, shift assignment, and communication routing.
Facility Spaces	Facility Spaces	Name, slug, sort order, active	Feeds the location pickers in Incident, Accident, and Air Quality reports.

Setting	Where	Valid values	Effect
Lists (custom dropdowns)	Lists	Per-domain options (currently Timezones)	Customizes dropdown options used elsewhere (today: the facility timezone picker).
Export branding/layout/columns	PDF/Export Settings	Logo URL, header/footer text, paper size, date format, CSV delimiter, per-module column checkboxes	Controls how exported PDFs and CSVs look and which columns appear.
Data retention	Data Retention	Keep-days (0 = forever, or ≥ 30) + nightly auto-purge on/off per module	Permanently deletes records older than the threshold. Destructive.

5. Screen-by-screen walkthrough

Dashboard (/admin)

The landing screen. It shows a **Setup checklist** card (Facility info, Canonical roles seeded, Role permission defaults, First admin linked, Staff added, Invites sent) — each item has a status icon and a button that jumps you to the right screen. A badge reads "Ready" when all are done or "N left" otherwise. Below the checklist are overview cards (Active employees, and for super admins, Total facilities), a **Recent report activity** grid showing submission counts per module for the last 7 and 30 days, and an **Offline sync queue** card showing pending / synced / failed counts for offline submissions. A super admin sees a **facility switcher** dropdown in the header to pick which facility the dashboard reflects.

Facility (/admin/facility)

This screen looks different depending on who you are.

Regular admin (facility_manager): you get a **read-only** view of your facility. A "Facility settings" card states "Only super admins can edit facility settings. Contact your administrator to make changes." It lists Name, Slug, Timezone, Status, Address, City, State, Zip code, Phone, Email, Created, and Last updated, plus an "At a glance" card counting Employees / Departments / Roles. You cannot change anything here.

Super admin: you get a full "Facilities" management screen — a card per facility (name, Inactive pill if applicable, slug · timezone · created date) with **Manage employees** and **Edit** buttons, plus a **New Facility** button at the top. Editing opens the facility form; saving requires super admin.

Facility form fields: Name (required, 2+ characters), Slug (required, lowercase letters/numbers/hyphens, must be unique — typing the name auto-suggests a slug until you edit it), Timezone (dropdown, options come from the Lists screen), Address, City, State (2-letter), Zip code, Phone number, Email, and (edit only) an **Active** checkbox ("Inactive facilities are hidden from most views"). The submit button reads **Create facility / Save changes**. There is no separate deactivate button — activation is the Active checkbox in the edit form.

Modules (/admin/modules)

A single **on/off switch per module**, under a "Staff modules" card. The description spells it out: "This is a visibility switch only — it does not change per-employee permissions, and disabled modules remain protected by their own access rules." Turning a module off hides it from the staff navigation for

this facility; it does not revoke anyone's underlying permissions.

The thirteen toggles are: Daily Reports, Ice Depth, Ice Operations, Refrigeration, Air Quality, Incidents, Accidents, Scheduling, Communications, Facility Paperwork, Dasher Boards, Rink Scheduling, and Insights. (Dashboard and the Admin Center itself are never toggleable. Insights has no dedicated admin console or training chapter of its own — it's a cross-module analytics view, not a submission module.) Flipping a switch is immediate — no Save button — and shows a confirmation toast such as "Daily Reports enabled for this facility." If a save fails, the switch rolls back and shows an error.

People / Employees (/admin/employees)

Titled "Employee / User Setup." A search box (name, email, role, code) and a status filter (**Active** / **Inactive** / **All**) sit above the employee table (columns: Name, Role, Email, Phone, Status, Actions). Two top buttons: **Bulk add** and **Add employee**.

Per-row actions: **Edit** (opens the form), **Invite** (only when the person is active, has an email, and has no login yet — sends a magic-link sign-in invite), **Preview** (view the app as that person), **Deactivate** / **Reactivate**, and **Delete** (shown only to super admins).

Add / Edit employee form (slides in from the right) — see the field reference in section 7 for the full list. Key behaviors:

- **Role** is required; the form shows a live **Role permissions** preview of what that role grants. A regular admin can only assign a role at or below their own rank.
- **Create a login & apply these permissions** (create mode, default on) sends an email invite and seeds the role's permissions. Requires an email. Uncheck it for schedule-only staff (e.g. minors) who do not log in.
- **Job areas:** pick up to 4; one can be marked **Primary**. You can create a new job area inline.
- **Employee is a minor** checkbox: when checked, the emergency contact name and phone become optional; otherwise both are required.

Deactivate opens a confirmation: "{name} will lose access immediately and stop appearing in shift assignments and routing rules. You can reactivate them later from this same list." Deactivation is a soft delete — the record and permissions are kept. **Delete** (super admin only) confirms "Delete {name}? This cannot be undone." and is a permanent hard removal.

Employee detail page (open a name) has tabs: **Profile** (read-only; edit via the list), **Certifications** (add/edit/delete certs with issuer and expiry, status badges warn when expiring or expired), **Module Access** (per-module effective level, its source, a "set override" dropdown, and a "Clear override" button), **Communication Groups** (add/remove the person from groups), and **Activity** (a read-only audit list for that person).

Bulk add (/admin/employees/bulk): a grid where you type or paste rows (First name, Last name, Email, Hire date, Role, Job areas), up to **100 at a time**, with live validation. Buttons: Add row, Paste from spreadsheet, Download template, Clear all. A **"Send login invites & apply role permissions"** switch (default on) controls whether each person gets an invite. Submitting reports each row as Added, Partial, or Failed.

Departments (/admin/departments)

Departments power the schedule's department filter, shift assignment, and communication routing. The table shows Name (with a color dot), Slug, Order (with up/down reorder arrows), Status, and Actions. **Add department** opens a form: Name (required), Slug (optional, auto-generated, lowercase/digits/hyphens), Color (default green #4DFF00), Sort order, and — in edit mode — Active.

Rows can be **Edited** and **Deactivated** / **Reactivated**. There is **no delete** for departments; deactivating hides a department from new assignments while existing shifts keep theirs.

Facility Spaces (/admin/spaces)

The shared list of physical locations that feed the location pickers in Incident, Accident, and Air Quality reports. **Seed defaults** creates a standard set (Main Rink, Lobby, Locker Room, Pro Shop, Parking Lot, and more). You can also **Add space** manually or **bulk import** from pasted CSV. Each space has Name, Slug, Sort order, and (edit only) Active. Rows offer **Edit**, **Deactivate** / **Reactivate**, and **Delete**. Delete confirms "Delete space \"{name}\"? This cannot be undone." — but a space that is referenced by existing reports cannot be deleted; the app tells you to deactivate it instead.

Permissions (/admin/permissions)

Titled "Module Access Control." It lists every active user; pick one to open their **permission matrix**. The matrix is a grid of **modules (rows) × four actions (columns): View, Submit, Edit, Admin**. The ten module rows are Daily Reports, Ice Depth, Ice Operations, Incident Reporting, Accident Reporting, Refrigeration, Air Quality, Employee Scheduling, Communications, and Admin.

- **What each action means:** View = can see the module; Submit = can create and submit entries; Edit = can modify their own or facility submissions; Admin = can change, approve, or configure that module's settings.
- **Each checkbox is a live write** — clicking it saves immediately and sets a per-person override. There is no Save button. If a save is rejected, the box rolls back.
- **Presets** (top-right buttons): **Full Access**, **Submitter Only** (View + Submit), **Viewer Only** (View), **No Access** (clears everything). A preset writes all module/action cells at once.
- **Admin guard:** only a super admin can grant the Admin module's Admin action (Admin Center access). For a regular admin, that one cell stays off even under Full Access, and they can only edit permissions within their own facility.
- A **Bulk CSV import** panel lets you paste rows with the header `user_id,facility_id,module,action,enabled`.
- A user with no facility assigned cannot have permissions set until a facility is assigned (the screen says so and links to People). A super admin user bypasses module permissions automatically; a notice explains the toggles only matter if their super-admin flag is removed.

You can also set a person's per-module override from their **employee detail → Module Access** tab.

Roles (/admin/roles)

Define reusable permission templates for your facility. **Seed default roles** (from the People area) creates the four canonical roles: Super Admin, Administrator, Manager, Staff. The Roles screen has two parts:

- **Role Manager:** rename, add, deactivate, reactivate, and reorder roles. **New role** asks for Display name, Key (lowercase letters/digits/underscores, e.g. `rink_lead`), Hierarchy level (0 = highest), and an optional Description. System roles cannot be deactivated. A **Copy permission defaults** control bulk-copies one role's module defaults onto another. Deactivating a role that still has assigned employees prompts a confirmation explaining those employees keep only their explicit overrides.
- **Role permission defaults matrix:** for each role, toggle the default **V / S / E / A** (View / Submit / Edit / Admin) per module. Changing a default re-applies to that role's current staff, but any per-person overrides set in Module Access Control are preserved.

A regular admin cannot create or modify a role that outranks their own hierarchy level.

Lists (/admin/lists)

Per-facility custom dropdown option lists. Today there is exactly one list domain: **Timezones**, which feeds the Facility settings timezone picker. **Seed defaults** loads the canonical set (safe to run repeatedly; never overwrites your edits). You can **Add**, **Edit**, **Deactivate** / **Reactivate**, and **Delete** options. Each option has a Key (for timezones, a valid IANA identifier such as *America/New_York*), a Display name, a Color, a Sort order, and (edit only) Active. An option that is in use cannot be deleted — deactivate it instead.

PDF / Export Settings (/admin/exports)

Two parts.

Run an export: pick a **Module**, a **Format** (CSV or PDF), and a **From** / **To** date range, then **Download**. The columns, delimiter, date format, and branding come from the settings below.

Export settings (one **Save export settings** button):

- **Branding** — Logo URL (recommended 300×80 px), Header text, Footer text (shown on exported PDFs).
- **Layout & Format** — Paper size (Letter / A4), Date format (MM/DD/YYYY, DD/MM/YYYY, or YYYY-MM-DD), CSV delimiter (comma / tab / semicolon), and an "Include on every export" group (Facility name, Export date, Submitted-by name).
- **Column Visibility per Module** — checkboxes per module choosing which columns appear in its export. Unchecked columns are hidden from exports but stay in the database.

Data Retention (/admin/retention)

Controls how long submitted data is kept per module, measured from each record's submission date. Summary cards show modules configured, auto-purge enabled count, and when a purge last ran. Each module row shows its current keep period and an **Edit** button.

- **Keep for (days):** a number, with preset buttons (30 days, 90 days, 180 days, 1 year, 2 years, 3 years, 5 years, Forever). Each module shows a recommended minimum (e.g. 365 for Incident/Accident/Audit Log). The value must be 0 (forever) or at least 30.
- **Enable nightly auto-purge:** when on, records older than the threshold are permanently deleted each night.
- **Manual Purge:** a "Run purge now" button reveals an inline confirmation ("Are you sure? This will permanently delete records.") before deleting immediately.

A red **Auto-purge warning** card stresses that deletion is permanent and irreversible — review your legal and regulatory retention obligations, especially for incident, accident, and workers' compensation records, before enabling it.

Audit Log (/admin/audit-log)

A **read-only**, filterable trail of create / update / delete / login / logout events. Filters: **Action**, **Entity type**, **Actor**, **From** / **To** dates, and a search by entity ID. The table columns are When, Actor, Action (color-coded badge), Entity type, and Entity ID. Selecting an entry's timestamp opens a detail panel with the full record, IP address, user agent, and collapsible **Before** / **After** snapshots. It shows up to 300 entries (newest first, last 30 days by default); narrow with filters. There is nothing to edit here.

Super Admin (/admin/super-admin)

Super_admin only — anyone else is sent to a Forbidden screen. Platform-wide management: stat cards (Total facilities, Active facilities, Total users, Super admins); a **Facilities** panel to **Activate** /

Deactivate facilities (paginated); a **Super admins & users** panel to **Reset password** (creates a recovery link), and **Promote / Revoke** super-admin status for other users (you cannot revoke your own); and an **Email invite service** health card with a **Run health check** button that verifies the deployment can send sign-in invites.

6. Step-by-step: common tasks

Turn a module on or off

1. Open **Modules** from the Admin sidebar.
2. Find the module row.
3. Click its switch. It flips immediately and shows a toast confirming the change. (Off hides it from staff navigation; permissions are unaffected.)

Add an employee and assign a role

1. Open **People**, then click **Add employee**.
2. Enter First name and Last name.
3. Pick a **Role** (you can only pick roles at or below your own rank). Review the permission preview.
4. Decide on **Create a login & apply these permissions** — leave it on (and enter an Email) for someone who logs in; uncheck it for schedule-only staff.
5. Optionally add Job areas (up to 4, mark one Primary), Employee code, Hire date, Max weekly hours, Phone.
6. If the person is a minor, check **Employee is a minor** (emergency contact becomes optional); otherwise enter the emergency contact name and phone.
7. Click **Create employee**. If a login was requested, an invite email goes out and the role's permissions are seeded.

Override one person's permissions

1. Open **Permissions** and pick the user (or open the person in **People** and go to the **Module Access** tab).
2. In the matrix, tick or untick the **View / Submit / Edit / Admin** box for the module you want to change. It saves instantly.
3. To reset to the role default, untick the overrides (or use the **No Access** preset / **Clear override**).
Note: only a super admin can grant the Admin module's Admin action.

Create a custom role

1. Open the **Roles** screen (from the Admin Dashboard setup checklist).
2. In Role Manager, click **New role**.
3. Enter a Display name, a Key (lowercase letters/digits/underscores), a Hierarchy level (0 = highest; you cannot create a role above your own rank), and an optional Description. Click **Create**.
4. In the role permission defaults matrix, toggle the **V / S / E / A** boxes per module to set what the role grants. Changes apply to staff with that role (their manual overrides are kept).

Configure data retention

1. Open **Data Retention** and click **Edit** on a module.
2. Set **Keep for (days)** (use a preset or type a number; 0 = forever, or at least 30).
3. Optionally turn on **Enable nightly auto-purge**, then **Save**.

- To delete old records now, use **Run purge now** and confirm. This is permanent — heed the warning, especially for incident, accident, and workers' comp data.

Export records

- Open **PDF/Export Settings**.
- In "Run an export," pick the **Module**, **Format** (CSV or PDF), and the **From / To** dates.
- Click **Download**. The file uses the branding, layout, and column settings configured below.

7. Field reference

Employee form (People → Add/Edit employee):

Field	What it means	Valid values/units	Required?	Notes
First name	Given name	Text, up to 100 chars	Yes	
Last name	Family name	Text, up to 100 chars	Yes	
Role	The person's role	Any facility role at/below your rank	Yes	Seeds that role's permission defaults; shows a live preview.
Create a login & apply these permissions	Whether to provision a sign-in	Checkbox	No (create only, default on)	Requires an email; sends an invite. Uncheck for schedule-only staff.
Job areas	Cross-trained work areas (scheduling)	Up to 4 selections	No	One can be marked Primary; can create new areas inline.
Employee code	Internal staff code	Text	No	
Hire date	Start date	Date	No	
Max weekly hours	Per-person scheduling cap	Number 1-168	No	Blank uses the facility default; the shift grid warns when exceeded.
Email	Contact / login address	Email	No*	*Required if a login is being created.
Phone	Contact number	Phone	No	
Employee is a minor	Flags an under-age worker	Checkbox	No	When on, emergency contact becomes optional.
Emergency contact name	Who to call	Text	Required unless minor	
Emergency contact phone	Emergency number	Phone	Required unless minor	

Facility form (super_admin only):

Field	What it means	Valid values/units	Required?	Notes
Name	Facility name	Text, 2+ chars	Yes	

Field	What it means	Valid values/units	Required?	Notes
Slug	URL identifier	Lowercase letters, numbers, hyphens; unique	Yes	Auto-suggested from the name.
Timezone	Local time zone	IANA zone from the Lists picker	No	Defaults if blank/invalid.
Address / City / State / Zip code	Mailing address	Text (State is 2-letter)	No	
Phone / Email	Facility contact	Phone / email	No	Email must be valid if provided.
Active	Whether the facility is live	Checkbox (edit mode)	No	Inactive facilities are hidden from most views.

Retention row (Data Retention):

Field	What it means	Valid values/units	Required?	Notes
Keep for (days)	Retention period	0 (forever) or ≥ 30 days	Yes	Per-module recommended minimums shown (e.g. 365 for incident /accident/audit).
Enable nightly auto-purge	Automatic deletion	Checkbox	No	Permanently deletes records older than the threshold each night.

8. Locking, saving & offline

Admin configuration is online-only. The Admin Control Center is a desk task and is not part of the offline submission queue. You must be connected to use it; there is no "save on this device" here.

Saving behaves in two patterns. Some screens save **instantly on click** — module switches, the permission matrix checkboxes and presets, and the role default matrix all write immediately with no Save button (a failed write rolls the control back and shows an error). Forms (Facility, Employee, Department, Space, List option, Retention, Export settings) collect your input and save when you press the submit button (**Create...** / **Save changes** / **Save export settings**).

What "locked" means here. Admins do not edit staff submissions in place. The Audit Log is permanently read-only — it is an immutable record. Facility records and employees use soft deactivation (kept for history) rather than deletion wherever possible; true deletes are reserved for super admins and, for facilities and referenced spaces/list options, are blocked or steered to "deactivate instead." Data Retention purges are the one deliberately destructive action, gated behind a warning and an inline confirmation.

9. Troubleshooting & FAQ

Why can't I edit my facility's name or address? Editing facility settings is reserved for super admins. As a facility admin you get a read-only view. Ask your super admin to make the change.

Why don't I see the Super Admin screen working? It only functions for super admins. If you open it without that flag you are sent to a Forbidden screen. This is by design.

Why can't I grant someone Admin Center access? Only a super admin can enable the Admin module's Admin action. For a regular admin, that one cell stays off even when you apply Full Access.

Why can't I assign the Administrator role to someone? You can only assign roles at or below your own rank. A manager can't create or promote into an admin-tier role; a super admin is unrestricted.

Why can't I find the Roles screen in the sidebar? It has no sidebar entry. Open it from the Admin Dashboard's setup checklist ("Canonical roles seeded" or "Role permission defaults"). ⚠ VERIFY — this is the only built-in path besides the direct address.

Why can't I delete this employee / space / list option? Hard-deleting employees is super-admin only — others should use Deactivate. A facility space or list option that is referenced by existing records can't be deleted at all; deactivate it instead so history stays intact.

I turned a module off but the person can still reach it by URL — why? The Modules switch only controls **navigation visibility**. The module is still protected by its own access rules; to actually remove someone's access, change their permissions in the Permissions screen.

I deactivated someone but they say they can still see things. Deactivation takes effect immediately for new sign-ins and blocks access, but an already-open session may need a refresh. A deactivated account cannot sign in, submit, or view data until reactivated.

Why is the permission matrix empty / greyed out for a user? A user with no facility assigned has nowhere to store permissions yet — assign them a facility from People first. A super-admin user bypasses module permissions entirely, so the toggles are informational unless their super-admin flag is removed.

Why did my export download the wrong columns? Column visibility is per module in PDF/Export Settings. Unchecked columns are intentionally hidden from exports (the data still exists in the database).

Source (footnote)

Key files verified for this chapter (paths for maintainers only; not part of the user-facing content):

- Access gating: `src/lib/auth/require-admin.ts`, `src/lib/auth/get-is-admin.ts`, `src/proxy.ts`, `src/lib/supabase/session.ts`
- Permission model: `src/lib/permissions/actions.ts` (modules × View/Submit/Edit/Admin, presets), `src/lib/permissions/levels.ts`, `src/lib/permissions/index.ts`
- Navigation & shell: `src/components/admin/nav-config.ts`, `src/components/admin/sidebar-nav.tsx`, `src/app/admin/layout.tsx`, `src/app/admin/page.tsx`
- Modules: `src/app/admin/modules/page.tsx`, `_components/module-toggles.tsx`, `actions.ts`, `src/lib/modules/module-keys.ts`
- Facility: `src/app/admin/facility/page.tsx`, `_components/{facility-form,read-only-view,facilities-table,new-facility-button,edit-facility-section}.tsx`, `actions.ts`, `types.ts`
- People/Employees: `src/app/admin/employees/{page.tsx,actions.ts,types.ts}`, `_components/{employee-form,employees-client,role-permission-preview,seed-roles-button}.tsx`, `[id]/_components/employee-detail.tsx`, `[id]/actions.ts`, `bulk/_components/bulk-add-client.tsx`, `bulk/{actions.ts,_lib/validation.ts}`

- Permissions: `src/app/admin/permissions/{page.tsx,user-permission-actions.ts,validators.ts}, [userId]/page.tsx, _components/permission-matrix.tsx`
- Roles: `src/app/admin/roles/{page.tsx,actions.ts}, _components/{role-manager,roles-matrix}.tsx`
- Departments / Spaces: `src/app/admin/departments/**, src/app/admin/spaces/**`
- Lists: `src/app/admin/lists/**, _lib/facility-dropdowns.ts`
- Exports: `src/app/admin/exports/{page.tsx,actions.ts,types.ts}, _components/{export-settings-form,run-export-panel}.tsx`
- Retention: `src/app/admin/retention/{page.tsx,actions.ts,types.ts}, _components/retention-row.tsx`
- Audit Log: `src/app/admin/audit-log/**`
- Super Admin: `src/app/admin/super-admin/{page.tsx,actions.ts,types.ts}, _components/{facilities-panel,super-admin-users-panel,invite-service-health-card}.tsx`

Daily Reports

1. What this module is for

The Daily Reports module is RinkReports' digital replacement for the paper "shift checklist" clipboard. Your facility is divided into **work areas** — front-of-house and operational zones such as Front Desk, Pro Shop, Custodial, Skate Sharpening, Concessions, Event Set Up, Learn to Skate, Public Skate, Locker Rooms, and Building Services. (Those ten are the example areas Tenuity Ice Pavilion uses; your facility's list is whatever your administrator has set up.)

For each area, an administrator builds one or more **checklists** (called "shifts" or "templates"). When you work a shift, you open Daily Reports, pick your work area and shift, tick off each task as you complete it, optionally leave a note, and submit. Managers can then review every area's reports for the day in the Admin Center.

Each work area is logged **independently** — submitting the Concessions checklist has nothing to do with whether the Locker Rooms checklist has been done. Every submission is a separate, permanent record (see Section 8 for exactly how saving and "locking" work — there is no single end-of-day report that fills up across areas).

You only see data for your own facility — this is automatic. There is no way to view or submit reports for another rink.

2. Who can use it

Daily Reports access is **permission-driven**, not strictly tier-driven: an administrator grants each person the ability to **view**, **submit**, **edit**, or **administer** the module, and — uniquely for this module — grants submit access **area by area** (the "Area Access" matrix). Two people with the same job title can therefore have different access if an admin has customized it.

The five-tier vocabulary used across this documentation maps onto the live app's roles as below. Because the real model is permission-based, treat the "Can do" column as the typical default, not a hard rule.

Role tier	Can see	Can do
super_admin	Everything, platform-wide	Full configuration plus delete; always has area access.
org_admin	△ VERIFY — no separate org-admin role exists in the live app; treat as super_admin.	△ VERIFY — same as super_admin in practice.
facility_manager (the live admin role)	All areas, all submissions, and the full admin console for their own facility	Create/edit areas, templates, checklist items; grant area access; review, edit, annotate, and delete submissions. Always has submit access regardless of the Area Access matrix.

Role tier	Can see	Can do
supervisor (the live manager role, or a custom role)	What their permissions allow	⚠ VERIFY — has no special daily-reports powers by default; sees and submits only where granted, exactly like staff unless an admin has enabled the module's admin action for them.
staff	The areas they've been granted submit access to, plus recent History for the facility	Submit daily reports for their granted areas; view History (read-only). No access to the admin console.

Anyone whose account is **deactivated** is denied everywhere. A signed-in user with **no area access granted** sees a "No areas assigned" message and can submit nothing until an admin grants access.

3. How to get there

Staff: In the left sidebar (or the bottom **Reports / Menu** tab on a phone), tap **Daily Reports**. This opens your reporting console at the Daily Reports page. From there, **View history** (top-right) opens the read-only History page.

The Daily Reports menu item only appears if your administrator has the module turned on for the facility and you have view/submit permission.

Administrators: Open the **Admin Center** (the "Admin Center" link appears in the staff sidebar for admins), then choose **Daily Reports Admin** under the "Module Admin" group.

4. Setup & configuration (admins)

The Daily Reports admin area is organized into five tabs: **Areas**, **Templates**, **Checklist Items**, **Area Access**, and **Submissions**. The first four are setup; Submissions is review (covered in Sections 5 and 6). A reminder banner notes that reports auto-delete after 14 days.

Daily Reports configuration is a facility_manager-level (admin) task. It is also cross-referenced from the **Admin Control Center**, where the **Modules** page carries the master ON/OFF switch that shows or hides Daily Reports in staff navigation.

Areas (the "tabs" staff see)

An **area** is one of the work zones staff pick from. On the **Areas** tab you can:

- **Add area** — give it a **Name** (required), an optional **Slug** (a URL-safe short name; auto-generated from the name if left blank — lowercase letters, digits, and hyphens), a **Color** swatch, and a **Sort order**.
- **Edit** an area, including toggling it **Active**.
- **Reorder** with the up/down arrows (this controls the order staff see).
- **Deactivate / Reactivate** — deactivating hides the area from staff without deleting its history.
- **Delete** — only succeeds if the area has no submissions; otherwise you're told to deactivate instead.
- **Bulk upload areas** — import many at once from a file.

The area cap. A facility may have at most **30 active areas** at one time. This limit is enforced by the database, not just the screen. The Areas tab shows a live "X / 30 active" badge; once you hit 30, the

Add area button reads "Cap reached" and reactivating another area is blocked until you deactivate one. (Note: some sales material mentions "up to 20 tabs" — the actual enforced cap is **30 active areas**.)

Templates (the "shifts" / checklists)

A **template** is a named checklist that belongs to a single area — for example "Opening checklist" or "Hourly inspection." On the **Templates** tab, first pick an area from the **Area** dropdown, then:

- **Add template** — **Name** (required), optional **Description**, and **Sort order**.
- **Edit**, **Deactivate / Reactivate**, and **Delete** (delete is blocked if any submission references it — deactivate instead).

An area can have several templates; the staff member chooses which one applies to the shift they're logging.

Checklist Items

A **checklist item** is a single tickable task within a template. On the **Checklist Items** tab, pick the **Area** and then the **Template**, then:

- **Add item** — **Label** (required; the text staff see, e.g. "Inspect ice surface for cracks") and an optional **Description** (a sub-line of guidance).
- **Edit**, reorder (up/down), **Deactivate / Reactivate**, and **Delete**.
- **Bulk upload** items into a template from a file.

A template with no items is allowed — staff can still submit it (it just records the area/shift and any note).

Area Access

The **Area Access** tab is a grid of **staff (rows) × areas (columns)** with a checkbox in each cell. Checking a box lets that staff member submit daily reports **in that area**. Notes:

- Changes save immediately as you toggle each box (no separate Save button).
- **Admins and super admins always have access**, regardless of the boxes.
- You can **Download CSV** (a template of the current grid) and **Import CSV** to set access in bulk; the import reports counts of granted / revoked / skipped rows and any issues.

5. Screen-by-screen walkthrough

Staff console (the Daily Reports page)

When you open Daily Reports you see a single scrolling page:

- **Header** — a breadcrumb (Reports > Daily Reports), the page title, and a **View history** button.
- **Meta-chip strip** — your name, your facility, today's date, and a **live clock** that updates every second.
- **Shift setup card** — two dropdowns:
 - **Work area** — lists only the areas you're allowed to submit to, each shown with its color swatch.

- **Shift** — lists the templates for the area you picked, with each one's item count beside it (e.g. "Opening checklist · 8 items"). If an area has exactly one shift, it is **auto-selected** for you; otherwise you'll see "Choose shift type..." until you pick.
- **Checklist card** — once a shift is chosen, its items appear as a list of large checkboxes (each with its optional description line). At the top is a "**X / Y complete**" counter and a **progress bar** that fills (and tints to the area's color) as you tick items. A template with no items shows "No checklist items on this template. You can still submit."
- **Note card** — an optional free-text box ("Anything to flag for managers?").
- **Sticky submit bar** — pinned to the bottom of the screen. It shows your progress ("X/Y complete") and a **Submit** button. The button stays disabled until a shift is selected. When you're offline it adds "· offline — will sync when reconnected."

After you submit, you land on a "**Submitted!**" confirmation screen with a green check, the area and shift name, the timestamp, an "X of Y items checked" summary, and **Submit another** plus a sign-out option.

History page

View history opens **Daily Report History** — a read-only list of recent daily reports for your facility (newest first). Each row shows the area (with its color dot), the shift/template name, the timestamp, who submitted it, and an "X/Y complete" badge. A note at the bottom reminds you reports auto-delete after 14 days. You cannot edit anything from here.

Admin tabs (Submissions review)

In **Daily Reports Admin → Submissions**, managers get a filterable table of submissions:

- **Filters** — by area, by employee, and by date range (defaults to the last 14 days).
- **Table** — Submitted time, Area, Template, Employee, Items (checked/total), Notes count, and an **Open** link.
- **Detail panel** (after clicking Open) — shows the full checklist with each item's checked/unchecked state, the list of notes, and admin actions: **toggle any item**, **add an admin note**, **edit or delete existing notes**, and **delete the submission**. There is a **Back to list** link to return to the filtered table.

6. Step-by-step: common tasks

Submit a daily report (staff)

1. Open **Daily Reports** from the sidebar (or Reports tab).
2. In **Work area**, pick the zone you're logging.
3. In **Shift**, pick the checklist (it may already be chosen if there's only one).
4. Tick each task as you complete it; watch the progress bar.
5. Optionally type a **Note** for managers.
6. Tap **Submit** in the sticky bar.
7. Confirm on the **Submitted!** screen, or tap **Submit another** to log a different area.

Add a new work area (admin)

1. Go to **Daily Reports Admin → Areas**.
2. Click **Add area** (if the "X / 30 active" badge shows you're at the cap, deactivate one first).

3. Enter a **Name**, optionally adjust the **Slug**, pick a **Color**, set a **Sort order**.
4. Click **Create area**. It now appears as an option in the staff Work area dropdown for anyone granted access.

Build a checklist template (admin)

1. Go to **Daily Reports Admin → Templates** and choose the **Area** from the dropdown.
2. Click **Add template**, give it a **Name** (and optional description), then **Create template**.
3. Switch to the **Checklist Items** tab, choose the same **Area** and your new **Template**.
4. Click **Add item** for each task (Label required), or use **Bulk upload** to paste many at once.
5. Reorder items with the up/down arrows so they read top-to-bottom the way staff work.

Grant area access to an employee (admin)

1. Go to **Daily Reports Admin → Area Access**.
2. Find the employee's row and the area's column.
3. **Check the box** — it saves immediately. (To grant many at once, use **Download CSV**, edit the `can_submit` column, and **Import CSV**.)

The employee will now see that area in their Work area dropdown the next time they open Daily Reports.

Review or annotate a submission (admin)

1. Go to **Daily Reports Admin → Submissions** and use the filters to narrow the list.
2. Click **Open** on a row.
3. Add an **admin note**, adjust a checkbox if needed, or **Delete submission**. Notes you add are visible to admins and to staff on the report.

7. Field reference

Staff console

Field	Type	Required	Notes
Work area	Dropdown	Yes	Only areas you have submit access to; shows a color swatch.
Shift	Dropdown	Yes	Templates within the chosen area; shows each one's item count; auto-selected if only one exists.
Checklist items	Checkboxes	No	One per active item on the template; drive the progress bar. A template may legitimately have zero items.
Note	Text area	No	Free text for managers; saved with the submission.

Area form (admin)

Field	Type	Required	Notes
Name	Text	Yes	What staff see in the Work area dropdown.
Slug	Text	No	Auto-generated from the name if blank; lowercase letters, digits, hyphens only.
Color	Color picker	No	Accents the area in dropdowns, the checklist, and History.
Sort order	Number	No	Lower numbers appear first.
Active	Checkbox	No (edit only)	Inactive areas are hidden from staff but keep their history; counts against the 30-active cap only while active.

Template form (admin)

Field	Type	Required	Notes
Name	Text	Yes	E.g. "Opening checklist."
Description	Text	No	Optional sub-line shown in the admin table.
Sort order	Number	No	Order of shifts within the area.
Active	Checkbox	No (edit only)	Inactive templates are hidden from staff.

Checklist item form (admin)

Field	Type	Required	Notes
Label	Text	Yes	The task text staff tick off.
Description	Text	No	Optional guidance line under the label.
Sort order	Number	No (edit only)	New items are auto-placed at the end.
Active	Checkbox	No (edit only)	Inactive items don't appear on new submissions.

8. Locking, saving & offline

There is no end-of-day "report" that locks. Unlike a single paper sheet that fills up over a shift, RinkReports treats each submit as its own complete record. Specifically:

- **Saved vs submitted.** While you're filling in the checklist, nothing is stored on the server — your ticks live in the page until you tap **Submit**. Pressing Submit creates the report.

- **Append-only.** Every Submit creates a **new, permanent record**. Staff cannot go back and edit a submitted report; if something was wrong, you simply submit again — the new submission is a separate row, and both are kept for the audit trail. Each report is automatically tagged with your facility's local calendar date so managers can group a day's reports, but a new day doesn't "close" the previous one — there is no lock step.
- **What admins can change.** Managers don't reopen a staff report for the staffer to edit; from the Submissions detail panel they can toggle items, add/edit/delete admin notes, or delete the whole submission.
- **Auto-delete.** Daily reports are kept for **14 days** and then automatically purged. History and the admin Submissions list only show what's still within that window.

⚠ **VERIFY** — "the report locks/submits at end of day." The training package described an end-of-day lock. The code does **not** implement a per-day lock or a single rolling report; it uses an append-only model (one immutable record per Submit, tagged with the facility-local date, auto-deleted after 14 days). Describe it that way.

Offline ("Save on this device"). Daily Reports works without a connection:

- If you tap Submit while offline, the report is **saved on your device** and queued. The confirmation reads "**Saved on this device**" and explains it will submit automatically once you're back online — and that the same checks run then. You can keep working.
- The sticky bar shows an "offline — will sync when reconnected" hint so you know you're in offline mode.
- When your connection returns, the queued report submits on its own. Each queued report carries a unique identifier so a retry can't create a duplicate, and the same area, shift, and permission checks that run online are re-applied when it syncs. If access was revoked while you were offline, the queued report can be rejected at sync time.

9. Troubleshooting & FAQ

"No areas assigned" / the Work area dropdown is empty. You haven't been granted submit access to any area. Ask an admin to check the **Area Access** grid in Daily Reports Admin.

"Account not ready" or "Your account isn't fully set up yet." Your employee record isn't active or fully provisioned. Contact your administrator.

I can't find Daily Reports in my menu. Either the module is turned off for your facility (an admin controls this on the **Modules** page) or you lack view permission.

The Shift dropdown says "No shifts available." The selected area has no active templates yet. An admin needs to add at least one template to that area.

I ticked the wrong box / submitted by mistake. You can't edit a submitted report. Submit a corrected one — both are kept — and, if needed, ask a manager, who can adjust or delete the submission from the Submissions tab.

Admin: "Cap reached" when adding an area. Your facility already has 30 active areas (the database limit). Deactivate an area you no longer use, then add the new one.

Admin: "Cannot delete area/template with existing submissions." Deletion is blocked once reports reference the item. Use **Deactivate** instead — it hides the area or template from staff while preserving history.

My report didn't show up. If you submitted offline, it's queued and will appear once it syncs. Also remember History only shows the last **14 days**; older reports are auto-deleted.

Can I see another facility's reports? No. You only see data for your own facility — this is automatic.

Source

```
src/app/reports/daily/ (page.tsx, actions.ts, _components/daily-report-console.tsx,
history/page.tsx, [areaSlug]/[templateId]/done/page.tsx, _lib/submit.ts, _lib/compute.ts,
_lib/offline.ts); src/app/admin/daily-reports/ (page.tsx, actions.ts, area-access-actions.ts,
types.ts, _components/areas-tab.tsx, templates-tab.tsx, items-tab.tsx, area-access-tab.tsx,
submission-detail.tsx, area-form.tsx, item-form.tsx);
supabase/migrations/000000000000007_daily_reports_schema.sql (30-active-area cap trigger, 14-day
retention). Cross-referenced with docs/training/00-MANIFEST.md.
```


Refrigeration Logs

RinkReports training guide — Refrigeration Logs module

1. What this module is for

The **Refrigeration Logs** module is where staff record the readings for your rink's refrigeration plant — compressors, pumps, condensers, brine/supply-return temperatures, machine hours, alarms, and anything else your facility has set up to track. Each time someone walks the plant and takes a "round" of readings, they open this form, enter the numbers, and submit.

The module does three things for you:

- **Captures readings** for each piece of equipment in a structured, repeatable way, so every round looks the same and nothing gets skipped.
- **Checks each reading against a normal range** as it's entered, and shows a "Normal: min – max" hint right under the field so staff know immediately whether a value looks right.
- **Flags out-of-range readings** and — for the most serious ("critical") ones — requires the person to write down what corrective action they took before they can submit. If your facility has turned on out-of-range alerts, managers also get notified.

Every submitted report is a permanent record. You don't edit a past report — you submit a new round. Admins can review every submission and add follow-up notes, but the original reading is never changed.

You only see data for your own facility — this is automatic. There's no facility switcher, and you can't see another rink's refrigeration logs.

2. Who can use it

Access is **permission-driven**, not strictly tied to a job title. The table below shows the typical default for each tier, but an admin can grant or remove any of these on a per-person basis. The four actions that matter are **View**, **Submit** (file a reading), **Edit**, and **Admin** (configure the module).

Tier	Submit a reading	Review history / add follow-up notes	Configure (Setup & Settings)
super_admin	Yes	Yes	Yes
org_admin △ VERIFY	No exact equivalent in the live app — treat as super_admin.	—	—
facility_manager (the admin role)	Yes	Yes	Yes
supervisor (the manager role, or a custom role) △ VERIFY	Typically yes (needs the submit permission)	Only if granted the admin action	Only if granted the admin action
staff	Yes (needs the submit permission)	No access to the admin History/Setup screens	No access

Notes that hold true in code:

- To open and submit the staff form, you need the `submit` permission for the refrigeration module **and** an active employee account assigned to your facility. Without it, you see a "No permission" message.
- The whole **Refrigeration Admin** area (Setup / History / Settings) requires **admin-level access** (the global super-admin flag, or the `admin` permission for your facility). Non-admins are shown a "Forbidden" message rather than the screens.
- A **deactivated** account is denied everywhere — it can't sign in, submit, or view.

⚠ **VERIFY** — the rows for supervisor and the `org_admin` tier are mapped to the live `super_admin` / `admin` / `manager` / `staff` model from the manifest. Because access is per-person and per-action, confirm against your configured facility before treating any of these as a hard rule.

3. How to get there

Staff (submitting a reading):

- In the left sidebar (or the mobile **Menu** tab), choose **Refrigeration**. The address is `/reports/refrigeration`.
- If the module is turned off for your facility, or you lack the submit permission, the menu item won't be there (or you'll get a "Not available" message).

Admins (configuring or reviewing):

- Open the **Admin Center** (the "Admin Center" link appears in the staff sidebar for admins), then choose **Refrigeration Admin** under **Module Admin**. The address is `/admin/refrigeration`.
- The admin screen opens on the **Setup** tab by default, with **History** and **Settings** tabs alongside it.

4. Setup & configuration (admins)

All refrigeration configuration lives on the **Refrigeration Admin** page (`/admin/refrigeration`), split across three tabs: **Setup**, **History**, and **Settings**. Changes here change what staff see on the reading form. For the broader admin picture — modules on/off, permissions, exports, retention — see the **Admin Control Center** chapter.

Getting started fast: Seed defaults

If your facility has **no sections yet**, the Setup tab shows a "**No sections yet**" card with a **Seed defaults** button. Clicking it creates six standard sections — **Compressors, Pumps, Condensers, Supply / Return, Machine Hours, Alarms** — plus a default settings row. You can then rename, add to, or delete any of them. You can also skip seeding and build everything by hand with **Add section**.

Sections

A **section** is a card on the staff form (for example, "Compressors" or "Brine"). On the Setup tab, sections are listed down the left; each entry shows its name and a count of "N equipment, N fields." Pick one to manage it.

For each section you can:

- **Add section** — name (required) and an optional slug (auto-generated from the name).
- **Rename** — change name, slug, and sort order (controls the order cards appear in).
- **Deactivate / Activate** — a deactivated section is hidden from the staff form but kept for history.

- **Delete** — removes the section. This is blocked if equipment, fields, or reports still reference it; the app suggests deactivating instead.

Equipment

Within a section, **Equipment** is the list of individual units — e.g. "Compressor 1," "Compressor 2." The count of compressors (or any equipment) is entirely admin-configurable: add as many as your plant has. Each piece of equipment gets its own labeled group of fields on the staff form.

For each section you can **Add equipment** (name, optional slug) and, per item, **Edit** (name, slug, sort order), **Deactivate / Activate**, or **Delete**.

Fields

Fields are the actual reading inputs. A field can be **section-level** (not tied to a specific unit) or attached to a **specific piece of equipment**. For each field you set:

- **Label** — what staff see (e.g. "Suction pressure").
- **Key** — an internal identifier (auto-generated from the label if left blank).
- **Type** — one of:
 - **numeric** — a number entry; the only type that supports thresholds and the "Normal:" range hint.
 - **text** — free text.
 - **boolean** — a yes/no checkbox.
 - **select** — a dropdown; you supply the options, one per line as `key|Label`.
 - **computed** — a read-only value calculated automatically on submit from other numeric fields in the same section. Staff see "Calculated automatically on submit" and never type into it.
- **Unit** — e.g. psi, °F. (Temperature units drive the °F/°C toggle behavior — see §5.)
- **Sort order** — reorder fields with the ↑ / ↓ buttons on each row.

Each field can be **Deactivated / Activated** or **Deleted**, and fields are also marked required where configured (a required field shows a red mark to staff and blocks submit if empty).

Thresholds (normal ranges)

A **threshold** defines the normal range for a numeric field and how serious it is to fall outside it. Open a numeric field's **Thresholds** panel (the "Thresholds" button on the field row) to add one. Each threshold has:

- **Min** and/or **Max** — the normal range. You can set just one side. This is exactly what staff see as the "Normal: min - max" hint.
- **Severity** — **warn**, **high**, or **critical**.
- **Scope** — for a section-level field, you can scope a threshold to **All equipment** or to one specific unit. An equipment-specific threshold wins over a section-wide one for that unit.

Severity matters at submit time:

- A reading outside **any** active threshold is flagged **out-of-range** and counted on the confirmation screen and in History.
- A reading outside a **critical** threshold additionally **forces the staff member to write a corrective-action note** before they can submit (see §5).

Thresholds can be **Edited**, **Deactivated / Activated**, or **Deleted**.

Settings (the Settings tab)

There is one settings row per facility, controlling alerting and the per-shift cap:

- **Enable out-of-range alerts** — a checkbox. When on, submitting a report with any out-of-range reading raises an alert to managers (a banner reminds staff: "Out-of-range readings will trigger an alert to managers.").
- **Default alert severity** — **warn / high / critical** — used when a triggered threshold doesn't carry its own severity.
- **Readings per shift** — a number from **1 to 99**, or **blank for unlimited**. This is the expected number of reading rounds per shift. When set, it (a) caps the Round number staff can enter and (b) shows staff the "of N" indicator and a "This facility logs N readings per shift" hint. The server rejects a round number above this cap.

Click **Save settings** to apply.

5. Screen-by-screen walkthrough

5.1 Staff reading form (/reports/refrigeration)

Header. At the top: a "Reports > Refrigeration" breadcrumb, the page title **Refrigeration**, and two buttons — **Back** (returns to the previous page) and **Dashboard** (jumps to your dashboard).

Meta-chip strip. A row of small chips just below the header, each with an icon: **your name**, **facility name**, **today's date**, the **live time** (it ticks each second), and the current **outdoor temperature** for your facility (e.g. "72°F · location"). If weather isn't available it reads "Temp unavailable."

Offline banner. If you're offline, a small note appears: "You're offline. Your report will be saved on this device and submitted automatically when you reconnect." (See §8.)

Alerts banner. If your facility has out-of-range alerts turned on, a note reminds you that out-of-range readings will alert managers.

Log Information card. This card holds the round's basics and the unit toggle:

- **°F / °C toggle** — a switch in the card header. It changes only how temperatures are **displayed**; values are always stored canonically in **°F**. Flipping it converts any temperature values you've already typed, once, to the new display unit. Non-temperature fields (like *psi*) are unaffected.
- **Facility** and **Employee** — shown read-only.
- **Reading time** — a date-and-time picker, pre-filled with now. Adjust it if you're logging a round you took earlier.
- **Shift (optional)** — free text, e.g. "AM / PM / Overnight."
- **Round # (optional)** — the round number for this shift. If your facility set a readings-per-shift cap, the label reads **"Round # (optional) — of N"** and a hint below says how many readings the facility logs per shift. Entering a round above the cap is rejected on submit.

Section cards. One card per active section, in the order admins set. Within a card:

- **Section-level fields** appear first in a responsive grid.
- Each **piece of equipment** gets its own sub-heading (e.g. "Compressor 1") with its fields grouped beneath it.

Field types as they appear:

- **numeric** — a number box (decimal keypad on mobile) with the unit shown beside it and appended to the label, e.g. "Suction pressure (psi)." Temperature fields show "(°F)" or "(°C)" matching the toggle.
- **text** — a single-line text box.
- **boolean** — a checkbox with the field label.
- **select** — a dropdown of the admin-defined options.
- **computed** — a greyed, read-only box reading "Calculated automatically on submit."

Normal-range hints. Under each numeric field that has a threshold, a muted line reads "**Normal: min - max unit**" (or " $\geq$ min" / " $\leq$ max" if only one bound is set). When you flip the °F/°C toggle, temperature ranges convert to match.

Corrective-action box. If a numeric value breaches a **critical** threshold, a highlighted **Corrective action** box appears under that field with a "(required)" mark. You must describe the action taken; submit is blocked until you do. (Warn/high out-of-range readings are still flagged, but don't force a note.)

Notes. A general **Notes (optional)** card near the bottom for any overall comments about the round.

Submit. A full-width button at the bottom:

- Online: "**Submit refrigeration report.**"
- Offline: "**Save on this device.**"
- While saving: "**Submitting...**"

If you try to submit with a missing required field, an unparseable number, or a missing critical corrective-action note, the form highlights the problem and jumps you to the first one.

Your recent submissions. Below the form, if you've submitted in the last 30 days, a read-only list shows each report's time, a "N values" badge, and an "N out-of-range" badge where applicable.

5.2 Confirmation screen (/reports/refrigeration/done)

After a successful online submit you land on a "**Submitted!**" screen with a green checkmark, the submission timestamp, a "N values recorded" chip, and — if any reading was out of range — an "N out-of-range" chip. Two buttons: **Submit another** and **Back to home**.

(When you submit **offline**, you instead see an inline "**Saved on this device**" confirmation with a **Back to dashboard** button — the report syncs later.)

5.3 Admin — Setup tab

Covered in detail in §4. Left column: the section list (with equipment/field counts and an "off" badge for inactive sections) plus an **Add section** card. Right column: the selected section's **Equipment**, **Section-level fields**, per-equipment field groups, and inline **Thresholds** panels — each with Add / Edit / Deactivate / Delete controls.

5.4 Admin — History tab

A filterable, read-only table of submitted reports. Filters include **employee**, a **date range** (defaults to the last 14 days), an **out-of-range** filter (yes/no), and a **notes search**. Columns: **Submitted**, **Submitter**, **Values**, **OOR** (out-of-range count badge), **Notes** (excerpt), and a **View** link. **Load more** appears when there are additional rows. There's also an **Export** button in the page header.

Clicking **View** opens the **report detail**: who submitted it and when, the reading-taken time / shift / round, the submitter's notes, then **Recorded values** grouped by equipment (each value shown with

its unit and an **Out of range / OK** status), and a **Follow-up notes** section. Admins can **Add follow-up note** here — notes are append-only and the original report is immutable (you cannot edit a submitted reading).

5.5 Admin — Settings tab

The single settings form described in §4: out-of-range alerts on/off, default alert severity, and readings-per-shift.

6. Step-by-step: common tasks

Log a refrigeration reading (staff)

1. Open **Refrigeration** from the menu.
2. In **Log Information**, confirm or adjust the **Reading time**; optionally type the **Shift** and **Round #**.
3. (Optional) Flip the **°F / °C** toggle to your preferred temperature display.
4. Work down each section card, filling in the fields for every piece of equipment. Watch the **"Normal: ..."** hints.
5. If a reading turns **critically** out of range, fill in the **Corrective action** box that appears.
6. Add any overall comments in **Notes**.
7. Tap **Submit refrigeration report** (or **Save on this device** if offline).
8. On the **Submitted!** screen, choose **Submit another** or **Back to home**.

Add a compressor (or any equipment) / add a section (admin)

1. Go to **Admin Center > Refrigeration Admin > Setup**.
2. To add a unit: pick the section (e.g. "Compressors"), then in **Equipment** use **Add equipment** — type the name (e.g. "Compressor 3") and **Add equipment**.
3. To add a whole new section: use **Add section** in the left column.
4. Add the **fields** each new section/equipment needs (see next task).

Add a reading field

1. On **Setup**, select the section.
2. In the **Section-level fields** block (or a specific equipment's "Fields for ..." block), use **Add field**.
3. Enter the **Label**, choose the **Type** (numeric / text / boolean / select / computed), set the **Unit** if any, and — for **select** — list options as key|Label, one per line.
4. **Add field**. Reorder with **↑ / ↓** as needed.

Set a normal range / threshold

1. On **Setup**, find the **numeric** field and click **Thresholds**.
2. In **Add threshold**, set **Min** and/or **Max**, pick the **Severity** (warn / high / **critical**), and (for a section-level field) choose the **Scope** — All equipment or one unit.
3. **Add threshold**. Staff will now see "Normal: min – max" under that field, and critical breaches will demand a corrective-action note.

Set readings per shift

1. Go to **Refrigeration Admin > Settings**.
2. In **Readings per shift**, enter a number from **1-99** (or leave blank for unlimited).

3. **Save settings.** Staff now see "Round # — of N" and can't submit a higher round number.

Turn on out-of-range alerts (admin)

1. Go to **Settings**, check **Enable out-of-range alerts**, and pick a **Default alert severity**.
2. **Save settings.** Submissions with out-of-range readings now raise a manager alert that requires acknowledgement.

Review a report and add a follow-up note (admin)

1. Go to **History**, filter as needed, and click **View** on a report.
2. Read the recorded values and any corrective-action notes.
3. In **Add follow-up note**, type your note and **Add note**. The original report stays unchanged.

7. Field reference

Round / log fields (staff form, Log Information card):

Field	Type	Required	Notes
Reading time	Date & time	No (defaults to now)	When the round was taken.
Shift	Text	No	Free text, e.g. AM / PM / Overnight.
Round #	Whole number	No	Capped at the facility's readings-per-shift value; shows "of N" when set.
Notes	Long text	No	General comments about the round.
Corrective action	Long text	Yes, when a value breaches a critical threshold	Appears inline under the offending numeric field.

Reading fields are entirely admin-defined per facility, so there is no fixed list. Each follows its configured type:

Field type	What staff enter	Normal/expected range shown?
numeric	A number, in the field's unit	Yes — "Normal: min – max unit," from the field's threshold(s)
text	Free text	No
boolean	A yes/no checkbox	No
select	One option from a dropdown	No
computed	(nothing — read-only, calculated on submit)	Shows a range only if a threshold is set on the computed value

Default seeded sections (if an admin clicks Seed defaults): Compressors, Pumps, Condensers, Supply / Return, Machine Hours, Alarms. Specific equipment and fields within them are configured per facility.

The exact reading labels, units, and normal ranges at your rink depend on how your administrator configured Setup and Thresholds. Use the on-screen "Normal: ..." hints as the source of truth for what's expected.

8. Locking, saving & offline

Append-only. A submitted refrigeration report is a permanent record. There is **no edit** of a past report — you submit a **new** round instead. Admins don't change submitted values either; they **add follow-up notes**, which are themselves append-only. The detail screen states plainly: "The original report is immutable."

Saving online. Tapping **Submit refrigeration report** files the report, records every value (with out-of-range flags), saves any corrective-action notes against the exact readings they address, optionally raises a manager alert, and takes you to the **Submitted!** screen.

Offline ("Save on this device"). RinkReports is a progressive web app, so the form works without a connection:

- When you're offline the Submit button reads **"Save on this device,"** and an on-screen note explains the report is queued locally.
- After saving offline you see a **"Saved on this device"** confirmation. The report is held on your device and **submits automatically when you reconnect**, re-running the same checks (required fields, critical notes, and the readings-per-shift cap) on the server.
- A pending/failed indicator and the offline-queue page let you see anything still waiting to sync.
- Because each queued report carries a unique id, a retry can't create a duplicate.

Don't close the tab with unsaved readings still on screen — the app warns you before a refresh or close would discard them (until the report has been queued or submitted).

9. Troubleshooting & FAQ

"No permission" when I open Refrigeration.

You don't have the **submit** permission for this module. Ask an admin to enable it for you in the Permissions area.

"Account not set up."

Your sign-in isn't linked to an active employee record at a facility. Contact your administrator.

"Not configured yet" / "Refrigeration reporting isn't configured yet."

No active sections or fields exist for your facility. An admin needs to seed defaults or build sections, equipment, and fields on the Setup tab.

The Refrigeration menu item is missing.

Either the module is switched off for your facility (admins control this on the Modules page) or you lack the view/submit permission.

It won't let me submit — a corrective-action box is red.

A numeric reading is **critically** out of range. Fill in the **Corrective action** box for that field, then submit.

My round number was rejected.

It exceeded the facility's **readings-per-shift** cap. Use a number within "of N," or ask an admin to raise the cap on the Settings tab.

I flipped to °C and my entered numbers changed.

That's expected — the toggle converts displayed temperatures. Values are stored in °F regardless; only the display changes.

I made a mistake on a submitted report.

You can't edit it. Submit a corrected new round. For the record, an admin can add a follow-up note explaining the correction.

Do out-of-range readings always alert managers?

Only if an admin has enabled **out-of-range alerts** on the Settings tab. When enabled, an alert is raised (requiring acknowledgement) summarizing the out-of-range readings.

Can I see another rink's logs?

No. You only see data for your own facility — this is automatic.

Source

Staff flow: `src/app/reports/refrigeration/page.tsx`,
`src/app/reports/refrigeration/_components/submission-form.tsx`,
`src/app/reports/refrigeration/done/page.tsx`, `src/app/reports/refrigeration/actions.ts`,
`src/app/reports/refrigeration/_lib/compute.ts`, `src/app/reports/refrigeration/_lib/submit.ts`.
Admin flow: `src/app/admin/refrigeration/page.tsx`, `src/app/admin/refrigeration/types.ts`,
`src/app/admin/refrigeration/_components/setup-tab.tsx`, `.../settings-tab.tsx`, `.../history-tab.tsx`,
`.../report-detail.tsx`, `.../seed-defaults-card.tsx`. Role/permission context:
`docs/training/00-MANIFEST.md`.

Incident Reporting

Product: RinkReports — the Next.js progressive web app (PWA) for ice-rink operations. **Brand colors:** #4DFF00 (primary green) · #002244 (navy). **Data scoping:** You only see data for your own facility — this is automatic. You never switch facilities or see another rink's incidents; RinkReports keeps each facility's records separate on its own.

1. What this module is for

Incident Reporting is how staff record something that went wrong on the ice or in the building — a slip, a collision, an altercation, an equipment problem, a guest who needed medical help. A report captures **when and where** it happened, **what happened**, **how serious it was**, whether an **ambulance was called**, who was **involved**, and any **witnesses**. Once filed, the report goes to managers/admins to review, track follow-ups, and mark resolved.

Two things make this module different from most other RinkReports forms:

- **You can correct your own report for 24 hours** after submitting. After that window closes, it locks and becomes read-only (see §8). Most other report types are one-and-done with no edit window.
- There is **no photo or file upload** anywhere in this module. You describe what happened in words; you do not attach images, video, or documents.

2. Who can use it

Access is **permission-driven**, not strictly tier-driven: an administrator can turn each person's access on or off per action (view / submit / edit / admin). The table below shows the typical default mapping. Where a label feels rigid, treat it as the default and confirm against your configured facility.

Role tier (doc vocabulary)	Typical access to Incident Reporting
super_admin	Full: file reports, edit own within 24h, and full admin (History, Types, Severities, Activities, status changes, follow-up notes).
org_admin	⚠ VERIFY — the live app has no separate org_admin tier . Treat as super_admin.
facility_manager (live role: admin)	Full admin for their own facility: review every report in History, change status, add follow-up notes, and configure Types / Severities / Activities. Can also file reports.
supervisor (live role: manager or a custom role)	⚠ VERIFY — typically can file reports and edit their own within 24h. Admin review/config only if granted the admin action for this module.
staff (live role: staff or a custom role)	File incident reports and edit their own within the 24-hour window. No access to the admin console (History/Types/Severities/Activities, status, follow-up notes).

Notes confirmed in code:

- To even open the staff incident page, you need the **submit** permission for Incident Reporting. Without it you see a "No access" message ("You don't have permission to submit incident reports.").
- The admin console requires admin-level access; staff who try to reach it are shown a Forbidden message, not the admin screens.
- A **deactivated** account cannot file, view, or edit anything.

3. How to get there

Staff (to file or view your own reports):

- Open the staff app and choose **Incidents** from the sidebar (desktop) or the **Menu** tab (mobile bottom bar: Home · Reports · Menu · Account).
- This opens **Incident Reports** (/reports/incidents): the new-report form, followed by **Your recent reports** (your own submissions from the last 30 days).
- The Incidents menu item only appears if your facility has the module turned on (admins control this on the Modules page) and you have permission.

Admins (to review and configure):

- Open the **Admin Center**, and under **Module Admin** choose **Incident Reports Admin** (/admin/incident-reports).
- The admin page opens on the **History** tab, with tabs across the top for **History** · **Incident Types** · **Severity Levels** · **Activities**.

4. Setup & configuration (admins)

Configuration lives in the **Incident Reports Admin** screen (/admin/incident-reports). Original submitted reports are **immutable** — admins never edit a staffer's report in place; they change its **status** and add **follow-up notes** (see §5).

The header carries two shortcuts: **Manage locations** (jumps to the shared Facility Spaces list) and an **Export** button for this module.

Severity Levels tab

Severity levels rank how serious an incident was (e.g. Critical, High, Medium, Low). This is the **required** picker on the staff form — a facility must have at least one active severity or staff see "Not configured yet."

- **Add severity** opens a side panel with: **Key** (lowercase letters, digits, underscores — e.g. high, very_high; must be unique in your facility), **Display name** (what staff see), **Color**, and **Sort order**.
- Each row can be **Edited**, **Deactivated** / **Reactivated**, or **Deleted**. Deactivating hides it from the staff picker while keeping it on past reports.
- **Delete is blocked** if any reports already use that severity — you'll be told "Cannot delete; in use by N reports. Deactivate instead."
- If no severities exist yet, a **Seed defaults** card offers the four standard levels: **Critical**, **High**, **Medium**, **Low**. Seeding is safe to run more than once (it skips duplicates).

Activities tab

Activities describe what was going on when the incident happened ("Activity at the time" on the staff form) — e.g. Public Skating, Hockey, Figure Skating. This picker is **optional** for staff, and staff can

always choose "**Other...**" and type a free-text activity instead.

- **Add / Edit / Deactivate / Reactivate / Delete** work the same way as severities (Key + Display name + Color + Sort order). Delete is blocked when an activity is in use; deactivate instead.
- **Seed default activities** offers Public Skating, Hockey, Figure Skating, Learn to Skate, and Maintenance.
- A **bulk import** card accepts a simple CSV (display_name, key, color, sort_order, one per line). Duplicate keys are skipped, not overwritten.

Incident Types tab

Incident Types categorize reports (e.g. Slip / Fall, Equipment, Altercation).

- **Add incident type** captures **Name**, **Slug** (auto-derived from the name if left blank), **Color**, and **Sort order**, with the same Edit / Deactivate / Reactivate / Delete controls. Delete is blocked when a type is in use.
-  **VERIFY** — **Type is shown and filtered on in the admin History view, but the current staff submission form does not present an incident-type picker.** Type appears as a column/filter and on the report detail; it is not collected from staff in the live form. Confirm whether your facility expects staff to pick a type before training on it as a staff field.

Facility Spaces (the "where" list) — lives elsewhere

The **Facility space** picker on the staff form is **not** configured inside this module. It is fed by the shared **Facility Spaces** list in the Admin Center (**Setup** → **Facility Spaces**, /admin/spaces), which is the same list used by Accident and Air Quality. Use the **Manage locations** button on the Incident Reports Admin header to jump straight there. Add, rename, reorder, or deactivate spaces there and they appear (or disappear) in the staff picker automatically. Staff can always add an "**Other**" free-text space when the right location isn't listed.

Cross-reference — Admin Control Center. Module visibility (whether Incidents shows in staff navigation at all) is set on the **Modules** page; who can file or administer is set under **Permissions / Roles**; the location list is **Facility Spaces**. See the Admin Control Center chapter for those shared screens.

5. Screen-by-screen walkthrough

A. The staff incident form (/reports/incidents)

The page opens with an "Incident Reports" header, then the form, then **Your recent reports** below. The form is grouped into cards:

Card 1 — When & where

- **When did it happen?** (required) — a date-and-time picker, pre-filled with the current date/time. Adjust it to when the incident actually occurred.
- **Facility space** (required) — a dropdown you can **multi-select**; pick one or more spaces. If there are many spaces a **search box** appears. At the bottom is an **Other** option — tick it and a free-text box appears to **describe the space**. You must pick at least one space or fill in an "Other" space.

Card 2 — What happened

- **Description** (required) — a free-text box, up to **500 characters** (a live counter shows how many you've used). Describe what happened in detail.

- **Activity at the time** (optional) — a dropdown of admin-defined activities, plus "**Other...**" which reveals a free-text box.
- **Severity** (required) — a dropdown of admin-defined severity levels.
- **Immediate actions taken** (optional) — a free-text box for what was done right after.
- **Was an ambulance called?** — a Yes/No toggle (defaults to No). Turning it on raises a higher-priority alert to the facility (see §8).
- **Number of people involved** (optional) — a numeric field; accepts a whole number (0 or more) only.
- **Follow-up required** — a Yes/No toggle (defaults to No).

Card 3 — Witnesses (optional)

- Add up to **3 witnesses**. Use **Add a witness / Add another witness**; each row has **Name**, **Phone**, **Email**, and a **Brief statement**, and a **Remove** button. A counter shows n/3.
- Rule per witness: if you start a witness you must give a **name** and **at least one contact** (phone or email). Empty witness rows are simply ignored.

Reporter identity is automatic. The form does **not** ask for your name or phone — RinkReports fills the reporter from your login so it can't be spoofed.

Submit & the "Are you sure?" dialog. The submit button reads **Submit incident report**. Tapping it first checks the required fields, then opens a confirmation dialog: "**Submit this incident report?**" with the reminder that you can edit it for 24 hours, after which it becomes read-only. Choose **Confirm & submit** to file it, or **Go back** to keep editing.

B. The "Reported" confirmation screen (.../done)

After submitting you land on a confirmation screen with a green check and "Reported — Thank you, [your name]. Your report has been submitted." It shows a summary (Status, Severity, when it happened, when submitted, and any "Other space"), a note that "You can edit this report for 24 hours," and three buttons: **Edit report**, **Submit another**, **Back to home**.

C. Your recent reports list

Below the form, **Your recent reports (Last 30 days)** lists your own submissions. Each entry shows the time submitted, a colored **severity** pill, a **status** badge (Submitted / In review / Resolved / Archived), any "Other" location, and a short excerpt of the description. Click **Incident** on a row to open that report.

D. The single-report view / 24-hour edit (/reports/incidents/[id])

Opening one of your reports does one of two things:

- **Within 24 hours, and it's yours:** the **Edit Incident Report** form opens, pre-filled with everything you entered, with an "Editable until [date/time]" note. Make changes and **Save changes**; a confirmation dialog ("Save changes to this report?") reminds you the change is recorded in the report's history. After saving you see a "Changes saved" banner and can keep editing until the window closes.
- **After 24 hours, or if it isn't yours:** a **read-only** view of the report. If it's yours, it says "The 24-hour edit window for this report has closed, so it's now read-only." A **Back to incident reports** button returns you to the list.

E. Admin: Incident Reports Admin (/admin/incident-reports)

History tab — a filterable table of every incident at your facility (most recent first). Columns: Submitted, Reporter, Type, Severity, Location, Status, and a **View** link. Filters let you narrow by **status, type, severity, employee, location, and a date range (from/to)**. Click **View** to open the detail panel.

Report detail panel — opens from History → View. It shows:

- The **Original report** (read-only, never editable here): Type, Severity, Activity (or the "(other)" text), Occurred-at, Reporter name & phone, **Ambulance called** (highlighted when Yes), People involved, Follow-up required, Facility spaces (including any "other"), Description, and Immediate actions.
- **Witnesses** — each name with phone/email and statement.
- **Status** — a **Change status** dropdown (Submitted → In review → Resolved → Archived) and a timeline of when each status was reached (Submitted / Reviewed / Resolved / Archived).
- **Follow-up notes** — an append-only thread. Type into **Add follow-up note** and **Add note**. Notes show author and time and **cannot be edited or deleted** once added.
- **Change log** — an append-only audit trail of edits (create/update) with who and when. Visible to admins only.

Incident Types · Severity Levels · Activities tabs — the configuration lists described in §4.

6. Step-by-step: common tasks

File an incident report (staff)

1. Sidebar/Menu → **Incidents**.
2. Set **When did it happen?**
3. Open **Facility space** and tick one or more spaces (or tick **Other** and describe it).
4. Write the **Description** (required, up to 500 characters).
5. Optionally pick **Activity** (or "Other..."), set **Severity** (required), add **Immediate actions**, toggle **Ambulance called** and **Follow-up required**, and enter **people involved**.
6. Optionally add up to **3 witnesses** (each needs a name + phone or email).
7. Tap **Submit incident report**, then **Confirm & submit** in the dialog.
8. You land on the **Reported** screen.

Edit an incident within 24 hours (reporter)

1. From the **Reported** screen tap **Edit report**, or open the report from **Your recent reports**.
2. If you're inside the window, the edit form opens pre-filled. Change anything (spaces, severity, witnesses, etc.).
3. Tap **Save changes** → **Save changes** in the confirmation dialog.
4. You'll see "Changes saved" and can keep editing until the window closes. After 24 hours the report locks to read-only.

Change an incident's status (admin)

1. Admin Center → **Incident Reports Admin** → **History**.
2. **View** the report.
3. In the **Status** section, pick a new value (Submitted / In review / Resolved / Archived). It saves immediately and stamps the matching timestamp.

Add a follow-up note (admin)

1. Open the report detail (History → View).
2. Scroll to **Follow-up notes**, type your note in **Add follow-up note**, and click **Add note**.
3. The note is permanent — it can't be edited or deleted.

Add a new severity (admin)

1. Admin Center → **Incident Reports Admin** → **Severity Levels**.
2. Click **Add severity**.
3. Enter a **Key** (e.g. high), a **Display name** (e.g. High), a **Color**, and a **Sort order**.
4. **Create severity**. It appears in the staff Severity dropdown right away. (To get four standard levels at once, use **Seed defaults** when the list is empty.)

Add an activity (admin) — same flow on the **Activities** tab (**Add activity**), or use **Seed default activities** / the **CSV bulk import**.

7. Field reference

Field	Where	Required?	Notes
When did it happen?	Staff form	Yes	Date & time picker; defaults to now.
Facility space	Staff form	Yes	Multi-select from Facility Spaces; or tick Other + free text. At least one space or an "Other."
Other space	Staff form	Conditional	Free text; required if Other is ticked.
Description	Staff form	Yes	Free text, max 500 characters (live counter).
Activity at the time	Staff form	No	Admin-defined list, plus "Other..." free text.
Severity	Staff form	Yes	Admin-defined list; facility must have at least one.
Immediate actions taken	Staff form	No	Free text.
Was an ambulance called?	Staff form	No (defaults No)	Yes raises a critical, acknowledgement-requiring alert to the facility.
Number of people involved	Staff form	No	Whole number, 0 or more.
Follow-up required	Staff form	No (defaults No)	Yes/No toggle.
Witnesses (up to 3)	Staff form	No	Each: Name + (Phone or Email) required; Statement optional.
Reporter name / phone	(Automatic)	—	Filled from your login; not asked on the form, not editable.
Status	Admin detail	—	Submitted / In review / Resolved / Archived; admin-set.

Field	Where	Required?	Notes
Follow-up notes	Admin detail	—	Append-only; can't be edited or deleted.
Incident Type	Admin History/detail	—	Admin-configurable category; △ VERIFY whether staff pick it (not in the live staff form).
Severity / Activity — Key	Admin config	Yes	Lowercase letters, digits, underscores; unique per facility.
Display name	Admin config	Yes	What staff see in the dropdown.
Color / Sort order	Admin config	No	Color swatch and ordering in the list.

8. Locking, saving & offline

Submitted status. A new report is saved with status **Submitted**. Admins later move it through **In review** → **Resolved** → **Archived** as they handle it.

The 24-hour edit window (confirmed). When a report is created, RinkReports stamps an edit-window end time of **submission + 24 hours**. While the current time is at or before that mark, **the reporter (and only the reporter) can edit their own report** — change details, spaces, severity, witnesses, etc. Every change is written to the report's change log. **Once 24 hours pass, the report locks:** the reporter sees a read-only view ("The 24-hour edit window for this report has closed"), and any attempt to save is rejected ("The edit window for this report has closed."). After that, only admins act on the report — and even they **never edit the original**; they change **status** and add **follow-up notes** instead. The original report and its follow-up notes/change log are immutable.

Ambulance alerts. Filing a report with **Ambulance called = Yes** raises a high-priority, acknowledgement-requiring alert so it surfaces above routine notifications. This is best-effort and never blocks your submission.

Offline. RinkReports is a PWA, so you can file an incident with no connection. When you're offline:

- The submit button changes to **Save offline** and a banner notes "This report will be saved on your device and submitted automatically when you reconnect."
- After confirming, you see **"Saved on this device"** with a **Submit another report** button. The report is queued locally and submitted automatically once you're back online — re-running the same checks as an online submission, with no duplicates.
- **Editing an existing report is online-only.** The offline queue handles new incident reports; the 24-hour edit/update path runs against the live record and isn't queued.

9. Troubleshooting & FAQ

"You don't have permission to submit incident reports." Your account lacks the **submit** permission for this module. Ask an admin to enable it (Permissions / Roles).

"Account not ready" / "Your account isn't fully set up yet." Your login isn't linked to an active employee record. Contact your administrator.

"Not configured yet" — no severities. A facility needs at least one active **Severity Level** before staff can file. An admin should add one (or **Seed defaults**) on the Severity Levels tab.

I can't find the right location in the Facility space list. Tick **Other** in the dropdown and type the space. Admins can add it permanently in **Facility Spaces** (Admin Center → Setup → Facility Spaces).

The activity I need isn't listed. Choose **"Other..."** and type it in. Admins can add activities on the Activities tab.

I need to fix my report but the edit form won't open. The **24-hour window has closed** — the report is now read-only. Ask a manager/admin to add a **follow-up note** with the correction; the original can't be changed.

Can I attach a photo? No. This module has no photo or file upload — describe the incident in the Description and Immediate-actions fields.

Why can't I edit someone else's report? You can only edit your **own** report, and only within its 24-hour window. Others' reports are read-only to you.

My note/status change disappeared from the dropdown. Follow-up notes are append-only and permanent (they can't be edited or deleted). A status change saves immediately and stamps its timestamp; if it didn't apply you'll get an on-screen error.

I submitted while offline — did it go through? It's saved on your device and will submit automatically when you reconnect. You'll find it in **Your recent reports** once it syncs.

Source (footnote)

Staff flow: `src/app/reports/incidents/page.tsx`,
`src/app/reports/incidents/_components/submission-form.tsx`,
`src/app/reports/incidents/[id]/page.tsx`, `src/app/reports/incidents/done/page.tsx`,
`src/app/reports/incidents/actions.ts`, `src/app/reports/incidents/_lib/compute.ts`,
`src/app/reports/incidents/_lib/submit.ts`, `src/app/reports/incidents/types.ts`.

Admin flow: `src/app/admin/incident-reports/page.tsx`, `src/app/admin/incident-reports/actions.ts`,
`src/app/admin/incident-reports/types.ts`, and `_components/` (`history-tab.tsx`, `history-filters.tsx`,
`report-detail.tsx`, `types-tab.tsx`, `severities-tab.tsx`, `severity-form.tsx`, `activities-tab.tsx`,
`activity-form.tsx`, `type-form.tsx`, `seed-defaults-card.tsx`, `status-badge.tsx`).

Schema / 24-hour window:

`supabase/migrations/000000000000103_incident_reports_redesign_columns.sql` (`edit_window_ends_at ...`
`default (now() + interval '24 hours')`), `...104_incident_report_children.sql`,
`...102_incident_activities.sql`, `...105_facility_spaces_incident_admin_write.sql`.

Accident Reports

Product: RinkReports — the Next.js progressive web app (PWA) for ice-rink operations. **Brand colors:** #4DFF00 (primary green) · #002244 (navy). **Data scoping:** You only see data for your own facility — this is automatic. You never switch facilities or see another rink's accidents; RinkReports keeps each facility's records separate on its own.

1. What this module is for

Accident Reports is how staff document an injury — a guest or employee who got hurt on the ice or in the building. A report captures **who was hurt** and how to reach them, **when and where** it happened, **what happened**, how **severe** it was, what kind of **injury** and **which body parts** were affected (on an interactive body diagram), whether **medical attention** was needed, any **witnesses**, and whether it's a **Workers' Compensation** claim.

Accident Reports is the close sibling of **Incident Reporting** (see that chapter) — same submitter/admin shape, same 24-hour self-edit window, same append-only follow-up notes and change log. The differences worth knowing up front:

- **There's an interactive body diagram.** Staff mark exactly which body parts were hurt — front, back, or both — on a two-view (front/back) figure, independently for the left and right side of paired regions. Incident Reporting has nothing like it.
- **A dedicated Workers' Comp tab.** Admins write one block of instructions text that appears to staff the moment they flag a report as a workers'-comp claim, with a required "I have read and understand" acknowledgement.
- **One dropdown value can raise a Communications alert.** Admins can mark a Medical Attention value (e.g. Emergency Room) as "Triggers communication alert" — selecting it on a submission fires a facility-wide alert into the Communications module (see §8).
- **No status lifecycle.** Unlike Incident Reporting's Submitted → In review → Resolved → Archived pipeline, an accident report has no status field at all. Admins track follow-up entirely through **follow-up notes** and the **change log** — there's no status to change. If you came from the Incident Reporting chapter expecting a status dropdown here, there isn't one; that's intentional, not an oversight.

2. Who can use it

Access is **permission-driven**, not strictly tier-driven: an administrator can turn each person's **submit** and **admin** access on or off per person, per facility, independent of their role label.

Role tier (doc vocabulary)	Typical access to Accident Reports
super_admin	Full: file reports, edit own within 24h, and full admin (History, Dropdowns, Workers' Comp, follow-up notes).
org_admin	⚠ VERIFY — the live app has no separate org_admin tier . Treat as super_admin.
facility_manager (live role: admin)	Full admin for their own facility: review every report in History, add follow-up notes, and configure Dropdowns / Workers' Comp instructions. Can also file reports.

Role tier (doc vocabulary)	Typical access to Accident Reports
supervisor (live role: manager or a custom role)	△ VERIFY — typically can file reports and edit their own within 24h. Admin review/config only if granted the module-scoped admin action for Accident Reports.
staff (live role: staff or a custom role)	File accident reports and edit their own within the 24-hour window. No access to the admin console (History/Dropdowns/Workers' Comp). Can only ever see their own submitted reports — the database's row-level security scopes a non-admin's read access to <code>employee_id = you</code> , so there's no staff-facing way to browse coworkers' reports even by guessing a URL.

Notes confirmed in code:

- To open the staff accident page, you need the **submit** permission for Accident Reports (`accident_reports / submit`). Without it you see "No access — You don't have permission to submit accident reports."
- The admin console requires **both** general Admin Center access (`requireAdmin`) **and** the module-scoped `accident_reports / admin` grant (`requireModuleAdmin`). An admin with console access but not this specific grant is blocked at every action with: "Your account has admin console access but not the accident reports module's admin permission. Ask an administrator to grant it under Admin → Permissions."
- A **deactivated** employee account cannot file, view, or edit anything.
- Filing requires an active `employees` row linked to your login; if your account isn't linked yet you'll see "Account not ready — Your account isn't fully set up yet. Contact your administrator."

3. How to get there

Staff (to file or view your own reports):

- Open the staff app and choose **Accidents** from the sidebar (desktop) or the **Menu** tab (mobile bottom bar: Home · Reports · Menu · Account).
- This opens **Accident Reports** at `/reports/accidents` — note the URL says "accidents," not "accident-reports." The new-report form comes first, followed by **Your recent submissions** (your own filings from the last 30 days).
- The Accidents menu item only appears if your facility has the module turned on and you have the **submit** permission.

Admins (to review and configure):

- Open the **Admin Center**, and under **Module Admin** choose **Accident Reports Admin** at `/admin/accident-reports` — here the folder is "accident-reports," unlike the staff route above.
- The admin page opens on the **History** tab, with tabs across the top for **History** · **Dropdowns** · **Workers' Comp**.

4. Setup & configuration (admins)

Configuration lives in the **Accident Reports Admin** screen (`/admin/accident-reports`). Original submitted reports are **immutable** — admins never edit a staffer's report in place; they can only add **follow-up notes** (see §5). The header carries two shortcuts: **Manage locations** (jumps to the shared

Facility Spaces list) and an **Export** button for this module.

Dropdowns tab

Five configurable vocabularies drive the staff form, confirmed directly from `DROPDOWN_CATEGORIES` in `types.ts`:

Category	Shown to staff as	Required on staff form?
Injury Type	"Primary injury type"	No
Body Part	(feeds the body diagram, not a dropdown)	Yes — facility must have at least one active value
Activity	"Activity at time of accident"	No
Medical Attention	"Medical attention"	Yes — facility must have at least one active value
Severity	"Severity" (pill buttons)	Yes — facility must have at least one active value

Location is deliberately not one of the five. As of migration 142, accident locations come from the shared **Facility Spaces** list (`/admin/spaces`) — the same list Incident Reporting and Air Quality use — not from `accident_dropdowns`. Use the **Manage locations** header button to jump there.

Each value has a **Key** (lowercase letters, digits, underscores; unique within its category per facility), **Display name**, **Color**, and **Sort order**, plus **Edit / Deactivate / Reactivate / Delete** controls (the same pattern as Incident Reporting's Severity/Activity/Type tabs). Deactivating hides a value from the staff picker while keeping it on past reports. **Delete is blocked** if a value is already in use — for Body Part specifically the error reads "Cannot delete; in use on **existing reports**. Deactivate instead," and for every other category it's the generic "in use by existing reports" wording.

The "Triggers communication alert" checkbox — Medical Attention only. When editing or creating a **Medical Attention** value, the form shows an extra checkbox: **"Triggers communication alert."** Its description in the app reads: "When the staff selects this medical-attention level, fire a comms alert (e.g. ER, Hospitalization). Stored as `metadata.triggers_alert`." This is a real, working wire, not just a label:

- The flag is stored as `{"triggers_alert": true}` in that dropdown row's `metadata` JSON column.
- When a staff member submits (or edits) a report and picks a Medical Attention value with this flag set, the submission form shows an inline warning right under the dropdown: "Selecting this option will alert managers."
- On successful submission, the server looks up the chosen Medical Attention row's metadata and, if `triggers_alert` is true, inserts a row directly into the shared `communication_alerts` table — the same table the Communications module reads from — with `source_module = "accident_reports"`, `requires_acknowledgement = true`, title **"Accident report requires medical attention follow-up,"** and a body summarizing who was hurt and why (truncated description). Its severity is mapped from the report's chosen accident-severity **key**: `critical`→critical, `high`→high, `medium`→warn, `low`→info (defaulting to `high` if severity wasn't set). This is best-effort: if the alert insert fails, the accident report itself is still saved.
- Separately (and always, regardless of the checkbox), every accident submission also runs the facility's general notification **dispatch rules** with the subject "Accident report submitted" — that's the same fan-out every module's submissions go through, distinct from the medical-attention alert above.

The seed defaults mark **Medical Office Visit**, **Emergency Room**, and **Hospitalization** as triggering an alert out of the box (**None** and **First Aid** do not).

Seed defaults. If a facility has zero dropdown rows in any category, a **Seed defaults** card appears and seeds **all five categories at once** in one click (not one category at a time): the canonical Body Part set (19 keys — two legacy ones, Arms and Head/Neck, seed **inactive** since new submissions use the split Upper Arms/Lower Arms and Head/Face-Jaw/Neck regions instead), four Severity levels (Low/Medium/High/Critical, each with a default color), five Medical Attention values, ten Injury Types, and eight Activities. Seeding is idempotent (safe to re-run; it skips duplicates on (facility_id, category, key)).

Bulk import. Each category tab has a **Bulk upload values** CSV import (columns: Category, Key, Display Name, Color, Active, Triggers Alert). Triggers Alert is accepted for any row but silently forced to false unless the row's category is medical_attention.

Workers' Comp tab

This is **not** a workflow — it's a single admin-editable block of instructions text (a Textarea, up to a page or so, newlines preserved) shown to staff at the moment they toggle **"Workers' comp claim?"** on the submission (or edit) form. There's no approval step, no separate claim record, and no routing: it's purely informational copy an admin writes once (e.g. "Call HR at ext. 204 and complete form WC-1 within 24 hours"), with a "Last updated [date]" stamp. Staff must tick a required **"I have read and understand the workers' comp instructions above"** checkbox before they can submit a workers'-comp-flagged report; ticking it stamps workers_comp_acknowledged_at on the report.

Facility Spaces (the "where" list) — lives elsewhere

Same as Incident Reporting: the **Location** picker on the staff form is fed by the shared **Facility Spaces** list (Admin Center → Setup → Facility Spaces, /admin/spaces), reached via the **Manage locations** button on this module's header. It is optional on the staff form.

Cross-reference — Admin Control Center. Module visibility is set on the **Modules** page; who can file or administer is set under **Permissions / Roles**; the location list is **Facility Spaces**. See the Admin Control Center chapter for those shared screens.

5. Screen-by-screen walkthrough

A. The staff accident form (/reports/accidents)

The page opens with an "Accident Reports" header, then the form (five numbered section cards), then **Your recent submissions** below.

1 • Person involved

- **Injured person's name** (required), **Contact — phone or email** (required, one free-text field, not split), **Age** (required, whole number 0–120).

2 • What happened

- **Severity** (required) — rendered as a row of colored pill buttons (SeverityRadioPill), not a dropdown; the admin-defined Severity list. Because this pill group isn't a native form control, the browser's required-field check can't catch a missing pick — the app enforces it in JavaScript right before the confirmation dialog opens ("Please select a severity before submitting").
- **Primary injury type** (optional) — admin-defined Injury Type dropdown.

- **Medical attention** (optional, but the facility must have at least one configured value or the form won't load) — admin-defined dropdown; shows the "Selecting this option will alert managers" notice when the chosen value has `triggers_alert` set (see §4).
- **Body parts affected** — the interactive body diagram (see below). Not required; you can submit with none selected.
- **What happened?** (required) — a free-text description, no character cap enforced in the UI (unlike Incident Reporting's 500-character limit).

3 • Where & when

- **When did it happen?** (required) — a date-and-time picker defaulting to the current moment.
- **Location** (optional) — a dropdown of active Facility Spaces.
- **Activity at time of accident** (optional) — admin-defined Activity dropdown.

4 • Witnesses (optional, up to 5)

- **Add witness** / each row has **Name**, **Phone or email (optional)**, and a **statement** textarea, plus **Remove**. Unlike Incident Reporting, only the **name** is required per witness here — contact and statement are both optional.

5 • Workers' comp

- A **"Workers' comp claim?"** checkbox. Turning it on reveals the admin-configured **instructions** block (read-only) and a required **"I have read and understand..."** acknowledgement checkbox. The submit button is disabled until that box is checked.

The interactive body diagram. Two SVG figures render side by side — a **Front View** and a **Back View** — built from simple shapes per region (head, torso, limbs, hands, feet, etc.). Tapping any region cycles it through a simple state machine per the view you're tapping in: **none** → **this view** → **both** → **the other view** → **none**. So tapping a leg on the Front View the first time marks it "front only"; tap it again on the Back View and it becomes "both"; tap either view a third time and it clears. **Paired regions** — shoulders, upper/lower arms, elbows, wrists, hands, fingers, upper/lower legs, knees, ankles, feet — are independently selectable per **left vs. right** side, so marking the right knee never touches the left. **Midline regions** — head, face/jaw, neck, torso, hips — have a single front/back/both state with no left/right split (face/jaw is front-only; back and both are disabled for it). A **"Selected body parts"** panel beneath the diagram lists every current pick with a colored Front/Back/Both badge and a **Remove** button, plus a **Clear all** link. An **"Add by list (accessible alternative)"** collapsible panel offers the same picks as explicit front/back/both/none buttons per region, for anyone who can't or doesn't want to tap the SVG. Two legacy body-part keys (a combined "Arms" and a combined "Head/Neck") exist only to keep historical pre-migration reports rendering correctly and are never offered on new submissions.

Submit & the "Are you sure?" dialog. The button reads **Submit report** (or **Save offline** — see §8). Tapping it validates the form, then opens: **"Submit this accident report?"** — "Accident reports can only be edited within 24 hours of submission. Make sure all details are accurate before confirming." — **Confirm & submit** or **Go back**.

B. After submitting (/reports/accidents/[id]?submitted=1)

You land on the single-report page with a green "Submitted. Thank you. You can review or edit this report below." banner, followed immediately by either the editable form or the read-only view (see below) for that same report.

C. Your recent submissions

Below the form on the main accidents page, **Your recent submissions · last 30 days** lists your own filings — a severity dot, the injured person's name, a severity pill, the medical-attention value (if any), and the submission time. Clicking a row opens that report at `/reports/accidents/[id]`.

D. The single-report view / 24-hour edit (`/reports/accidents/[id]`)

- **Within 24 hours, and it's yours:** the **edit form** opens, pre-filled with everything you entered (including the body diagram and witnesses), inside a warning banner reading "**Editable for N more hour(s).**" Change anything and **Save changes**; every change is written to the report's change log.
- **After 24 hours (or someone else's report — which RLS won't even let a non-admin fetch):** a **read-only view** renders instead, showing every field, the body diagram (non-interactive), and witnesses. If the window has closed on your own report it adds: "The 24-hour edit window has closed. Contact a manager if you need to change this report."

E. Admin: Accident Reports Admin (`/admin/accident-reports`)

History tab — a filterable table of every accident at your facility (most recent first, up to 200 shown with a **Load more** link). Columns: Submitted, Injured person, Severity, Medical, Location, Activity, W/C (workers' comp), Body parts (count), and a **View** link. Filters: date range (defaults to the last 30 days), employee (reporter), severity, body part, location, activity, medical attention, and workers'-comp yes/no.

Report detail panel — opens from History → View. It shows:

- The **Original report** (read-only, never editable here): injured person, contact, age, occurred-at, submitted-at, severity, primary injury type, location, activity, medical attention, workers'-comp status and acknowledgement time, and the full description.
- The **body diagram**, rendered read-only from the report's saved selections.
- **Witnesses** — name, contact, statement.
- **Follow-up notes** — an append-only thread (**Add follow-up note** → **Add note**); notes show author and time and cannot be edited or deleted once added.
- **Change log** — an append-only audit trail (create/update) with an expandable before/after JSON diff. Visible to admins only.

There is **no status field or status control anywhere on this screen** — that's the sharpest contrast with Incident Reporting's History/detail view.

Dropdowns · Workers' Comp tabs — the configuration screens described in §4.

6. Step-by-step: common tasks

File an accident report (staff)

1. Sidebar/Menu → **Accidents**.
2. Fill in the injured person's **name**, **contact**, and **age**.
3. Pick a **Severity** pill (required), optionally an **injury type** and **medical attention**, mark any injured **body parts** on the diagram, and describe **what happened**.
4. Set **when it happened**, and optionally a **location** and **activity**.
5. Optionally add up to **5 witnesses** (name required; contact/statement optional).
6. If this is a workers'-comp claim, toggle it on, read the instructions, and check the acknowledgement box.

7. Tap **Submit report**, then **Confirm & submit** in the dialog.

Edit an accident report within 24 hours (reporter)

1. Open the report from **Your recent submissions**.
2. If you're inside the window, the edit form opens pre-filled — change any field, including body-diagram selections and witnesses.
3. Tap **Save changes**. The change is written to the report's change log; you can keep editing until the window closes.

Add a follow-up note (admin)

1. Admin Center → **Accident Reports Admin** → **History** → **View** on the report.
2. Scroll to **Follow-up notes**, type the note, and click **Add note**.
3. The note is permanent — it can't be edited or deleted, and there's no status to change alongside it.

Add a new dropdown value (admin)

1. Admin Center → **Accident Reports Admin** → **Dropdowns**.
2. Pick the category tab (Injury Type / Body Part / Activity / Medical Attention / Severity) and click **Add [category] value**.
3. Enter a **Key**, **Display name**, **Color**, and **Sort order**. For Medical Attention, optionally check **"Triggers communication alert."**
4. **Create value**. It's live on the staff form immediately. (Use **Seed defaults** to populate all five categories at once instead.)

Write or update the Workers' Comp instructions (admin)

1. Admin Center → **Accident Reports Admin** → **Workers' Comp**.
2. Edit the instructions textarea and **Save**. Staff see the new text the next time they toggle a workers'-comp claim on.

7. Field reference

Field	Where	Required?	Notes
Injured person's name	Staff form	Yes	Free text.
Contact (phone or email)	Staff form	Yes	Single free-text field, not split by type.
Age	Staff form	Yes	Whole number, 0–120.
Severity	Staff form	Yes	Pill-button picker; admin-defined; client-enforced (not a native required field).
Primary injury type	Staff form	No	Admin-defined dropdown.
Medical attention	Staff form	No (list itself must be non-empty)	Admin-defined dropdown; a value can carry <code>triggers_alert</code> .

Field	Where	Required?	Notes
Body parts affected	Staff form	No	Interactive front/back diagram; paired regions track left/right independently.
What happened? (Description)	Staff form	Yes	Free text, no enforced character cap.
When did it happen?	Staff form	Yes	Date & time picker; defaults to now.
Location	Staff form	No	From the shared Facility Spaces list.
Activity at time of accident	Staff form	No	Admin-defined dropdown.
Witnesses (up to 5)	Staff form	No	Each: Name required; Phone/email and statement optional.
Workers' comp claim?	Staff form	No (defaults off)	Reveals instructions + a required acknowledgement checkbox when on.
Reporter identity	(Automatic)	—	Filled from your login; not asked on the form.
Follow-up notes	Admin detail	—	Append-only; can't be edited or deleted.
Status	—	—	Does not exist for this module (contrast with Incident Reporting).
Dropdown — Key	Admin config	Yes	Lowercase letters, digits, underscores; unique per category per facility.
Dropdown — Display name	Admin config	Yes	What staff see.
Dropdown — Color / Sort order	Admin config	No	Swatch and list ordering.
Dropdown — Triggers communication alert	Admin config	No	Medical Attention category only; ignored elsewhere.
Workers' Comp instructions	Admin config	No	Free text shown to staff on a workers'-comp-flagged submission.

8. Locking, saving & offline

The 24-hour edit window (confirmed, same mechanism as Incident Reporting). Every accident report is stamped with `edit_window_ends_at = submitted_at + 24 hours` at creation. While `now() <= edit_window_ends_at`, **the reporter (and only the reporter) can edit their own report** — every field, the body diagram, and witnesses. Each save writes a before/after snapshot to the report's change log. **Once the 24 hours pass, the report locks:** even the reporter now sees the read-only view, and the update database policy itself refuses the write ("The edit window for this report has

closed."). After that, admins can still add **follow-up notes**, but the original report and its history are immutable at the database level, not just hidden in the UI — there's no admin override that reopens editing (see the RLS excerpt in 00000000000010_accident_reports_schema.sql: UPDATE: super_admin OR module admin OR (own row AND now() <= edit_window_ends_at) — note that even a module admin cannot update someone else's original report after the fact, only add notes).

No status lifecycle. As covered in §1/§5, this module has no equivalent of Incident Reporting's Submitted → In review → Resolved → Archived states. "Handling" an accident report administratively means adding follow-up notes; there is nothing to mark resolved.

Medical-attention alerts. Submitting (or editing into) a Medical Attention value flagged "**Triggers communication alert**" inserts a high-priority, acknowledgement-requiring row into the shared communication_alerts table for the Communications module to surface — see §4 for exactly how the severity mapping and messaging work. This is best-effort and never blocks your submission. Separately, every accident submission (regardless of medical attention) also runs the facility's general notification dispatch rules.

Offline. RinkReports is a PWA, so you can file an accident report with no connection:

- The submit button changes to **Save offline**, and the sticky bar footnote reads "offline — will sync when reconnected."
- After confirming, you see "**Saved on this device**" — the report (including the body-diagram JSON and witnesses) is queued locally via the service worker's sync queue (moduleKey: "accident_reports") and submitted automatically once you reconnect, running through the exact same persistence pipeline — including the medical-attention alert check — as an online submission. The replay is idempotent (a local-id claim token prevents duplicate rows if retried).
- **Editing an existing report is online-only.** The offline queue handles new accident reports only; the 24-hour edit path runs against the live record and isn't queued. ⚠ VERIFY — this parallels Incident Reporting's documented behavior; the accidents edit-form component itself doesn't call the offline queue, so this reads as intentional rather than confirmed by an explicit code comment the way the submission path is.

9. Troubleshooting & FAQ

"You don't have permission to submit accident reports." Your account lacks the **submit** permission for this module. Ask an admin to enable it (Permissions / Roles).

"Account not ready" / "Your account isn't fully set up yet." Your login isn't linked to an active employee record. Contact your administrator.

"Not configured yet" — accident reporting isn't set up. Your facility needs at least one active value in **Severity**, **Medical Attention**, and **Body Part** before staff can file. An admin should add values (or use **Seed defaults**) on the Dropdowns tab.

I can't find the right location. The **Location** field is optional — leave it blank, or ask an admin to add the space under **Facility Spaces** (Admin Center → Setup → Facility Spaces).

Why did my submission trigger a notice about alerting managers? The Medical Attention value you picked (typically Emergency Room, Hospitalization, or a Medical Office Visit) is configured by an admin to raise a Communications alert. This is expected — it gets the right people looking at a serious injury faster.

I need to fix my report but the edit form won't open. The **24-hour window has closed** — the report is now permanently read-only, even to admins. Ask a manager/admin to add a **follow-up note**

with the correction.

Can I mark a report as "Resolved"? No — this module has no status field. Follow-up notes are how admins record what happened next.

Why is the Workers' Comp checkbox grayed out until I check a box? Turning on "Workers' comp claim?" requires you to also check "I have read and understand the workers' comp instructions above" before the form will submit — this is a required acknowledgement, not a bug.

Why can't I edit someone else's report? You can only edit (or even open) your **own** report, and only within its 24-hour window — the database itself won't return another employee's accident report to a non-admin account.

I submitted while offline — did it go through? It's saved on your device and will submit automatically when you reconnect, including any medical-attention alert. You'll find it in **Your recent submissions** once it syncs.

Source (footnote)

Staff flow: `src/app/reports/accidents/page.tsx`, `src/app/reports/accidents/actions.ts`, `src/app/reports/accidents/types.ts`, `src/app/reports/accidents/_components/submission-form.tsx`, `src/app/reports/accidents/[id]/page.tsx`, `src/app/reports/accidents/[id]/_components/edit-form.tsx`, `src/app/reports/accidents/[id]/_components/read-only-view.tsx`, `src/app/reports/accidents/_lib/compute.ts`, `src/app/reports/accidents/_lib/compute.test.ts`, `src/app/reports/accidents/_lib/submit.ts`, `src/app/reports/accidents/_lib/offline.ts`, `src/app/reports/accidents/_lib/dropdowns.ts`.

Body diagram: `src/components/staff/body-diagram/body-diagram.tsx`, `src/components/staff/body-diagram/types.ts`, `src/components/staff/body-diagram/lazy.tsx`.

Admin flow: `src/app/admin/accident-reports/page.tsx`, `src/app/admin/accident-reports/actions.ts`, `src/app/admin/accident-reports/types.ts`, and `_components/` (`history-tab.tsx`, `history-filters.tsx`, `report-detail.tsx`, `dropdowns-tab.tsx`, `dropdown-form.tsx`, `dropdowns-import.ts`, `workers-comp-tab.tsx`, `seed-defaults-card.tsx`).

Schema / 24-hour window / alerts: `supabase/migrations/00000000000010_accident_reports_schema.sql` (`edit_window_ends_at` ... default (`now()` + interval '24 hours'), RLS policies), `00000000000051_accident_witnesses_and_age.sql`, `000000000000142_accidents_use_facility_spaces.sql` (location moved to `facility_spaces`), `00000000000009_communications_schema.sql` (`communication_alerts` shared table).

Navigation: `src/components/app/sidebar-nav.tsx` (staff "Accidents" nav entry, `moduleKey: "accident_reports"`).

Ice Operations

1. What this module is for

Ice Operations is the digital logbook for the day-to-day maintenance your crew performs on the ice and on the machines that maintain it. Instead of paper sheets pinned by the resurfer, your team logs each job on a phone or tablet — and every entry is automatically stamped with who did it and when.

The module covers four kinds of jobs ("operation types"):

- **Ice Make** — a resurfacing run: which rink, which machine, water used, machine hours, and how much snow was taken.
- **Edging** — time spent running the edger around the boards.
- **Blade Change** — swapping the blade on a machine, recording the old blade's hours and the new blade's ID.
- **Circle Check** — a structured pre-use safety/condition inspection of a machine, with a pass/fail checklist. Failed items raise an alert so a manager knows right away.

Two important things to understand up front:

- **Your facility chooses which of these four operation types it uses.** An admin can hide any of them, so your tabs may show all four or just one or two. (The four types themselves are built in — they can't be renamed or added to — but visibility is yours to control.)
- **Equipment is your own list.** The machines you pick from (resurfacers, edgers, blade sets, etc.) and their "types" are set up by your admin for your facility — nothing is hard-coded.

You only see data for your own facility — this is automatic.

2. Who can use it

Access is permission-driven, not a fixed rank. The table shows the typical default for each tier; an admin can widen or narrow any one person's access per action (view / submit / configure). Treat the tiers as defaults, not hard rules.

Tier	Typical access to Ice Operations
super_admin	Full access everywhere, plus all admin configuration.
org_admin	△ VERIFY — the live app has no separate org-admin tier; treat as super_admin.
facility_manager (the admin role)	Submit reports and run all admin setup, history, and settings for their facility.
supervisor (the manager role, or a custom role)	Submit reports if granted the submit permission. Admin setup only if also granted the module's admin permission. △ VERIFY whether your facility grants supervisors config rights by default.
staff (or a custom role such as driver)	Submit reports if granted the submit permission. No access to the admin setup/history/settings screens.

To log an operation you must have an active employee account at the facility and the `submit` permission for Ice Operations. If you lack it, the form shows a "No permission" notice instead of the tabs. If your account isn't fully set up, you'll see an "Account not set up" notice — contact your

administrator.

3. How to get there

- **Staff side (logging jobs):** open **Ice Operations** from the left sidebar (or the **Menu** tab on mobile). The link only appears if the module is turned on for your facility and you have access. The URL is `/reports/ice-operations`; it lands you on the first enabled operation tab automatically.
- **Admin side (configuring):** admins open the **Admin Center**, then **Ice Operations Admin** under Module Admin (`/admin/ice-operations`). This is the only way in — form configuration lives entirely in the admin area, so there is no configuration shortcut on the staff screens.

4. Setup & configuration (admins)

Everything in this section lives in **Admin Center → Ice Operations Admin**, which has three tabs: **Setup**, **History**, and **Settings**. The page header reminds you that submitted reports are immutable — admins review and annotate, they don't rewrite. For broader admin topics (employees, permissions, exports, retention, the Modules on/off switch), see the **Admin Control Center** chapter.

If your facility has nothing configured yet, the Setup tab offers a **Seed defaults** button. It creates a starter settings row plus five starter circle-check items (four for ice resurfacers, one for the edger). You can edit or add to everything afterward.

Setup tab — what you configure

The Setup tab is a stack of cards. Each card lists existing items with **Edit / Deactivate / Delete** buttons and an "Add" form at the bottom. "Deactivate" hides an item from staff without erasing its history; "Delete" removes it (with a confirmation prompt).

- **Rinks** — the surfaces ice operations are performed on (name and an optional slug, with a sort order). Rinks are required for Ice Make.
- **Fuel types** — power sources for resurfacers (e.g. Electric, Gas, Propane). Each fuel type can anchor one circle-check template.
- **Equipment** — your machines, grouped by **equipment type**. The built-in types are **Ice Resurfacer, Edger, Blade Set, Hand Edger, and Other**. For each machine you can set a name, type, slug, model, serial number, an **hours count** (admin-maintained — it does **not** auto-update from submissions), an optional fuel type, and sort order. The equipment type is what decides which operation tab a machine shows up under (resurfacers → Ice Make / Circle Check, edgers → Edging, blade sets → Blade Change).
- **Circle-check items** — the individual inspection items staff see during a circle check. Each item has a label, optional description, and an "applies to" scope (All equipment, or one equipment type); staff answer every item as **Pass / Fail**. Items can be reordered with up/down arrows, toggled active, deleted, and **bulk-imported** (up to 50 per facility).
- **Circle-check templates** — an optional, more structured alternative to the flat item list. You create **one template per fuel type** (up to **4 templates** per facility), give it a name/description, and add its own list of checklist **fields** (also bulk-importable). When an operator picks a resurfacer whose fuel type matches a template, that template's fields replace the generic item list for that check.

Settings tab

- **Visible operations** — checkboxes for Ice Make, Circle Check, Edging, Blade Change. Check the ones your facility uses; **leaving all unchecked shows every operation** (the default). This is what controls which tabs staff see.

- (Removed) **Temperature unit** — the staff forms no longer collect a temperature, so the °F/°C selector was removed from Settings. Legacy temperature readings on older submissions still display in the facility's previously saved unit (default °F).
- **Enable ice operations alerts** — when on, a failed circle check raises a facility alert.
- **Default alert severity** — warn / high / critical, used when a circle-check failure alert is raised without its own severity.

5. Screen-by-screen walkthrough

The Ice Operations shell

Every staff screen sits inside a shared shell:

- A **page header** with the "Ice Operations" title and breadcrumb (Reports > Ice Operations).
- Header buttons: **Show Activity Feed** (toggles a "Recent activity" panel of the last few submissions — each row shows the operation type, the rink/machine, a "failed" badge if a circle check had failures, and the time), **Back**, and **Dashboard**.
- A **tab row** with one tab per **enabled** operation type. The tabs follow a fixed order: Ice Make, Circle Check, Edging, Blade Change.

If a needed prerequisite is missing, you'll see a friendly notice inside the shell instead of a form — e.g. "No rinks configured" (Ice Make needs at least one rink), "No machines configured" (no equipment of the right type), or "No checklist items" (Circle Check with no items and no template). These point you to ask an administrator.

Auto-timestamp & operator: you never type who you are or stamp the time by hand. Each form captures the current date/time when it loads (shown as "When it happened"), and the submission is recorded against **your** logged-in employee account. The confirmation screen shows "Submitted by" with your name.

Ice Make form

Fields: **Rink** (required), **Machine** (required — only resurfacers appear), **Water Used (gallons)**, **Machine Hours**, **Snow Taken (%)** (0-100), **Time On**, **Time Off**, and **Notes**. Submit button reads **Submit resurface**.

Edging form

Fields: **Machine** (required — only edgers appear), **Hours Run**, and **Notes**. Submit button reads **Submit edging report**.

Blade Change form

Fields: **Machine** (required — only blade sets appear), **Old Blade Hours**, **New Blade ID** (free text — serial number or ID), and **Notes**. The person who performed the change is recorded automatically as you (the logged-in user); there is no "who did it" dropdown. Submit button reads **Submit blade change**.

Circle Check form

1. Pick a **Machine** (required). Until you do, the checklist area says "Select a machine to view the checklist."
2. The app loads the right checklist:

- If the machine has a fuel type whose **template** exists, that template's fields are shown ("Using template: ...").
 - If the machine has **no** fuel type assigned but templates exist, a **Fuel type** dropdown appears so you can pick one for this check.
 - Otherwise, the generic **circle-check items** that apply to that machine's type are shown.
3. For each item, tap **Pass** (green) or **Fail** (red). When you mark an item **Fail**, a required "**What's wrong?**" note box appears beneath it.
 4. Add optional **General Notes**.
 5. **Submit circle check**. The button stays disabled until every item has been answered (a counter shows how many remain) and every failed item has a note — items are never recorded as passed by default.

A failed circle check is flagged on the confirmation screen, recorded with a failed-item count, and — if alerts are enabled — raises a manager alert listing the failed items.

Confirmation ("done") screen

After any submit you land on a **Submitted!** screen with a green checkmark, the operation label, and (for a circle check with failures) a "_N_ failed items" badge. Below it: when it happened, when it was submitted, the rink, the equipment, and who submitted it. Buttons: **Submit another** and **Back to home**.

6. Step-by-step: common tasks

Log an Ice Make (resurface)

1. Open **Ice Operations** → **Ice Make** tab.
2. Choose the **Rink** and the **Machine**.
3. Fill in water used, machine hours, snow %, and time on/off as appropriate (these are optional).
4. Add notes if needed → **Submit resurface**.

Run a Circle Check

1. Open **Ice Operations** → **Circle Check** tab.
2. Select the **Machine** (and a **Fuel type** if prompted).
3. Tap **Pass** or **Fail** on every checklist item (all must be answered); write a note for every **Fail**.
4. Add general notes if needed → **Submit circle check**. Failures notify your managers.

Log a Blade Change

1. Open the **Blade Change** tab → select the **Machine**.
2. Enter the **Old Blade Hours** and the **New Blade ID** → **Submit blade change**.

Enable or disable an operation type (admin)

1. Admin Center → **Ice Operations Admin** → **Settings**.
2. Under **Visible operations**, check the types you want staff to see (uncheck to hide). Leaving all unchecked shows every type.
3. **Save settings**. The staff tabs update accordingly.

Add a resurfer (or other machine) to equipment (admin)

1. Admin Center → **Ice Operations Admin** → **Setup**.

2. In the **Equipment** card's "Add equipment" form, enter a **Name**, choose the **Type** (e.g. Ice Resurfacer), optionally set model/serial/hours and a **Fuel type**, then **Add equipment**.
3. The machine now appears in the matching operation tab for staff.

Build a circle-check template (admin)

1. Setup tab → add a **Fuel type** (if you haven't).
2. In **Circle check templates**, add a template for that fuel type, then add its **fields** (typed in, or bulk-uploaded).
3. Assign that fuel type to the relevant resurfacers in the **Equipment** card so operators load the template automatically.

7. Field reference

Operation type	Field	Required	Notes
Ice Make	Rink	Yes	Active rinks only
	Machine	Yes	Ice Resurfacer equipment only
	Water Used (gallons)	No	Decimal
	Machine Hours	No	Decimal
	Snow Taken (%)	No	0–100
	Time On / Time Off	No	Time of day
	Notes	No	Free text
Edging	Machine	Yes	Edger equipment only
	Hours Run	No	Decimal
	Notes	No	Free text
Blade Change	Machine	Yes	Blade Set equipment only
	Old Blade Hours	No	Decimal
	New Blade ID	No	Free text (serial/ID)
	Performed by	(auto)	Recorded as the logged-in user
	Notes	No	Free text
Circle Check	Machine	Yes	Ice Resurfacer equipment only
	Fuel type	Only if machine has none	Loads the matching template
	Checklist items	Per template/config	Pass/Fail each
	Failure note ("What's wrong?")	Yes for each failed item	Blocks submit if missing
	General Notes	No	Free text

All four types also capture an automatic **occurred-at timestamp** and the **operator** (you).

8. Locking, saving & offline

- **Append-only / no editing.** Every submission is a permanent record. The form itself says "You can't edit this after submitting." There is no edit button — to correct something, submit a new entry. Admins can review entries and add **follow-up notes** from the History tab, but they cannot alter the original.
- **Offline.** All four forms work offline. If you submit while your device is offline, the report is **saved on this device** (you'll see a "Saved on this device" confirmation) and **queued**. When you reconnect, it syncs automatically and the exact same checks run server-side then — including the rink/machine and failed-note validation. Each queued item is tracked so a retry can't create a duplicate. You can keep working in the meantime; pending items also appear in your offline queue.
- **Validation is the same online and offline.** Required machine, a valid occurred-at time, and a note on every failed circle-check item are enforced either way.

9. Troubleshooting & FAQ

A tab I expected is missing. Your facility has that operation type unchecked in Settings → Visible operations. An admin can re-enable it.

"No machines configured" / "No rinks configured." No equipment of the needed type (or no rink, for Ice Make) has been set up yet. Ask an admin to add it in Setup.

"No checklist items" on Circle Check. No circle-check items and no template fields are configured. An admin adds items (or a template) in Setup.

The Submit button on a circle check is greyed out. You marked an item **Fail** but didn't write a note. Add a note in the "What's wrong?" box under each failed item.

The Circle Check shows no checklist after I pick a machine. The machine's fuel type may have no template yet, or no generic items apply to its type. Pick a fuel type if prompted, or ask an admin to configure items/templates.

I'm offline — will my report be lost? No. It's saved on your device and syncs when you're back online.

I made a mistake after submitting. You can't edit a submitted entry. Submit a corrected new entry; for context, an admin can add a follow-up note to the original in History.

"You don't have permission to submit." Your account doesn't have the Ice Operations `submit` permission. An admin grants it from Permissions.

Who got my circle-check failure alert? Failed circle checks raise a facility alert (if alerts are enabled in Settings) at the configured default severity, listing the failed items — your managers see it in the Communications/alerts area.

Source

- Staff: `src/app/reports/ice-operations/page.tsx`,
`src/app/reports/ice-operations/[operationType]/page.tsx`, `src/app/reports/ice-operations/[operationType]/_components/{ice-make-form,edging-form,blade-change-form,ice-ops-shell,offline-queued-card,use-offline-submit,shared}.tsx|ts`,
`src/app/reports/ice-operations/[operationType]/done/page.tsx`,
`src/app/reports/ice-operations/actions.ts`,
`src/app/reports/ice-operations/_lib/{submit,compute,offline}.ts`,
`src/app/reports/ice-operations/types.ts`.
- Admin: `src/app/admin/ice-operations/page.tsx`, `src/app/admin/ice-operations/_components/{setup-tab,settings-tab,history-tab,history-filters,submission-detail,seed-defaults-card,circle-check-import}.tsx|ts`, `src/app/admin/ice-operations/{actions,types}.ts`.
- Manifest: `docs/training/00-MANIFEST.md`.

Air Quality

Part of the **RinkReports** training package. RinkReports is a progressive web app (PWA) for ice-rink operations. Brand colors: **#4DFF00** (primary green) and **#002244** (navy). **You only see data for your own facility — this is automatic.** You never switch facilities or see another rink's air-quality readings; the app keeps each facility's records separate on its own.

1. What this module is for

The Air Quality module records the air-quality readings staff take around the rink — typically carbon monoxide (CO), nitrogen dioxide (NO₂), and carbon dioxide (CO₂), measured in parts per million (ppm) — and checks each reading against your facility's compliance thresholds the moment you type it.

What makes Air Quality different from a plain log:

- It is **threshold-aware and live**. As you enter a number, the form shows a colored badge — **Within range**, **Corrective action**, **Notification**, or **Evacuation** — based on the regulatory profile your administrator selected.
- It tracks a **regulatory jurisdiction**. The thresholds, the sampling cadence ("how often must we test this week?"), and the escalation instructions all come from a compliance profile, so the form reflects the rules your facility is actually held to.
- A reading that crosses a threshold is flagged as an **exceedance**, recorded on the report, and (if your facility has alerts turned on) raises an alert for managers.
- It is **append-only**. Once you submit, the report is a permanent record — you can't edit it, you submit a new reading instead. Admins add follow-up notes rather than changing the original.

Temperature is one of several reading types defined in the database (alongside CO, NO₂, etc.), each carrying its own unit and decimal precision. Because the readings list is metadata-driven, there is **no single global °F/°C toggle** here — each reading type simply shows the unit your administrator configured for it.

2. Who can use it

Access is **permission-driven**: what you can do depends on the per-module permissions an admin has set for you, not strictly on your job title. Two people with the same title can have different access if an admin customized it. The table below shows the typical default mapping.

Role tier	Typical Air Quality access
super_admin	Full access to everything, across the platform. △ VERIFY — there is no separate "org_admin"; treat as super_admin.
org_admin	No access as a distinct tier — the live app has no org_admin role. △ VERIFY .
facility_manager (the admin role)	Full module admin: Setup, Compliance, History, Settings, follow-up notes, exports. Requires the air_quality admin permission (or the super-admin flag).

Role tier	Typical Air Quality access
supervisor (the manager role, or a custom role)	Submit and view readings; configure only if granted the admin permission. ⚠ VERIFY — exact rights depend on the permission matrix, not a fixed tier.
staff (or a custom role)	Submit air-quality readings (needs the air_quality submit permission). No access to the admin Setup/Compliance/History/Settings screens.

Notes that hold true in the code regardless of title:

- To open the staff reading form at all, you need the **submit** permission for **air_quality**. If you don't have it, the page shows "No permission."
- If your account isn't linked to an active employee record, the page shows "Account not set up."
- If the facility hasn't configured any locations yet, the page shows "Not configured yet — talk to your administrator."
- The whole admin area (/admin/air-quality) requires facility-manager-level admin access; staff who browse to it are shown a "Forbidden" message.

3. How to get there

Staff (submitting a reading):

- In the left sidebar (or the mobile **Menu** tab), choose **Air Quality**. This opens the reading form at /reports/air-quality.
- The menu item only appears if your facility has the Air Quality module switched on and you have permission to use it.

Admins (configuring the module):

- Open the **Admin Center** (the extra link in the staff sidebar, visible to admins), then under **Module Admin** choose **Air Quality Admin** (/admin/air-quality).
- The admin screen has four tabs across the top: **Setup**, **Compliance**, **History**, **Settings**.

4. Setup & configuration (admins)

All configuration lives in the **Air Quality Admin** screen and is a facility-manager task. See the **Admin Control Center** chapter for cross-cutting setup (Facility Spaces, Modules on/off, Permissions, Exports).

First-time setup: Seed defaults

If nothing has been configured yet, the Setup tab shows a **Seed defaults** card. Clicking **Seed defaults** creates the standard ice-rink reading types — **CO**, **NO₂**, and **CO₂** — so you have a working form in one click. You can edit or add to them afterward.

Setup tab — Equipment & Reading types

The Setup tab has two halves:

- **Locations** come from your shared **Facility Spaces** list (the same locations used by Incident and Accident reports). You don't create them here — you add, rename, or bulk-import them under **Facility Spaces** (/admin/spaces), and they appear here automatically. On this tab you click a location only to scope equipment to it.

- **Equipment** — monitors/sensors. You can add equipment that is **facility-wide** (not tied to a location) or **scoped to a specific location**. For each you can set a name, an optional slug, an optional model and serial number, and a sort order. Each item can be **Edited**, **Deactivated/Activated**, or **Deleted**. Equipment scoped to a location only appears as a choice on the reading form when that location is selected.
- **Reading types** — what staff actually measure (e.g. CO, NO₂, CO₂, temperature). For each reading type you set:
 - **Label** (e.g. "CO") and an optional **Key** (auto-generated from the label if blank).
 - **Unit** (e.g. ppm).
 - **Decimals** (0-6) — how many decimal places the input accepts.
 - **Required on reports** — whether staff must fill this reading in before they can submit.
 - **Sort order** — reading types can also be reordered with the ↑ / ↓ buttons.
 - Reading types can be **bulk-uploaded** and can be Edited / Deactivated / Deleted.
 - **Threshold tiers are not set here** — they come from the facility's compliance profile on the Compliance tab.

Compliance tab — profile, metrics, thresholds, escalation, roles

This is where the form's "live feedback" gets its rules.

- **Compliance profile (jurisdiction)**. Pick the regulatory profile for the facility from a dropdown. The reading form, the threshold tiers, the sampling cadence, and the escalation steps all derive from it. A badge shows whether it is **Binding regulation** or **Guidance**, and whether it uses the **Single-sample** or **1-hour TWA** measurement method.
- **Metrics tracked**. Check which of the profile's metrics (CO, NO₂, ...) are active for your facility.
- **Threshold overrides — tighten only**. Each metric shows the profile's regulatory ceiling for each tier (**Corrective**, **Notification**, **Evacuation**). You may enter a stricter (lower) value to hold your facility to a tighter standard, or leave a field blank to use the regulatory ceiling. **You can tighten a threshold but you can never loosen it**. The screen states this plainly ("A value must be at or below the floor"), and the app **enforces it**: if you enter an override above the regulatory floor, the save is rejected with a message like "...override is looser than the regulatory floor...; overrides may only tighten." Internally the engine also clamps any too-loose value back to the regulatory ceiling, so a looser number can never take effect even by accident.
- **Escalation steps**. For each tier you can type the contacts/actions operators should take (e.g. who to call at the Notification level). This text is shown in the reading form when a reading reaches that tier. Leave blank to use the built-in default guidance.
- **Role access**. Optional comma-separated role keys for "May submit readings" and "May view logs." This is **advisory only** — actual access is still governed by each user's module permissions, not by what you type here.
- **Save compliance profile** writes all of the above at once.

Below the profile panel, a **Compliance rules** section lists the facility's stored rules grouped by jurisdiction (these back the engine and any "sustained exceedance" evacuation criteria).

Settings tab

One row of facility-wide settings:

- **Testing frequency** — free text guidance shown to staff on the reading form (e.g. "CO every 2 hours during sessions").

- **Default jurisdiction** — pre-fills the jurisdiction field when adding a compliance rule (offers existing jurisdictions as suggestions).
- **Enable air quality alerts** — master on/off switch for the alerts raised when a reading exceeds a threshold.
- **Default alert severity** — warn, high, or critical; used when a triggered threshold has no severity of its own.

History tab (review)

A filterable, read-only list of submitted reports with a drill-down detail panel. Admins **cannot edit** a submitted report; they can **add follow-up notes**, which are themselves append-only ("Notes are append-only and cannot be edited or deleted. The original report stays intact"). Filters include date range, employee, location, equipment, reading type, and whether the report had an exceedance.

5. Screen-by-screen walkthrough

Staff reading form (/reports/air-quality)

From top to bottom, what you see:

1. **Page header** — breadcrumb (Reports → Air Quality), the eyebrow "Air quality reading," the title, and a reminder that after you submit, the report can't be edited.
2. **Location** (required) — a dropdown of your facility's spaces. Pick where the readings were taken. You can't submit until you choose one. Changing the location clears any equipment choice that doesn't belong to the new location.
3. **Compliance context panel** (shown when a compliance profile is configured) — displays the profile's name, a **Binding** or **Guidance** badge, and a **1-hr TWA** or **Single sample** method badge, plus any guidance note. It also includes:
 - A **frequency tracker**: a badge reading **On schedule** or **Behind by N**, and a line like "2 of 3 this week · weekend 0/1 (weekend sample needed)" — telling you how many readings are still required this period.
 - A **Reading type** selector — **Routine**, **Post-resurfacing**, or **Post-edging**. (This labels why the reading was taken and drives the frequency tracking; it is distinct from the "Reading Kind" / corrective concept below.)
4. **Equipment** (optional) — appears when monitors are configured for the selected location (or facility-wide). Pick the monitor you used.
5. **Readings** — one input per active reading type, in the admin-defined order. Each shows its **label**, its **unit** beside the box, and a **required** marker (*) where applicable. The numeric keypad opens on mobile, and the input respects the configured decimal precision.
 - For any reading tied to a tracked metric, a small grey hint shows the tier ceilings (e.g. "Corrective > 35 · Notification > 100 ppm"), and once you type a number a **live range badge** appears: **Within range** (green), **Corrective action** (amber), **Notification** (red), or **Evacuation** (red). This is the threshold-aware feedback — it updates as you type.
6. **1-hour TWA calculator** (only when the profile uses the TWA method) — a collapsible helper to enter the required samples (e.g. 13 readings, one every 5 minutes); it averages them and fills the reading for you when you click **Use average**.
7. **Compliance banner + corrective-action note** — if any reading reaches Corrective level or higher, a colored banner appears with the escalation instructions for that level, **and a required**

"Corrective action taken" note box appears. You cannot submit an over-threshold reading until you describe the corrective steps you took. (This is the "corrective reading kind" — an exceedance forces a corrective note.)

8. **Notes (optional)** — anything to flag for the manager.
9. **Monitoring log sections** (collapsible, optional) — supplementary record-keeping: **Equipment & tester info** (date of test, tester certification, CO/NO₂ monitor type/model/calibration date — with a warning if a calibration is over a year old, ventilation last-inspection date); **Section 1: General information & equipment status** (arena operating status, ice resurfacers and other fuel-burning equipment with fuel types, ventilation status, last-maintenance notes); **Section 2: Air quality measurements** (routine and post-edging measurement rows with location/time/CO/NO₂/temp/note); **Section 4: Additional recommendations & notes** (electric-equipment consideration, staff-trained and public-signage checkboxes, unusual observations).
10. **Submit** — the button reads **"Submit readings for [location]"** when online, or **"Save offline"** when offline. It stays disabled until a location is chosen, every required reading is filled, and any required corrective note is written; helper text below explains what's still missing.

Confirmation screen (/reports/air-quality/done)

A green checkmark and **"Submitted!"**, the location name and timestamp, a chip showing how many **readings were recorded**, and — if anything crossed a threshold — a chip showing the **exceedance count and severity**. Buttons: **Submit another** and **Back to home**.

Admin tabs (/admin/air-quality)

Setup, Compliance, History, Settings — covered in §4. The History detail panel shows the submitter's notes, each reading with its at-submit verdict, and an append-only follow-up-note thread; an **Export** action is available from the page header.

6. Step-by-step: common tasks

Record an air-quality reading (staff)

1. Open **Air Quality** from the sidebar/menu.
2. Choose the **Location**.
3. If a compliance panel appears, glance at the **frequency tracker** to see whether more readings are due this period, and set the **Reading type** (Routine / Post-resurfacing / Post-edging).
4. Optionally pick the **Equipment** (monitor) you used.
5. Enter each reading. Watch the **live range badge** — green means within range.
6. If a badge turns amber or red, read the **escalation banner**, take the action it describes, then fill in the required **Corrective action taken** note.
7. Add optional **Notes** (and any monitoring-log details your facility wants).
8. Tap **Submit readings**. You land on the **Submitted!** screen showing the reading and exceedance counts.

Tighten a threshold (admin)

1. Go to **Air Quality Admin → Compliance**.
2. Make sure the correct **jurisdiction profile** is selected.
3. Under **Threshold overrides (tighten only)**, find the metric and tier, and type a value **at or below** the shown regulatory ceiling. (Leaving it blank keeps the regulatory ceiling.)

- Click **Save compliance profile**. If you accidentally enter a value above the floor, the save is rejected — you can only tighten.

Add a reading type (admin)

- Go to **Air Quality Admin → Setup**.
- In the **Reading types** card, use the **Add reading type** form: enter a **Label**, optional **Key**, **Unit**, **Decimals**, and tick **Required on reports** if staff must always fill it.
- Click **Add reading type**. Reorder with ↑ / ↓ if needed. (Bulk upload is available for many at once.)

Set up compliance escalation (admin)

- Go to **Air Quality Admin → Compliance**.
- Select the **jurisdiction profile**.
- Under **Escalation steps**, type the contacts and actions for each tier — **Corrective**, **Notification**, **Evacuation**. (Blank fields fall back to built-in default guidance.)
- Click **Save compliance profile**. This text now appears in the staff reading form's banner whenever a reading reaches that tier.

7. Field reference

Staff reading form

Field	Type	Required	Notes
Location	Dropdown (facility spaces)	Yes	Must be chosen before you can submit.
Reading type	Dropdown: Routine / Post-resurfacing / Post-edging	Shown with a compliance profile	Labels why the reading was taken; drives frequency tracking.
Equipment	Dropdown	No	Only monitors valid for the chosen location (or facility-wide).
Each reading (e.g. CO, NO ₂ , CO ₂ , Temp)	Numeric, decimal	Per its "Required" flag	Unit and decimal precision come from the reading type. Shows tier-ceiling hint + live range badge.
Live range badge	Computed	—	Within range / Corrective action / Notification / Evacuation, per the effective (override-tightened) thresholds.
1-hour TWA calculator	Sample entry helper	No	Only for profiles using the TWA method; averages samples into the reading.
Corrective action taken	Text	Yes, when any reading is over threshold	You cannot submit an exceedance without it.
Notes	Text	No	For the manager.

Field	Type	Required	Notes
Date of test / tester certification / monitor info / sections 1, 2, 4	Mixed (collapsible)	No	Supplementary monitoring-log detail.

Thresholds shown (per metric, from the compliance profile):

Tier	Meaning on the form
Within range	Reading is at or below the corrective ceiling — green badge.
Corrective	Reading exceeds the corrective ceiling — amber badge; corrective note required.
Notification	Reading exceeds the notification ceiling — red badge; notify per escalation steps.
Evacuation	Reading exceeds the evacuation ceiling — red badge; follow evacuation protocol.

Admins set each tier's ceiling per metric on the Compliance tab. The **effective** ceiling is the regulatory floor, optionally **tightened** (lowered) by a facility override — never raised.

8. Locking, saving & offline

- **Append-only.** A submitted air-quality report is permanent. You cannot edit it — submit a new reading instead. Admins don't edit submissions either; they add **follow-up notes**, which are themselves append-only.
- **Exceedances.** On submit, the server re-evaluates every reading against the facility's effective thresholds (the same engine the form previews with). Readings over threshold are flagged as **exceedances**, the report records the highest severity, and — if alerts are enabled in Settings — an alert is raised for managers. A required corrective-action note must accompany any exceedance. A separate "sustained exceedance" rule can raise a critical, evacuation-level alert when readings stay high over time.
- **Offline.** Air Quality works offline. If you're offline when you submit, the button reads **Save offline**, the readings are queued on your device, and you see a "Saved on this device" confirmation. When you reconnect, the queued reading submits automatically and the **same exceedance checks run then** — so an over-threshold reading is flagged on sync, not silently accepted. Each queued item has a unique id, so a retry can't create a duplicate. (See the **Offline** chapter for the queue and pending/failed indicators.)

9. Troubleshooting & FAQ

The Air Quality menu item isn't showing. Either your facility has the module switched off (an admin controls this on the Modules page) or you don't have the `air_quality` view/submit permission. Ask your administrator.

I opened the form and it says "No permission" / "Account not set up" / "Not configured yet."
 "No permission" = you lack the submit permission. "Account not set up" = your login isn't linked to an active employee record. "Not configured yet" = the facility has no locations yet (an admin adds them in Facility Spaces).

The Submit button is greyed out. You must choose a **Location**, fill in **every required reading**, and — if any reading is over threshold — write the **Corrective action taken** note. The helper text under the button tells you which one is missing.

A reading turned red and now there's a note box I can't skip. That reading exceeded a threshold. Take the action in the escalation banner, then describe what you did in the **Corrective action taken** box. This is required before an over-threshold reading can be saved.

I can't raise a threshold above the regulatory number (admin). That's by design — **thresholds can be tightened but never loosened**. Enter a value at or below the regulatory ceiling, or leave it blank to use that ceiling.

Why is there no °F/°C switch? Temperature is just one of several database-defined reading types, each with its own configured unit. There's no single global temperature toggle; the unit shown next to each reading is whatever the admin set for that reading type.

Can I fix a reading I just submitted? No — reports are append-only. Submit a corrected new reading; an admin can add a follow-up note to the original explaining the correction.

Will my reading still get checked if I submit it offline? Yes. It's queued on your device and re-evaluated against the thresholds when it syncs, so exceedances are still flagged.

Source

Staff: `src/app/reports/air-quality/page.tsx`,
`src/app/reports/air-quality/_components/submission-form.tsx`,
`src/app/reports/air-quality/actions.ts`, `src/app/reports/air-quality/done/page.tsx`,
`src/app/reports/air-quality/types.ts`, `src/app/reports/air-quality/_lib/{compliance.ts,submit.ts,compute.ts,load-compliance.ts,sustained.ts}`. Admin: `src/app/admin/air-quality/page.tsx`,
`src/app/admin/air-quality/types.ts`, `src/app/admin/air-quality/actions.ts`, `src/app/admin/air-quality/_components/{setup-tab.tsx,compliance-tab.tsx,compliance-profile-panel.tsx,settings-tab.tsx,history-tab.tsx,report-detail.tsx,seed-defaults-card.tsx}`. Permission/auth:
`src/lib/permissions/check.ts`, `src/lib/auth`. Threshold tighten-only enforcement: `validateOverrides` and `effectiveMetricTiers` in `_lib/compliance.ts`, called from `saveComplianceProfileConfig` in `src/app/admin/air-quality/actions.ts`.

Ice Depth

1. What this module is for

Ice Depth is how RinkReports records **how thick the ice is** across a sheet — point by point, against a rink diagram, so you can see at a glance where the ice is at target, too thin, or too thick.

It is built for the way the job is actually done: you walk the rink with a depth gauge (manual or a Bluetooth caliper), tap each measurement point on a digital rink diagram, and enter the reading. Each point lights up in color the instant you enter a value — green when the depth is in range, red when it's below the minimum, amber when it's above target — so problem spots are obvious before you even finish.

This module measures **depth, not temperature**. There is no °F/°C toggle anywhere in it. Readings are recorded in the facility's chosen unit (inches or millimeters).

A few things this module deliberately does **not** do:

- **No photo documentation.** There is no camera, attachment, or image-upload step. The "documentation" is the annotated diagram and the per-point readings.
- It is a **two-phase flow** — first you measure, then you review and submit — not a single long form.

Once you submit, the session becomes a permanent, read-only record with a printable diagram, a PDF, and the option to send it to your facility's distribution list.

You only see data for your own facility — this is automatic.

2. Who can use it

Access is permission-driven, not strictly tied to a job title. The table below shows the typical default for each tier; an admin can grant or remove any of these per person.

Tier	Submit a depth session	Review history / analytics	Configure rinks, diagrams, points, settings
super_admin	Yes	Yes	Yes (plus the only tier that can permanently delete a submitted session)
org_admin	⚠ VERIFY — no separate org-admin tier exists in the app; treat as super_admin	⚠ VERIFY	⚠ VERIFY
facility_manager (admin)	Yes	Yes (Admin → Ice Depth)	Yes (Admin → Ice Depth)
supervisor (manager or custom role)	Yes, if granted the Ice Depth submit permission	Only if granted Ice Depth admin access	Only if granted Ice Depth admin access
staff	Yes, if granted the Ice Depth submit permission	No access	No access

Notes:

- To **submit** a depth session you need the Ice Depth submit permission. Without it, the page shows a "No permission" message instead of the form.

- The whole **Admin → Ice Depth** area (Rinks, Diagrams, History, Analytics, Settings) requires facility-admin access for the Ice Depth module.
- **Deleting** a submitted session is restricted to super_admin — the Delete button only appears for them, and the server re-checks. Everyone else, sessions are immutable.
- A deactivated or not-yet-set-up account is denied everywhere, with an "Account not set up" message.

3. How to get there

Staff (to submit a reading):

- In the staff app, open **Ice Depth** from the left sidebar (or the **Menu** tab on mobile). The menu item only appears if your facility has the Ice Depth module turned on and you have access.
- Opening Ice Depth takes you straight to a rink diagram — the app automatically lands you on your facility's **default rink and its default diagram**. If your facility has more than one rink or diagram, a **Rink** and **Diagram** picker appears at the top so you can switch.

Admins (to configure or review):

- Open the **Admin Center** (the "Admin Center" link in the staff sidebar), then choose **Ice Depth Admin** under Module Admin. This opens the tabbed admin screen at `/admin/ice-depth`.

4. Setup & configuration (admins)

All Ice Depth setup lives under **Admin → Ice Depth**, which has five tabs: **Rinks, Diagrams, History, Analytics, Settings**. Everything here applies only to your own facility. See the **Admin Control Center** chapter for how admin access, modules, and permissions are managed.

If Ice Depth has never been set up, the Diagrams and Settings tabs show a "**Seed defaults**" card to create the initial settings/rink/diagram for you to start from.

Rinks tab

A **rink** is a sheet of ice. Create one with **New rink** (name, optional slug, sort order). For each rink you can:

- **Rename**, change **sort order**.
- **Make default** — the rink staff land on first (only an active rink can be the default).
- **Deactivate / Activate** — hide a rink from staff without deleting it.
- **Delete** — only allowed once the rink has no diagrams; otherwise you're told to move or delete its diagrams first.

The **first rink you create automatically becomes the facility default**.

Diagrams tab (the diagram + measurement points)

A **diagram** (called a "layout" internally) belongs to a rink and defines the rink picture plus where the measurement points sit. There is a cap of **8 active diagrams** per facility (shown as "X / 8 active").

- **New diagram** — choose its rink (required), a name, optional slug/description, and an **aspect ratio** (width ÷ height; default 0.425, roughly a vertical NHL rink). You must create a rink first.
- Selecting a diagram opens the **point-placement editor**:
 - A **USA Hockey rink** drawing fills the panel. An optional **Logo URL** can be set per diagram (square PNG/SVG with transparent background) and renders at center ice.

- A mode toolbar with three modes:
 - **Place** — click anywhere on the rink to drop a new numbered point.
 - **Select** — click a point to open its editor (set a **Label**, fine-tune **X/Y** position as 0-1 fractions, move up/down in order, **Deactivate/Activate**, or **Delete**).
 - **Drag** — drag a point to reposition it; release to save.
- **Renumber 1..N** compacts the active points' numbers into order.
- A cap of **60 active points** per diagram is enforced.
- Per diagram you can also **rename**, set **description / logo / aspect ratio / sort order**, **Make default** (the diagram staff land on for that rink), **Deactivate/Activate**, or **Delete** (deleting cascades the points; if sessions reference it, deactivate instead). Deleting a point keeps the snapshots on any existing measurements.

Settings tab

One row per facility controls how depths are classified and displayed:

- **Measurement unit** — inches or mm. (⚠ VERIFY — the training brief said "in/cm," but the app's unit options are **inches** and **mm**, not centimeters.)
- **Low threshold** — at or below this value, a reading is flagged **low (below min)**.
- **High threshold** — above this value, a reading is flagged **high (above target)**.
- **Low / OK / High colors** — the color each severity shows in admin views (History detail, Analytics).
- **Enable alerts on submitted sessions, Alert on** (low / high / any), and **Default alert severity** — when on, a submitted session that contains the chosen severities can trigger a notification to your facility's configured recipients.

Existing sessions **snapshot** the unit and thresholds at the moment they were submitted. Changing settings later does **not** reclassify old history.

5. Screen-by-screen walkthrough

Layout (rink + diagram) picker

Opening Ice Depth redirects you onto the default rink's default diagram. If your facility has more than one rink or more than one diagram, a top bar shows a **Rink** dropdown and a **Diagram** dropdown. Switching either jumps you to that diagram's measure screen. A **sync chip** (online/offline indicator) sits in the header, and a back arrow returns to Ice Depth.

Measure phase (the interactive rink)

This is the first of two phases. The screen shows:

- A point-progress line — "**Point X of Y**" while you're entering, or "**Y points — tap to enter**" before you start — and a "**X recorded**" counter, plus a green progress bar.
- A collapsible "**Using a Bluetooth caliper?**" helper (see §6).
- The **USA Hockey rink diagram** with a numbered **chip at every measurement point**.

To record a point:

1. **Tap a chip.** A small **popover** opens anchored to that point, showing the point number, its label, a number field (with the facility's unit beside it), and **Skip / Save & Next** buttons.
2. **Type the depth** (or let a Bluetooth caliper type it). The field only accepts digits and a single decimal point, up to 3 decimal places.

3. As you type, the popover border, the point chip, and a small label update **live by severity** — **Optimal** (green), **Below min** (red), or **Above target** (amber).
4. **Press Enter** (or Tab) to save and jump to the next point. On the last point the button reads **"Save & Review"** and takes you to the review screen. **Esc** closes the popover without advancing. **Skip** moves to the next point without recording this one.

You can revisit any point at any time by tapping its chip again — entering a new value overwrites the old one. When at least one point has a value, the big **"Review & Submit"** button at the bottom becomes active and jumps you to the review phase. (Nothing is saved to the server during the measure phase — it's all on your screen until you submit.)

Review & Submit phase

"Step 3 of 3." This is your last chance to check the readings before they become a permanent record:

- A **summary card** with a small annotated rink thumbnail, the **average depth**, and **severity pills**: "N optimal," "N thick," "N below min," and "N skipped" as applicable.
- A **per-point list** — each point's number (colored by severity), its label, its status (Optimal / Below min / Above target / Not recorded), and the recorded value.
- A **Notes (optional)** box.
- A green **Submit** button. Its label reflects completeness — **"Submit Reading"** when every point is recorded, or **"Submit (X of Y recorded)"** otherwise. (When offline it reads **"Save Offline."**)
- A **"← Back to measure"** link to return and adjust before finalizing.

Done / confirmation screen

After submitting you land on the confirmation screen for that session:

- A green checkmark and a **Submitted** badge, the diagram name, and the submitted timestamp.
- **Stat pills**: Optimal, Below min, Above target, and Total measurements.
- The **annotated rink diagram** with every measured point colored by severity, plus any submitter **Notes**.
- Action buttons:
 - **Download PDF** — downloads a PDF of the session report.
 - **Print Diagram** — opens the browser print dialog; print styling hides the app chrome so only the rink diagram (with name and timestamp) prints, full-page on US Letter portrait.
 - **Send Report** — emails/sends the report to your facility's Ice Depth recipients (see §6). Ice Depth does **not** auto-send on submit; sending stays under your control.
 - **Submit Another** — returns to Ice Depth to start a new session.
 - **Back to Dashboard**.
 - **Back to Form** (top) returns to the measure screen for that diagram.

Admin tabs (Admin → Ice Depth)

- **Rinks** — manage sheets of ice (see §4).
- **Diagrams** — draw diagrams and place/label measurement points (see §4).
- **History** — a filterable list of submitted sessions (filter by diagram, employee, date range, and whether the session had low/high readings, with "Load more" paging). Click a session to open a **detail panel**: the annotated diagram at its snapshot severities, submitter info, unit and thresholds as snapshotted, low/high badges, submitter notes, and an **append-only follow-up notes** thread you can add to. Super admins also see a **Delete session** button (with a confirmation dialog).

- **Analytics** — per-diagram trends over a date range: summary stats (sessions, readings, average depth, % below min, % above target), a "**Problem spots**" heat-map of the rink colored by each point's most common condition, a **per-point breakdown** table (worst-first), and a **daily activity** strip.
- **Settings** — unit, thresholds, colors, alerting (see §4).

6. Step-by-step: common tasks

Measure ice depth across the rink

1. Open **Ice Depth**. Pick the **Rink** and **Diagram** at the top if prompted.
2. **Tap the first point** on the rink diagram.
3. **Type the depth** and press **Enter** to save and advance to the next point. Use **Skip** to leave a point unrecorded.
4. Watch each chip color in: green = optimal, red = below min, amber = above target.
5. When done (or after recording at least one point), tap **Review & Submit**.
6. On the review screen, check the per-point list and average, add **Notes** if needed, then tap **Submit Reading**.

Use a Bluetooth caliper

RinkReports doesn't pair the caliper for you — calipers in this class (e.g. iGaging Absolute Origin) connect at the OS level as a **Bluetooth keyboard**.

1. **Pair once** in your phone/tablet's Bluetooth settings: hold the caliper's **DATA** button until it appears (it shows up as a keyboard), then connect.
2. Set the caliper to the **same unit** as the form (shown in the in-app helper).
3. In the form, **tap a point** to open its popover, then press the caliper's **DATA** button — the reading fills in and the form jumps to the next point automatically. Keep pressing DATA to walk the whole rink hands-free.

Tip: the in-app "**Using a Bluetooth caliper?**" panel above the rink has these steps. The caliper's Enter (and Tab) terminator both save and advance, just like pressing Enter yourself.

Create a rink layout (admin)

1. **Admin → Ice Depth → Rinks → New rink**. Name it and save. (The first rink becomes the default.)
2. Go to the **Diagrams** tab → **New diagram**, pick the rink, name it, set the aspect ratio, and create.
3. Open the new diagram. In **Place** mode, **click the rink** to drop each measurement point.
4. Switch to **Select** mode to **label** points and fine-tune positions; use **Drag** mode to nudge them. Use **Make default** so staff land on this diagram for that rink.
5. Confirm in **Settings** that the unit and low/high thresholds are right for this facility.

Download, print, or send a report

On a session's **Done** screen (or reach a past session through Admin → History):

- **Download PDF** to save/share a PDF copy.
- **Print Diagram** to print the annotated rink (the app chrome is hidden automatically).
- **Send Report** to push it to the facility's configured Ice Depth recipients. If none are set up, you'll see "No recipients are configured for ice depth — set up a send list in Admin → Communications." The button confirms how many recipients it reached.

7. Field reference

Field	Where	What it is
Rink	Layout picker	Which sheet of ice you're measuring (only if more than one exists).
Diagram	Layout picker	Which point layout to use for this rink (only if more than one exists).
Measurement point	Measure phase	A numbered, optionally labeled spot on the rink. Tap to enter its depth.
Depth value	Point popover	The reading, in the facility's unit (inches or mm). Digits + one decimal point, up to 3 decimals; must be zero or positive.
Notes (optional)	Review phase	Free text for anything worth flagging (conditions, equipment, resurface schedule).
Severity (auto)	Everywhere	Computed from the reading vs. thresholds: Optimal (in range), Below min ($\leq$ low threshold), Above target ($>$ high threshold). Not editable.
Average depth (auto)	Review / Done	Average of all recorded readings in the session.
Stat pills (auto)	Review / Done	Counts of optimal / below-min / above-target / skipped / total points.
Admin — Settings		
Measurement unit	Settings	inches or mm.
Low / High threshold	Settings	Boundaries that classify each reading's severity.
Low / OK / High color	Settings	Colors used in admin History and Analytics.
Enable alerts / Alert on / Default alert severity	Settings	Whether and when a submitted session triggers a notification.
Admin — Rink	Name, slug, sort order, default, active	Defines a sheet of ice.
Admin — Diagram	Name, slug, description, rink, logo URL, aspect ratio, sort order, default, active	Defines a rink picture and its point set.
Admin — Point	Label, X (0-1), Y (0-1), order, active	A single measurement location on a diagram.

8. Locking, saving & offline

Adjust freely before you finalize. During the **measure** phase, nothing is sent to the server — every reading lives on your screen. You can re-tap any point to change its value, skip points, and bounce between **measure** and **review** ("← Back to measure") as many times as you like. The only thing that commits a session is the **Submit** button on the review screen.

A submitted session is immutable. Once submitted it becomes a permanent, read-only record. There is no "edit this session." If something was wrong, you record a **new** session. Admins don't edit submitted sessions in place — on the History detail panel they can add **append-only follow-up notes**, and that's it. The only way a session is removed is a **super-admin hard delete**, which is gated behind a confirmation dialog and cannot be undone.

Snapshots protect history. Each session stores the unit, thresholds, and each point's number/label/position at submit time. Editing a diagram or changing Settings afterward never changes how past sessions look or are classified — deleted points even keep their snapshots on old measurements.

Offline. Ice Depth works offline:

- If you're offline when you submit, the button reads **"Save Offline"** (or "Save on this device"), and the session is **queued on your device**. You'll see a "Saved on this device" confirmation; the session submits automatically when you reconnect, and the same severity checks and rules run then.
- A queued session is replayed exactly once (it can't create a duplicate) and lands the same record an online submit would.
- ⚠ VERIFY: if your device can't queue offline at all (no service worker active), the app blocks the submit and tells you plainly to reconnect or reload — your typed readings stay on screen rather than being lost.
- Note: offline covers **submitting** a session. Browsing previously submitted sessions or the Done page requires a connection.

9. Troubleshooting & FAQ

"No permission" when I open Ice Depth. Your account doesn't have the Ice Depth submit permission. Ask an admin to grant it (Admin → Permissions).

"Account not set up." Your login isn't linked to an active employee record yet. Contact your administrator.

"Not configured" / "No diagrams." Your facility hasn't created a rink and diagram yet, or no diagram is assigned to your rinks. An admin sets these up under Admin → Ice Depth → Rinks and Diagrams.

A point chip won't turn green even with a value in it. Green means "Optimal." If the chip is red or amber, the reading is below the low threshold or above the high threshold for your facility — that's expected behavior flagging a thin or thick spot, not an error.

My Bluetooth caliper types into the wrong place / nothing happens. Make sure you've **tapped a point first** so its popover (the input) is focused, then press the caliper's DATA button. The caliper must be paired as a keyboard in your device's Bluetooth settings, and set to the same unit as the form.

I entered the wrong depth. Before submitting, just tap that point again and type the correct value, or use "← Back to measure" from the review screen. After submitting, the session is locked — record a new one.

Can I add a photo? No. Ice Depth has no photo or attachment feature by design; the record is the annotated diagram and the readings.

Why is there no °F/°C switch like Refrigeration has? Because this module measures **depth**, not temperature. The unit (inches or mm) is set once by an admin in Settings.

"Send Report" says no recipients. No distribution list is configured for Ice Depth. An admin sets one up under Admin → Communications.

Do changes to thresholds change my old reports? No. Each session snapshots its unit and thresholds at submit time; later settings changes never reclassify history.

Why does the average / counts look off vs. what I measured? Skipped points aren't counted in the average or severity totals — only recorded readings are. The summary shows how many points were skipped.

Source

- Staff flow: `src/app/reports/ice-depth/page.tsx`, `src/app/reports/ice-depth/[layoutSlug]/page.tsx`, `src/app/reports/ice-depth/_components/{submission-form.tsx,diagram-nav.tsx,sync-chip.tsx}`, `src/app/reports/ice-depth/actions.ts`, `src/app/reports/ice-depth/_lib/{compute.ts,submit.ts,offline.ts}`, `src/app/reports/ice-depth/types.ts`
- Done / PDF / send / print: `src/app/reports/ice-depth/[layoutSlug]/done/page.tsx`, `src/app/reports/ice-depth/[layoutSlug]/done/_components/{send-report-button.tsx,print-diagram-button.tsx}`, `src/app/reports/ice-depth/[layoutSlug]/done/pdf/route.ts`
- Admin: `src/app/admin/ice-depth/page.tsx`, `src/app/admin/ice-depth/_components/{rinks-tab.tsx,layouts-tab.tsx,layout-editor.tsx,history-tab.tsx,session-detail.tsx,analytics-tab.tsx,settings-tab.tsx,seed-defaults-card.tsx}`, `src/app/admin/ice-depth/{actions.ts,types.ts}`, `src/app/admin/ice-depth/_lib/analytics.ts`
- Shared rink rendering: `src/components/ice-depth/usa-rink.tsx`, `src/components/ice-depth/rink-geometry.ts`

Dasher Boards

1. What this module is for

Dasher Boards is a **spatial condition tracker** for the physical perimeter around a sheet of ice — every board panel, glass panel, and door, in the order they actually run around the rink. It is not a report-writing form like Daily Reports or Incidents. It's a tap-a-segment-on-a-diagram tool: the diagram is the record.

Each segment carries its own **persistent condition**. Tap a panel, report what's wrong, and it stays flagged on the diagram — red, yellow, or coral depending on severity — until someone resolves it. There's no "submission" that closes the loop by itself; an open issue just sits there, visible to the next person who opens the map, until it's fixed.

Layered on top of that is an **optional inspection walk**: a facility can require a periodic (daily/weekly/monthly/yearly, per checklist item) sweep of the perimeter that ends in a sign-off. But the walk is an overlay, not a gate — tapping a bad panel and reporting an issue works identically whether or not a walk is open. The model is exception-based: when a walk is signed off, every segment nobody tapped is implicitly attested "OK."

The module is **per-rink**. A facility with multiple sheets of ice configures each rink's perimeter separately, and staff pick which rink's map they're looking at.

A few things worth knowing up front:

- **Labels are permanent identity.** Every board, glass, and door position gets a label (B1, G1, D1...) the first time it's created, and that label is never reused or reassigned to a different physical position — even if the position is later removed, converted, or the rink's whole layout is edited. This is deliberate: a door labeled "D4" stays "D4" in every issue report and every history row forever, so a supervisor reading last year's report can still tell exactly which physical door it was about.
- **Custom labels, glass numbers, and zones are a separate DISPLAY layer** on top of that permanent identity — a facility can rename what staff see ("Zam Gate Left," or "42" instead of "G42") without touching the identity issue history is keyed to.
- Dasher Boards writes queue through the same offline sync mechanism as the rest of the PWA (see §8) — a walk started in a dead zone under the stands still gets recorded.

You only see data for your own facility — this is automatic.

2. Who can use it

Access follows the app-wide permission model: four cumulative actions — **view, submit, edit, admin** — granted per user, per module, per facility (`src/lib/permissions/actions.ts`). An admin can grant or remove any of these independently of job title.

Action	What it lets you do in Dasher Boards
view	See the condition map, open issues, and walk history. No interaction.
submit	Tap a segment and report an issue; start/continue a walk; run the guided walkthrough; sign off a walk; mark a non-critical (B/C) issue fixed.

Action	What it lets you do in Dasher Boards
edit	Everything submit does, plus: acknowledge a severity-A issue, mark any issue (including A) fixed, set a glass/door replacement spec, and (in the admin console) manage custom labels, aliases, zones, and out-of-service flags.
admin	Everything edit does, plus the full Admin → Dasher Boards console: build/edit the perimeter, convert boards ↔ doors, configure glass numbering, manage the checklist, door subtypes, and issue categories.

The canonical role defaults (migration 193) map onto this as:

Role	Default action
super_admin / admin	admin
manager	edit
staff / driver (and similar custom roles)	submit

⚠ **VERIFY** — the seed migration also lists `supervisor` at `edit` and `gm` at `admin`, but the current live role model per this codebase's own guidance is `super_admin / admin / manager / staff` (plus custom roles like `driver`); treat `supervisor/gm` here as legacy entries carried in the canonical-grants table rather than roles you'll see on a new facility.

Notes that hold true in code:

- Reaching the **Admin → Dasher Boards** console requires both general Admin Center access (`requireAdmin()`) **and** the module-scoped `dasher_boards admin` grant — the two are checked separately (`requireModuleAdmin("dasher_boards")`), same pattern as Ice Operations. An admin console user without the module grant is turned back with an explicit message rather than a silent no-op.
- Setting a **glass or door replacement spec** is available to the **edit** tier even without admin console access (`ensureDasherBoardsEditor` — managers hold `edit` but not `admin`). Structural changes (relabeling, board↔door conversion, inserting/removing positions) require the **admin** tier.
- On the severity-A issue pipeline specifically: **submit** can report an A issue and later mark a **B or C** issue fixed, but only **edit** can acknowledge an A issue or close it out — a staff member can't self-close a safety-critical finding.

3. How to get there

Staff:

- In the sidebar (or the mobile **Menu** tab), choose **Dasher Boards**. This lands you directly on the condition map — `/reports/dasher-boards` resolves your facility's **default rink** (its `is_default` rink, or the first by sort order) and renders that rink's map immediately. There is no separate "pick a rink" screen.
- If your facility runs more than one rink, a **Rink** dropdown appears in the page header next to the title; switching it navigates to `/reports/dasher-boards/[rinkSlug]`, which is also the shareable/bookmarkable URL for a specific rink's map.

Admins (to configure):

- Open the **Admin Center**, then choose **Dasher Boards Admin** under Module Admin. This opens the tabbed console at `/admin/dasher-boards`.
- If your facility hasn't created a rink yet, the console skips straight to a "Set up your first rink" wizard (rink basics, then the perimeter sequence builder) instead of showing empty tabs.

4. Setup & configuration (admins)

Admin → Dasher Boards has five tabs: **Perimeter**, **Labels & Zones**, **Checklist**, **Lists**, **Walks**. A rink switcher (when there's more than one rink) sits above the tabs; each tab operates on whichever rink is selected. See the **Admin Control Center** chapter for how admin access and modules are managed generally.

Perimeter tab — building the physical sequence

This is where the board/glass/door sequence around one rink gets built. Nothing about panel counts, doors, or positions is hardcoded — you generate a starting sequence, then edit it by tapping the diagram.

Two ways to start (shown when the rink has no positions yet):

- **Standard rink template** — one click applies a pre-built NHL-shaped layout: 33 board panels with 1:1 glass, four 3-segment corner-radius groups, a Zamboni gate, two penalty gates, and two bench gates, with zones pre-assigned (North End, East Side, Penalty Boxes, South End, West Side, Visitor Bench, Home Bench). It's a starting point, not a lock-in — every position can be relabeled, inserted, removed, reassigned, or reconverted afterward.
- **Uniform sequence** — enter a board-position count (1–500) and every position is created as a plain board panel with a 1:1 glass row; you mark doors and exceptions afterward by tapping.

Either way, a **start point** (an anchor label like "Zamboni gate," set by clicking a spot on the diagram) and a **walk direction** (clockwise/counterclockwise) define where sequence position 1 begins and which way the sequence counts — set once, changeable anytime, and purely a rendering choice (it never rennumbers or relabels anything).

Tap a position on the diagram to select it and open its editing panel:

- **Mark as a door** (or **Convert to board** for an existing door), pick a **door subtype** (Bench, Scoreboard, Public Skate, Zamboni, or a facility-defined one from the Lists tab).
- **Relabel** — override the permanent label directly (rare; the display layer on Labels & Zones is the normal way to rename what staff see).
- **Insert a board/door after this position**, or **Remove** it.
- Toggle **glass on/off** at a board position (for sections with no shielding).
- Enter a **replacement spec** (width × height × thickness × material + notes) for the glass at this position, or for a door's own glass.

The board ↔ door conversion, precisely: converting a board panel to a door keeps the same underlying row and its issue history, but the position takes the **next available door label** — e.g. B12 becomes D5; B12 is retired forever and can never be reused, even by a later position. The position's 1:1 glass row is deactivated ("parked"), because the door now carries its own glass spec instead. Converting a door back to a board does the reverse — it takes the next available **board** label, and restores the parked glass row only if that specific park was caused by this door's original conversion (a glass row an admin turned off independently, before or since, stays off). Every conversion, relabel, insert, and removal is written to a permanent asset-event audit trail.

Removing a position: if the position (or its glass) has any issue history, it's **soft-retired** — the row and its label are preserved (so history stays intact) and the sequence gap closes; a position with zero history is hard-deleted instead. Either way you can't remove a position that still has **open** issues — resolve them first.

Glass numbering (its own card) is a separate, purely cosmetic overlay: a prefix, starting number, direction, and whether doors take a number — it changes what the app prints on the diagram and in reports, never the underlying permanent G-label. Off by default (shows the permanent G-labels). A **bulk glass spec** tool applies one width/height/thickness/material to a range of glass panels at once (e.g. every side-wall panel), with individual overrides afterward for exceptions like corner radius panels or doors.

Labels & Zones tab

A banner at the top restates the datum: every label, zone assignment, and walkthrough order below is measured from the Perimeter tab's start point and direction, with a shortcut back to change it.

- **Zones** — named groupings (North End, Home Bench, etc.) for faster lookup during a walk and as bulk-labeling targets. Add, rename, reorder, and deactivate (deactivating keeps the assignment history but hides the zone from new assignments).
- **Bulk labeling** — apply a custom **display** label pattern (prefix + start number + step + direction) across a whole zone or a contiguous perimeter range in one pass, e.g. every panel in Home Bench becomes HB1, HB2, HB3.... A live preview flags any collision (case-insensitive, per-rink) before you can apply it. This only ever writes the display-layer `custom_label` — the permanent `label` underneath is untouched, and issue history keeps following the permanent label regardless of what a segment is renamed to.
- **Segments list** — every active position in perimeter order, grouped by zone, with drag-and-drop (or up/down arrows) to **reorder the physical walk sequence itself**. This changes the order the sequence runs in, not any position's permanent label. Expanding a row lets you set its **custom label**, **search aliases** (up to 12, for finding a segment by a nickname), its **zone**, and its **out of service** flag (which flags the segment on walk checklists until cleared).

Checklist tab

Sets the **inspection weekday** (which day of the week weekly items come due) and manages **checklist items** — a facility-defined list of non-spatial things to check on a cadenced schedule: **daily**, **weekly**, **monthly**, or **yearly** (with a due month for yearly items). Weekly items come due on the configured weekday; monthly items are due until the month's first completed walk answers them; yearly items are due in their configured month until answered that month. The daily cadence ships with no items by design — daily coverage comes from the tap-a-segment model on the map itself, not a checklist.

Lists tab

Two facility-wide (not per-rink) lists that a walk needs in order to classify what it finds:

- **Door subtypes** — the picker shown when marking a position as a door (ships with Bench, Scoreboard, Public Skate, Zamboni; add your own).
- **Issue categories**, one managed list per asset type (board panel, glass panel, door) — the quick-pick category shown when reporting an issue on that kind of segment.

If a facility has neither list configured yet, a **Seed defaults** card offers to fill in a starter set (four door subtypes, a handful of issue categories per asset type) in one click — it only ever fills empty lists, so it's safe to click even after some manual customization, and it never touches rinks or the perimeter

(those are physical layout decisions with no sensible default).

Walks tab

A read-only drilldown into every inspection walk run on the selected rink — completed and in-progress, newest first, with the inspector, a pass/fail tally, and (when applicable) an "Annual contractor" badge. Selecting a walk opens its detail: every asset check on that walk (fails listed first), the walk's own notes, and — for an annual contractor inspection — the contractor's name and company. An **Export history (CSV)** button pulls the rink's complete inspection/issue history (every walk check and every issue ever logged, with resolution status) for offline recordkeeping — the kind of artifact an ice-rink safety audit (ORFA) wants to see.

5. Screen-by-screen walkthrough

The condition map (`/reports/dasher-boards` **or** `/reports/dasher-boards/[rinkSlug]`)

This single screen is the whole staff-facing tool:

- A status line: when the rink was last walked (and by whom), or "No completed walks yet," plus "Due today" once a day passes without one.
- The **perimeter diagram** — every board and door as a tappable segment in perimeter order, colored by condition: **red** = an open severity-A issue, **yellow** = an open B/C issue, **coral** = a flagged fail with no issue reported yet, **lime** = a door, **dashed outline + x** = out of service. A **Glass** toggle overlays the glass layer; a **search box** highlights segments matching a label, custom label, glass number, or alias.
- **Checklist — due today** card (only shown when items are due and you hold submit) — Pass/Fail each due item; a flagged item immediately opens its report-issue sheet, because a flag needs a linked issue before sign-off.
- The **walk bar** — starts collapsed with just a "Start inspection walk" button (plus a secondary "Annual contractor inspection..." option that asks for a contractor name and company) when no walk is open. Once a walk is active it becomes a sticky bar showing elapsed time, due-item progress, a **Guided walkthrough** launcher, and a **Sign off** toggle that expands into the completion form.
- Tapping **any segment** — walk open or not — opens its bottom sheet (see below). Tapping a checklist item opens a similar sheet scoped to that item.

The asset bottom sheet

Opens when you tap a board, glass, or door position:

- Header: the segment's resolved display label (custom label, or glass number, or the permanent label — whichever applies), its type/subtype badge, an out-of-service badge if flagged, and its zone.
- **Open issues** on this segment, each with severity, category, description, and (permission-gated) Acknowledge/Mark fixed buttons.
- **Report issue** — the default, expanded-by-default form: pick board vs. glass if the position has both, a severity (A/B/C, defaulting to B), a category, a description, and — only for severity A — a required action-taken note and a required supervisor. Submitting collapses the form to a confirmation with a "Report another" option.
- **Condition check** (only while a walk is open) — a separate Pass/Fail with its own optional note, saved immediately on tap.
- A read-only **replacement spec** (dimensions, thickness as a fraction, material, notes) when one is on file.

- A collapsed **issue history** (resolved issues only; online-only), each row flagging when it was logged under a different label than the segment currently shows.

Guided walkthrough

A full-screen, one-segment-at-a-time overlay for stepping through every active positioned segment in perimeter order — the same order the diagram itself walks. Each screen shows the segment's big display label, type, zone, and aliases, with two giant buttons: **OK** (records a pass and auto-advances to the next unchecked segment) and **Flag** (records a fail and opens that segment's report-issue sheet).

Previous and **Skip** step freely without the "already checked" filter. When every segment has been handled, it hands off straight to the sign-off panel. This is a second way to drive the exact same Pass/Fail persistence the free-tap flow uses — nothing about it is a separate code path.

Sign-off (walk completion)

Expanding "Sign off" in the walk bar shows an optional walk-notes box and a **Complete walk** button. Completing checks three gates server-side and rejects the sign-off (with a specific count in the error) if any fail:

1. any severity-A issue logged during this walk still lacks supervisor acknowledgment;
2. any checklist item due that day is unanswered;
3. any checklist item flagged "needs attention" has no linked issue report.

A signed-off walk is immutable — see §8.

Walk-complete screen (/reports/dasher-boards/[rinkSlug]/done?id=...)

A confirmation screen for one completed walk: pass/fail/issue/checklist stat pills, a read-only diagram colored by that walk's final conditions, the notable condition checks (every fail, plus any pass with a note), the issues logged, the checklist answers, and the walk notes. Action buttons: **Download PDF**, **Print Report** (browser print, chrome hidden automatically), **Send Report** (to the facility's configured Dasher Boards recipients — see §6), and links back to the rink map or the dashboard.

6. Step-by-step: common tasks

Report a problem panel (no walk needed)

1. Open **Dasher Boards** — you land on your rink's map.
2. Tap the board, glass, or door segment that's damaged.
3. In the bottom sheet, fill out **Report issue**: pick board or glass if relevant, choose a severity, a category, and describe it. Severity A also needs a supervisor and the action taken.
4. Tap **Submit issue**. The segment turns red or yellow on the map immediately for everyone who opens it next.

Run an inspection walk

1. From the map, tap **Start inspection walk** in the walk bar (or **Annual contractor inspection...** if this is an ORFA-style contractor visit — enter their name and, optionally, company).
2. Either tap segments directly as you walk the rink (Pass/Fail in the bottom sheet's Condition check block), or tap **Guided walkthrough** to step through every segment in order with big OK/Flag buttons.
3. Answer any **due checklist items** shown in their own card.

- When done, expand **Sign off**, add notes if you like, and tap **Complete walk**. Fix anything the sign-off gate flags (an unacknowledged A, an unanswered due item, a flag with no issue) and try again.
- You land on the walk-complete screen — download/print/send the report from there if needed.

Convert a board position to a door (admin)

- Admin → Dasher Boards → Perimeter**, tap the board position on the diagram.
- Pick a **door subtype** (or leave it unset), then tap **Mark as door**.
- The position keeps its history but takes the next door label; its glass row is parked automatically. Enter the door's own glass spec if it carries one.

Rename what staff see without losing history (admin)

- Admin → Dasher Boards → Labels & Zones**.
- Either open one segment in the list and set a **Custom label**, or use **Bulk labeling** to rename a whole zone or range at once with a prefix/start/step pattern.
- The permanent label (shown in parentheses wherever it diverges) keeps every past and future issue tied to the right physical position.

Seed a new facility's Dasher Boards config (admin)

- Admin → Dasher Boards → Lists** — click **Seed defaults** if shown, to get starter door subtypes and issue categories.
- Perimeter** tab — apply the **standard rink template** or generate a uniform sequence, then mark doors and adjust by tapping the diagram.
- Labels & Zones** — set up zones and any custom labels the facility uses.
- Checklist** — add any cadenced items beyond the spatial tap-and-flag coverage.

Send or download a walk report

On the walk-complete screen (or reached again from Admin → Walks → a walk → the walk detail,  VERIFY whether the admin detail view itself links back to this screen):

- **Download PDF** for a shareable copy.
- **Print Report** for a clean, chrome-free printout.
- **Send Report** to push it to the facility's configured Dasher Boards recipients (Admin → Communications routing rules). If none are configured, the button says so and names where to set one up.

7. Field reference

Field	Where	What it is
Rink	Header switcher	Which sheet of ice you're viewing (only shown with more than one active rink).
Segment (board / glass / door)	Condition map	A single perimeter position. Tap to open its sheet.
Severity	Report issue	A (safety-critical — requires a supervisor + action taken), B (needs repair), C (cosmetic). Defaults to B.

Field	Where	What it is
Category	Report issue	A quick-pick reason, scoped to the segment's asset type (board/glass/door).
Description	Report issue	Free text describing the problem. Required.
Action taken / Supervisor	Report issue (severity A only)	Both required before an A issue can be saved.
Condition check	Asset sheet, walk only	Pass/Fail (+ optional note) for this segment on the current walk.
Walk kind	Walk bar	Routine (default) or Annual contractor (records a contractor name + company).
Checklist item	Due-today card	A cadenced (daily/weekly/monthly/yearly) non-spatial check; Pass or Flag. A flag needs a linked issue before sign-off.
Walk notes	Sign-off	Optional free text saved with the completed walk.
Admin — Rink		Name, slug, template (NHL/Olympic/custom dimensions), perimeter anchor label + direction, inspection weekday, active/default.
Admin — Perimeter position		Type (board/door/corner/post-gap), permanent label, sequence position, subtype (doors), glass spec (width/height/thickness/material/notes).
Admin — Custom label / aliases / zone / out-of-service	Labels & Zones	Display-layer overrides; never affect the permanent label or issue history.
Admin — Glass numbering	Perimeter	Prefix, start number, direction, whether doors count; display-only.
Admin — Door subtype	Lists	The category shown when marking a position as a door.
Admin — Issue category	Lists	Per-asset-type quick-pick list for issue reports.
Admin — Checklist item	Checklist	Label, cadence (daily/weekly/monthly/yearly), due month (yearly only).

8. Locking, saving & offline

Open issues stay open until resolved — there's no auto-expiry. A reported issue persists on the diagram (coloring that segment red/yellow) until someone with the right tier acknowledges (severity A) and/or marks it fixed. Nothing about tapping a segment "closes" the loop by itself.

A signed-off walk is immutable. Once `completeInspection` succeeds, the walk row is locked — the three sign-off gates (unacked A issues, unanswered due items, unlinked flags) are the only thing

standing between "in progress" and "final," and once past them there's no editing a completed walk from either the staff or admin side. If something was logged wrong, the fix is a new issue or a new walk, not an edit to the old one.

Labels never get renumbered or reused, including on retired/removed positions — see §1 and the Perimeter tab section in §4. This is a deliberate design constraint enforced by the database, not just app-level convention: a relabel, conversion, or removal can shift sequence, but never identity.

Offline. Dasher Boards routes every staff write through the app's standard offline sync queue (`_lib/offline.ts`, mirroring the Ice Depth pattern) — this module is **not** one of the online-only exceptions called out for scheduling in the architecture notes:

- **Reporting an issue, starting a walk** (routine or annual contractor), **answering a checklist item, saving a Pass/Fail check**, and **signing off a walk** all queue on your device when you're offline and replay automatically once you reconnect.
- An offline-started walk has no server id yet; every queued action targets the rink, and replay resolves "my open walk on this rink" server-side — so a queued sequence (start → report issue → answer checklist → sign off) replays in order and lands correctly even though it was all typed with no connection.
- The **sign-off gates re-run at replay time**, not just when you tapped Complete — an offline sign-off that no longer satisfies them (say, an unacknowledged severity-A issue that's still unacknowledged once you're back online) is parked as a failed sync rather than being forced through.
- **Acknowledging or resolving an issue is not a queued offline action** — those are supervisor decisions made against live state and require a connection.
- Viewing walk history, the Admin console, and the walk-complete/PDF screens all require a live connection; only the act of submitting is offline-capable.

9. Troubleshooting & FAQ

I don't see Dasher Boards in my menu. Your facility hasn't turned the module on, or your account lacks the **view** permission. Ask an administrator.

"No access to Dasher Boards." You have view access to the app generally but not the Dasher Boards **view** grant specifically. Ask an admin to grant it under Admin → Permissions.

"Dasher Boards isn't set up yet." No rink/perimeter has been configured for your facility. An admin needs to build the Perimeter tab first (Admin → Dasher Boards). If you're an admin yourself, opening the module sends you straight into the setup wizard instead of this message.

I tapped a segment but there's no Report issue form. You need the **submit** permission to report issues. Viewing is unrestricted at the view tier; reporting requires submit.

A severity-A issue won't let me mark it fixed. Severity-A (safety-critical) issues require the **edit** tier to acknowledge or resolve — a staff member with only submit can close out B/C issues but not A. Ask a supervisor or admin.

Sign-off keeps getting rejected. Read the error — it names exactly what's blocking it: an unacknowledged severity-A issue, an unanswered due checklist item, or a flagged item with no linked issue. Fix that specific thing and try again.

Why does converting a board to a door change its label? Labels are permanent identity tied to what a position physically is. A door and a board panel are different kinds of asset, so the conversion assigns the next label in the door sequence (e.g. D5) and permanently retires the old board label (e.g.

B12) rather than reusing it — this keeps every prior issue report on that spot traceable.

I renamed a segment and now the diagram shows something different from an old report.

That's expected — custom labels and glass numbers are a display layer. Every issue and check keeps its `label_snapshot`, the label that was in effect when it was logged, so old reports still read correctly even after a rename; the asset sheet and history call this out explicitly when the two diverge.

"Send Report" says no recipients. No distribution list is configured for Dasher Boards. An admin sets one up under Admin → Communications.

Can I edit a signed-off walk? No. Completed walks are immutable by design (see §8) — log a new issue or start a new walk instead.

Does the daily checklist cadence do anything? Not until an admin adds items to it. It ships empty on purpose — daily coverage is meant to come from tapping problems on the map as you notice them, not from a separate daily list.

Source (footnote)

This chapter was written against the following files (not from memory):

- Admin: `src/app/admin/dasher-boards/{page.tsx,actions.ts,types.ts}`, `src/app/admin/dasher-boards/_components/{perimeter-tab.tsx,labels-tab.tsx,checklist-tab.tsx,lists-tab.tsx,walks-tab.tsx,walk-detail.tsx,rink-settings-card.tsx,seed-defaults-card.tsx,bulk-labeler.tsx}`
- Staff: `src/app/reports/dasher-boards/{page.tsx,actions.ts}`, `src/app/reports/dasher-boards/_components/{rink-screen.tsx,rink-switcher.tsx}`, `src/app/reports/dasher-boards/[rinkSlug]/page.tsx`, `src/app/reports/dasher-boards/[rinkSlug]/_components/{condition-map.tsx,guided-walk.tsx,walk-bar.tsx,due-card.tsx,asset-sheet.tsx,issue-form.tsx,item-sheet.tsx}`, `src/app/reports/dasher-boards/[rinkSlug]/done/page.tsx`, `src/app/reports/dasher-boards/[rinkSlug]/done/_components/{print-walk-button.tsx,send-report-button.tsx}`, `src/app/reports/dasher-boards/[rinkSlug]/done/pdf/route.ts`
- Logic: `src/app/reports/dasher-boards/_lib/{compute.ts,submit.ts,queries.ts,offline.ts,glass-numbering.ts,segment-labels.ts,perimeter-template.ts,display-label.ts}` and their companion `.test.ts` files
- Permissions: `src/lib/permissions/actions.ts`, `supabase/migrations/000000000000193_dasher_boards_module_registration.sql`

Facility Paperwork

RinkReports training guide — Facility Paperwork module

1. What this module is for

Facility Paperwork is a read-only document library for staff. Admins upload the facility's own policies, handbooks, and manuals — an Emergency Action Plan, an employee handbook, a safety document, whatever the facility needs everyone to be able to find — and staff browse and download them from their phone or tablet. It replaces a binder at the front desk or an email attachment that goes stale.

There is no submission workflow here — staff don't create anything in this module, they only read. Everything staff see is uploaded and organized by an admin.

You only see data for your own facility — this is automatic. There's no facility switcher, and you can't see another rink's documents.

Don't confuse this with the built-in Training & Documentation. The Rink Reports training manuals, onboarding guides, and PDFs (this document included) ship with the app itself, under **Admin → System → Training & Docs**. Facility Paperwork is for your facility's own documents — the ones only your staff need, that nobody at Anthropic or the app's maintainers ever sees.

2. Who can use it

Access is **permission-driven**, not strictly tied to a job title.

Tier	Browse & download documents	Upload / manage documents
super_admin	Yes	Yes, for any facility (picks one first if more than one exists)
facility_manager (the admin role)	Yes	Yes, for their own facility
supervisor / staff	Yes, if they have an active employee account	No

Notes that hold true in code:

- Any signed-in, active employee can open the staff Facility Paperwork page and see their facility's documents — there is no separate `submit/view` permission gate for browsing, unlike most other modules.
- Managing documents (upload, edit, hide, delete) requires the **facility admin** permission specifically — the module's own `admin/admin` action — not just general Admin Center access. An admin whose account has admin-console access but lacks that specific grant sees: "Your account has admin console access but not the facility admin permission required to manage documents. Ask an administrator to grant it under Admin → Permissions."
- A **deactivated** account is denied everywhere.

3. How to get there

Staff (browsing documents):

- In the sidebar (or the mobile **Menu** tab), choose **Facility Paperwork**. The address is `/reports/facility-paperwork`.

Admins (managing documents):

- Open the **Admin Center**, then choose **Facility Paperwork** under **Module Admin**. The address is `/admin/facility-documents`.
- A super admin who hasn't chosen a facility yet is shown a facility picker first (or is sent straight through if there's only one facility).

4. Setup & configuration (admins)

There are no tabs — the whole admin screen is one page: a **Bulk Upload** card at the top and the document list below it.

Bulk Upload

- **Category** (required) — one of six fixed categories: Emergency Action Plan, Employee Handbook, Staff Manual, Policy Document, Safety Document, or Other. Every file in one upload batch shares the category you pick.
- **Description** (optional) — applied to every file in this upload.
- **Files** — choose one or more files at once. Accepted types: PDF, Word (`.doc/.docx`), Excel (`.xls/.xlsx`), PowerPoint (`.ppt/.pptx`), plain text, CSV, RTF, and common image formats (PNG, JPG, GIF, WEBP). Each file is capped at **25 MB**.
- Each file's **title** defaults to its filename (underscores/hyphens turned into spaces); rename it afterward from the document list if you want something friendlier.
- If some files in a batch fail (wrong type, too large, a name collision) and others succeed, the upload still completes for the ones that worked and reports which ones didn't, rather than failing the whole batch.

The document list

- Documents are grouped by category, in the same fixed order as the category dropdown.
- Each row shows the title, file type/size, and (if set) the description, with **Edit**, **Deactivate**, and **Delete** actions.
- **Edit** lets you change the title, category, and description — not the underlying file. To replace a file's contents, delete it and upload the new one.
- **Deactivate** hides a document from staff without deleting it (it stays in the admin list, greyed out, and can be reactivated). **Delete** permanently removes the document record and its stored file.

5. Screen-by-screen walkthrough

Staff — Facility Paperwork (`/reports/facility-paperwork`)

A **Filter by Category** dropdown narrows the list to one category (only categories that actually have documents appear as options); "All Categories" is the default. Documents render as a grid of cards, each showing the title, category + file size, an optional description (truncated to three lines), and a **Download** link. Download links are short-lived signed URLs generated fresh each time the page loads — they aren't permanent links you can bookmark or share.

If the facility hasn't uploaded anything yet, staff see "No documents available." If a filter matches nothing, they see "No documents in this category."

Admin — Facility Paperwork (/admin/facility-documents)

One page: the Bulk Upload card, then the categorized document list described in §4. A card at the top links out to the in-app **Training & Docs** area as a reminder that the built-in Rink Reports manuals don't need to be re-uploaded here.

6. Step-by-step: common tasks

Upload your Emergency Action Plan (admin).

1. Go to **Admin → Facility Paperwork**.
2. In Bulk Upload, choose **Emergency Action Plan** as the category.
3. Choose the file (PDF works well) and, optionally, add a description.
4. Click **Upload**. The document appears in the list below immediately.

Find a document (staff).

1. Open **Facility Paperwork** from the menu.
2. Optionally filter by category.
3. Tap **Download** on the document you need.

Retire an outdated policy without losing the history (admin).

1. Find the document in the list.
2. Click **Deactivate** rather than **Delete** — staff stop seeing it, but you can still find and reactivate it later if you need to reference the old version.

7. Field reference

Field	Notes
Category	One of: Emergency Action Plan, Employee Handbook, Staff Manual, Policy Document, Safety Document, Other. Fixed list — not facility-configurable.
Title	Defaults from the filename; editable afterward. Max 200 characters.
Description	Optional, shown to staff under the title.
File	Max 25 MB. Allowed types: PDF, DOC/DOCX, XLS/XLSX, PPT/PPTX, TXT, CSV, RTF, PNG, JPG/JPEG, GIF, WEBP.
Active	On by default. Deactivating hides a document from staff without deleting it.

8. Locking, saving & offline

- There is no draft/lock concept here — an upload either succeeds or reports which files failed, immediately.

- **△ VERIFY** — this module does not appear to use the offline submission queue described in the app's PWA architecture (there's nothing for staff to submit; browsing and downloading a document both require a live connection to fetch the file and its signed URL). Treat "download it while you have signal" as the practical guidance for staff working somewhere with unreliable wifi.
-

9. Troubleshooting & FAQ

I don't see Facility Paperwork in my menu. Your facility has turned the module off, or your employee account isn't active yet. Ask your admin.

A document won't download / the link says "Unavailable." Download links expire after a short time. Refresh the page to get a fresh link.

I'm an admin but I can't upload anything. You have general Admin Center access but not the specific facility-admin permission this module requires. Ask a super admin or another facility admin to grant it under **Admin → Permissions**.

I uploaded the wrong file. Delete the document and upload the correct one — there's no in-place file replacement, only title/category/description edits.

Can staff upload anything here? No. This module is admin-managed only; staff have read/download access exclusively.

Source (footnote)

This chapter was written against the following files (not from memory):

`src/app/admin/facility-documents/{page.tsx,actions.ts,types.ts},`
`src/app/admin/facility-documents/_components/facility-documents-client.tsx,`
`src/app/reports/facility-paperwork/page.tsx,`
`src/app/reports/facility-paperwork/_components/documents-browser.tsx,` and the shared helpers in `src/lib/facility-documents.ts`.

Communications

RinkReports training guide — Communications module

1. What this module is for

Communications is the facility's messaging and alerting hub. It does two distinct jobs under one roof:

- **Alerts.** Other modules — Ice Operations, Refrigeration, Air Quality, Accident Reports, Incident Reports, Scheduling, Rink Scheduling — automatically raise an **alert** when something trips a configured trigger (an out-of-range refrigeration reading, an air-quality exceedance, a severity-flagged incident, and so on). Alerts land in this module's inbox for everyone with access to see, and **routing rules** decide who additionally gets notified by in-app message and email.
- **Messaging.** Staff send in-app messages to recipient **groups** (and reply to messages they've received); admins can **broadcast** to groups, a role, or the whole facility, either immediately or on a schedule; and admins can set up **recurring reminders** that resend a template on a cron-like schedule.

Behind those two staff-facing jobs sits the admin configuration that makes them work: **Templates** (reusable subject/body presets), **Groups** (named recipient lists), **Routing** rules (which alerts notify whom, how, and when), **Reminders**, a **Deliveries** view for retrying failed sends, and an **Audit Log**.

How a submission becomes an alert, in five steps:

1. A staff member submits a report elsewhere in the app. If it trips a configured trigger, that module raises a **communication_alert**.
2. **Routing rules** match the alert (by source module, severity, and optionally area) and fan it out to their targets on the rule's timing — immediately, in an end-of-day digest, or weekly.
3. Matched recipients get an **in-app message** and an **email**; if the rule attaches a PDF, a rendered PDF of the source submission is linked in the message and attached to the email.
4. If the rule requires acknowledgement, recipients must acknowledge it from their inbox — admins can watch the acknowledgement list on the alert.
5. Background jobs run every few minutes to render PDFs, drain the notification queue, and send emails, retrying automatically; anything that fails permanently for good surfaces in **Deliveries** for a manual retry.

One thing that surprises new admins: **routing rules control who gets notified, not who can see an alert.** Every alert your facility generates shows up in the Alerts inbox for **every** employee with view access to Communications — routing only decides who additionally receives an in-app message + email for it. With no routing rules configured at all, alerts still appear in the inbox; they simply don't fan out to anyone's message list.

You only see data for your own facility — this is automatic. There's no facility switcher, and you can't see another rink's alerts, messages, groups, or routing rules.

2. Who can use it

Access is **permission-driven**, not tied to a job title: an administrator grants each person **view**, **submit**, and/or **admin** on the communications module independently, per facility (`user_permissions`). Note that "editing" someone else's sent message isn't a thing this module supports at all — messages

and alerts aren't edited in place by anyone, staff or admin (see §8).

Role tier (doc vocabulary)	Typical access to Communications
super_admin	Full: sees the facility's Alerts/Messages/Sent inbox, composes and sends messages, plus the full admin console (Broadcast, Templates, Groups, Routing, Reminders, Deliveries, Audit Log).
facility_manager (live role: admin)	Same, for their own facility — provided their account also holds the module-scoped communications admin permission (see note below); passing the general Admin Center gate is not by itself enough.
supervisor (live role: manager, or a custom role)	⚠ VERIFY — typically granted view and submit on Communications by default: sees the inbox, acknowledges alerts/messages, composes and replies. Reaches the admin console only if separately granted the module's admin permission.
staff (live role: staff, or a custom role)	Sees the inbox and can acknowledge alerts/messages if granted view ; can compose new messages and reply if granted submit . No access to the admin console (Broadcast, Templates, Groups, Routing, Reminders, Deliveries, Audit Log).

Notes confirmed in code:

- The staff inbox (/reports/communications) requires the **view** permission for communications. Without it: "You don't have permission to view communications."
- Composing or replying (/reports/communications/compose) requires **submit**. Without it, the "New message" button doesn't appear, and hitting the compose URL directly shows "You don't have permission to send messages."
- Staff sends are restricted to **groups flagged "Staff can message this group"** — an admin's own compose (Broadcast) isn't limited that way. A **reply** is the one exception to "groups only": a non-admin may reply directly to the sender of a message they actually received, and only to that sender.
- The **admin console** (/admin/communications) is gated **twice**: `requireAdmin()` (general Admin Center access — see the Admin Control Center chapter) **and** `requireModuleAdmin("communications")`, the module-scoped admin grant that the RLS write policies also key off (`has_module_admin_access('communications')`). Someone with general admin access but not this specific grant is sent to /forbidden before the console even renders. Every admin server action re-checks the same grant server-side and, if it's missing, returns: "Your account has admin console access but not the communications module's admin permission. Ask an administrator to grant it under Admin → Permissions."
- A **deactivated** employee account can't see or do anything in this module — the staff pages look for an active employees row and show "Account not set up" if there isn't one.

3. How to get there

Staff (inbox, compose, reply):

- Sidebar (desktop) or the **Menu** tab (mobile) → **Communications**. This opens /reports/communications, defaulting to the **Alerts** tab; append `?inbox=messages` or `?inbox=sent` to land on another tab directly (the in-page tab links do this for you).

- **New message:** the **New message** button on the inbox (visible only with submit) goes to `/reports/communications/compose`. A reply from a message's detail page goes to `/reports/communications/compose?replyTo=<message id>`.
- After sending, you land on `/reports/communications/compose/done?id=<message id>`.

Admins (configuration and oversight):

- Admin Center → **Module Admin** → **Communications**, at `/admin/communications`. It opens on the **Inbox** tab; the other seven — **Broadcast**, **Templates**, **Groups**, **Routing**, **Reminders**, **Deliveries**, **Audit Log** — are `?tab=` query params on the same URL (`/admin/communications?tab=routing`, etc.), so a tab's URL is bookmarkable and shareable.

4. Setup & configuration (admins)

Every admin action in this module is double-gated: `requireAdmin()` plus the module-scoped **communications** admin permission (`ensureCommsAdmin()` in `actions.ts`), so an admin who can reach other consoles but lacks this specific grant is turned back with the message quoted in §2. Configuration below is spread across five of the eight admin tabs — **Templates**, **Groups**, **Routing**, **Broadcast**, and **Reminders**. (Inbox, Deliveries, and Audit Log are operational views, covered in §5.)

Templates tab

Templates are reusable subject + body + acknowledgement presets, used by staff compose, admin Broadcast, and Reminders alike.

- **Add template** captures **Name** (required), **Slug** (auto-derived from the name if left blank, must be lowercase letters/digits/hyphens and unique per facility), **Category** (free text — e.g. "safety", "ops" — not a fixed list), **Subject** (optional), **Body** (required), and a **Requires acknowledgement** checkbox.
- Existing templates get **Edit**, **Deactivate/Activate**, and **Delete**. **Delete is blocked** if any active **Reminder** still references the template ("Template in use by N reminders; deactivate instead.").
- Picking a template on the staff compose form or the admin Broadcast form fills in its subject, body, and acknowledgement setting in one click — the sender can still edit any of it before sending.

Groups tab

Groups are named recipient lists — the unit that both staff compose and admin Broadcast (and Routing/Reminders targets) address messages to.

- **Add group:** **Name** (required), **Slug** (auto-derived, unique per facility), **Description** (optional), and the **"Staff can message this group"** checkbox — this is the switch that decides whether an ordinary staff member sees this group as a compose destination at all. Admins can always address any active group from Broadcast regardless of this flag.
- Selecting a group from the left-hand list opens its detail pane: **Edit** (name/slug/sort order/description/staff-can-message), **Deactivate/Activate**, **Delete**, and a **Members** panel where you add or remove employees one at a time from a dropdown of active, not-yet-member employees.
- **Delete is blocked** if the group is still referenced by a routing rule or a reminder ("Group is referenced by routing rules or reminders; deactivate instead.").

Routing tab

Routing rules are the heart of the alerting story: each rule decides which alerts get **fanned out as notifications** (in-app message + email), to whom, how urgently, and with what extras. Every rule has:

- **Name** (optional, for your own reference).
- **Source module** — which module's alerts this rule watches: Ice Operations, Refrigeration, Accident Reports, Air Quality, Incident Reports, Scheduling, Communications, Daily Reports, Ice Depth, Dasher Boards, or Rink Scheduling.
- **Severity filter** — `info / warn / high / critical`, or **Any** to match every severity from that module.
- **Area (optional)** — scopes the rule to one location/area within the source module. For modules with a real area list wired up — **Daily Reports** (its Areas) and **Air Quality** (its Facility Spaces/locations) — you get a proper dropdown of that module's areas, plus an "Any area" option. Every other module falls back to a raw-UUID text box, because there's no unified areas table to enumerate across modules; ⚠ VERIFY whether the module you're routing for actually stamps an `area_id` on its alerts at all — if it never does, an area-scoped rule for it will simply never match.
- **Target** — exactly one of:
 - **Group** — an existing communication group.
 - **Role** — every active employee holding that role key (`super_admin`, `admin`, `manager`, or `staff`).
 - **Employee** — one specific person.
 - **Department** — every active employee in that department.
- **Timing** — **Immediate** (send now), **End of day digest**, **Weekly (next Monday)**, or **Manual** (queued into `notification_outbox`, no automatic send — a scheduler drains due rows). End-of-day and weekly timings work the same way: they queue rather than send immediately.
- **Attach PDF of the submission** — renders a PDF of the underlying source record, links it from the in-app message, and attaches it to the outbound email.
- **Require recipient acknowledgement** — recipients must explicitly acknowledge the resulting message; use this for anything genuinely critical (e.g. a critical-severity accident report).
- **Priority** — a plain integer; higher-priority rules apply first when several rules could match the same alert.

Rules render as a list, newest-priority first, each with **Preview recipients**, **Edit**, **Activate/Deactivate**, and **Delete**. **Preview recipients** calls the `resolve_rule_recipients` database function live and shows exactly who the rule currently resolves to (name + email) — use it before saving a rule, or any time membership changes, to sanity-check that a rule targets who you think it does; an empty result (with the message "No active employees match this rule") usually means an empty group, a role nobody currently holds, or a department with no assigned employees.

With **zero routing rules**, alerts still land in every viewer's Alerts inbox (see §1) — they just never generate an in-app message or email to anyone specifically.

Broadcast tab

Broadcast is the admin's own compose tool — unlike staff compose, it isn't limited to `staff_can_message` groups, and it can also target a role or the whole facility.

- **Use a template (optional)** pre-fills subject/body/acknowledgement from any active template; you can still edit before sending.
- **Subject** (optional) and **Message** (required body).
- **Send to** — one of three mutually exclusive scopes:
 - **Groups** — check one or more active groups; delivered to the union of their members.
 - **Everyone with a role** — pick one role; delivered to every active employee holding it.

- **Whole facility** — every active employee in the facility.
- **Send later (optional)** — a date/time picker. Leave blank to send immediately. Set it, and the broadcast is instead queued as one `notification_outbox` row per resolved recipient with `scheduled_for` set to that time; the notifications-drain cron converts the batch into a single message + recipient fan-out once it's due. A scheduled broadcast **arrives as a system message** — it does not carry your name as sender, unlike an immediate send.
- **Requires acknowledgement** checkbox, same meaning as everywhere else in this module.

Scheduled broadcasts appear in a **Scheduled broadcasts** card above the compose form, each showing its subject, send time, and resolved recipient count, with a **Cancel** button. Cancelling flips every **still-pending** row of that batch to **cancelled** — rows the drain cron already converted (i.e., already delivered) are untouched, so cancelling only ever stops what hasn't gone out yet. If nothing is left to cancel you'll see: "Nothing left to cancel — it may have already been sent."

Sending (immediate or scheduled) reuses the exact same `persistMessage` pipeline staff sends use — the delivery mechanics, read/ack tracking, and email fan-out are identical either way; only the sender-restriction rules differ (admin broadcast has none).

Reminders tab

Recurring reminders resend a template to a target on a repeating schedule — think shift-change checklists, weekly safety reminders, or a recurring maintenance nudge.

- **Add reminder: Name** (required), a **5-field cron schedule** (minute hour day-of-month month day-of-week, e.g. `0 8 * * 1` = every Monday at 08:00, evaluated in the facility's timezone), a **Template** to send (only active templates are offered, unless the reminder already points at one that's since been deactivated — that one still appears, marked "(inactive)"), an optional **Next run** timestamp, and a **Target: Group or Role** (the same two target kinds Routing offers, minus Employee/Department).
- Each row shows its **Last run** and **Next run** timestamps and offers **Edit**, **Activate/Deactivate**, and **Delete**.
- The scheduler that actually fires reminders runs every few minutes in the background; it is app-level cron logic (△ VERIFY the exact cron/worker mechanism — the schema comment on `schedule_cron` notes it's "interpreted by the app worker, not `pg_cron`").
- With **no active templates** in the facility, the Reminders tab tells you to add one on the Templates tab first rather than showing an unusable form.

Cross-reference — Admin Control Center. Whether Communications shows in staff navigation at all is set on the **Modules** page; the four permission actions (view/submit/admin, and the module-scoped `admin` grant this console specifically requires) are set under **Permissions**. See the Admin Control Center chapter for those shared screens.

5. Screen-by-screen walkthrough

A. Staff inbox (/reports/communications)

A tab bar — **Alerts**, **Messages**, and **Sent** (Sent only shown if you hold submit) — each with an unread-count badge. **New message** appears top-right if you hold submit.

- **Alerts tab.** A search box, **Source** and **Severity** filter-pill rows (multi-select, click to toggle), and **Unread only** / **Hide resolved** checkboxes, all client-side over up to the 100 most recent alerts. Each row shows the source-module badge, severity badge, an "Ack required" badge if

unacknowledged and required, a "Resolved" badge if closed, an unread dot, the title, a relative timestamp, and a body excerpt. Tapping a row opens the alert.

- **Messages tab.** Search plus **Unread only**, over your received messages (most recent 100). Rows show sender, relative time, subject (or a body excerpt if no subject), an unread dot, and an "Ack required" badge if applicable.
- **Sent tab** (submit-holders only). Your own authored messages with **recipient / read / acknowledged** counts rolled up right on the row.

B. Alert detail (/reports/communications?alert=<id>)

Shows the module badge, severity, resolved/ack-required badges, title, full body, when it was created and by whom (if attributable), the area id if any (raw, not resolved to a name —  VERIFY whether staff ever see a human-readable area name here), and — if resolved — when and by whom. If the alert requires acknowledgement and you haven't yet, an **Acknowledge this alert** card appears with an optional notes field; once acknowledged, it instead shows your acknowledgement time and any notes you left, permanently.

C. Message detail (/reports/communications?message=<id>)

Shows sender, sent time, subject, body, and a **Download PDF attachment** link if the message carries one (a routing-rule PDF attachment or otherwise). Opening the page automatically marks the message read (fire-and-forget, no button — see §8). If acknowledgement is required and outstanding, an **Acknowledge this message** card appears the same way alerts' does. A **Reply** button appears only if the message has a real employee sender (system-authored messages, e.g. a drained scheduled broadcast, have none to reply to).

D. Sent-message receipts (/reports/communications?sent=<id> and the post-compose done screen)

Shows what you sent, then a **Receipts** list: every recipient with a **Read / Unread** badge and, if acknowledgement was required, an **Acknowledged / Awaiting ack** badge with timestamps. This is server-rendered and relies on an RLS extension (migration 170) that lets a message's sender read its own recipients' rows even though those rows otherwise belong to each recipient.

E. Compose / Reply (/reports/communications/compose)

Described field-by-field in §7. In **reply mode** (?replyTo=<id>), the recipient is locked to the original sender (shown as a read-only "To:" chip) and the group picker is hidden entirely — you cannot broaden a reply into a group send.

F. Admin — Inbox tab (/admin/communications?tab=inbox)

Two views via a toggle, **Alerts** and **Messages** (?inbox=alerts / ?inbox=messages), each with URL-driven filters (source module, severity, resolved/open, free-text search, and a date range defaulting to the last 30 days) and a **Load more** link once the current window (200 rows) is exhausted.

- **Alerts view:** each row is a link into a drilldown with the full alert plus its acknowledgement list, and **Resolve / Re-open / Delete** actions. Resolving stamps `resolved_at` + who resolved it; deleting an alert also deletes its acknowledgement records and prompts "Resolving is usually the better way to close out a real alert."
- **Messages view:** a table of every facility message with recipient/read/ack rollups and a **Delete** action per row (deletes read/ack records with it).

G. Admin — Deliveries tab (/admin/communications?tab=deliveries)

Two tables, shown only when non-empty: **Failed email deliveries** (`communication_recipients` rows whose `email_status` reached `failed` after exhausting the automatic retry backoff) and **Failed queued notifications** (`notification_outbox` rows the drain cron marked `failed`). Each row shows the recipient, subject, attempt count / error text, and queued/scheduled time, with a **Retry** button that resets the attempt counter and re-queues the row for the next scheduled cron run. If nothing has ever failed, the tab shows a single "No failed deliveries" card.

H. Admin — Audit Log tab (/admin/communications?tab=audit)

A filterable (entity type, action, actor, date range), paginated (500-row window, **Load more**) append-only log of every mutation across this module — template/group/routing-rule/reminder create/update/delete/activate/deactivate, alert resolve/reopen/delete, message delete, message sent, broadcast scheduled/cancelled, and failed-delivery retries. Each row can be expanded (**Diff**) to show its before/after JSON snapshots side by side.

6. Step-by-step: common tasks

Send a message to a group (staff).

1. **Communications** → **New message**.
2. Optionally pick a template to pre-fill subject/body/acknowledgement.
3. Write your **Message** (required).
4. Check one or more **Recipient groups** (only groups flagged staff-messageable appear).
5. Optionally check **Requires acknowledgement**.
6. Tap **Send message** (or, offline, **Save offline** — see §8).

Reply to a message (staff).

1. Open the message from **Messages** and tap **Reply**.
2. The recipient is locked to the original sender; write your reply and send.

Acknowledge a required alert or message (staff).

1. Open the alert/message; if acknowledgement is required and outstanding, an **Acknowledge** card appears.
2. Optionally add notes, then tap **Acknowledge**. This is permanent — there is no un-acknowledge.

Route a module's alerts to an on-call group (admin).

1. Admin → Communications → **Routing** → **Add routing rule**.
2. Pick the **Source module** and, if you want to narrow it, a **Severity** and/or **Area**.
3. Pick a **Target** — Group / Role / Employee / Department.
4. Pick a **Timing**, and check **Attach PDF** / **Require acknowledgement** as needed.
5. **Add rule**, then click **Preview recipients** on the saved rule to confirm it resolves to who you expect.

Send an immediate broadcast to the whole facility (admin).

1. Admin → Communications → **Broadcast**.
2. Write (or template-fill) a **Subject** and **Message**.
3. Under **Send to**, choose **Whole facility**.
4. Leave **Send later** blank and click **Send broadcast**.

Schedule and later cancel a broadcast (admin).

1. Compose the broadcast as above, but set **Send later** to a future date/time, then **Schedule broadcast**.
2. It appears under **Scheduled broadcasts**; click **Cancel** any time before it fires to stop delivery to whichever recipients haven't received it yet.

Set up a weekly reminder (admin).

1. Ensure an active **Template** exists (Templates tab).
2. Communications → **Reminders** → **Add reminder**.
3. Name it, set **Schedule** to something like 0 8 * * 1 (Mondays at 8 AM, facility time), pick the **Template** and a **Group** or **Role** target.
4. **Add reminder** — it fires automatically from its next scheduled run.

Retry a failed email (admin).

1. Communications → **Deliveries**.
2. Find the row under **Failed email deliveries**, click **Retry**.

7. Field reference

Field	Where	Required?	Notes
Subject	Staff/admin compose	No	Free text. Reply mode auto-prefixes "Re:" from the original subject.
Message (body)	Staff/admin compose	Yes	Free text; empty body is rejected with "Please enter a message."
Recipient groups	Staff compose	Yes , unless replying	Must pick ≥1 group flagged <code>staff_can_message</code> , or be in reply mode.
Requires acknowledgement	Compose / template / rule	No (default off)	Recipients must explicitly acknowledge before the message is considered handled.
Use a template	Compose	No	Pre-fills subject/body/ack from an active template; still editable after.
Send to (scope)	Admin Broadcast	Yes	Groups / Everyone with a role / Whole facility — exactly one.
Send later	Admin Broadcast	No	Future date/time; queues per-recipient outbox rows instead of an immediate send.
Name	Group / Template / Routing rule / Reminder	Varies (required for Group, Template, Reminder; optional for Routing rule)	Free text.

Field	Where	Required?	Notes
Slug	Group / Template	Auto from name if blank	Lowercase letters/digits/hyphens, unique per facility.
Staff can message this group	Group	No (default off)	Gates whether ordinary staff see the group as a compose destination.
Category	Template	No	Free text, not a fixed list (e.g. "safety", "ops").
Source module	Routing rule	Yes	One of the modules listed in §4.
Severity	Routing rule	No (default "any")	info / warn / high / critical.
Area	Routing rule	No	Real picker for Daily Reports / Air Quality; raw UUID box for other modules.
Target kind + value	Routing rule / Reminder	Yes	Group / Role / Employee / Department (Reminder: Group or Role only).
Timing	Routing rule	No (default "immediate")	Immediate / End-of-day digest / Weekly / Manual.
Attach PDF	Routing rule	No (default off)	Renders + links/attaches a PDF of the source submission.
Priority	Routing rule	No (default 0)	Integer; higher applies first among matching rules.
Schedule (cron)	Reminder	Yes	5 fields: minute hour day-of-month month day-of-week, evaluated facility-local.
Next run	Reminder	No	Optional override of the next scheduled fire time.

8. Locking, saving & offline

There's no draft/edit-window concept anywhere in this module — no message, alert, template, group, rule, or reminder is ever edited "in place" after the fact by its original author or a recipient. What each side can do instead:

- **Messages and alerts are immutable content.** A staffer can't edit a message after sending it, and can't edit an alert at all (alerts are system-generated). Admins can **resolve/re-open** an alert (a status flag, not a content edit) and **delete** either one outright; there is no partial edit.
- **Acknowledgements and the audit log are append-only** — the database has no UPDATE/DELETE policy on either table. Once you acknowledge something, that record and its timestamp are permanent; the only "undo" is that admins can still see and act on the underlying alert/message independently.

- **Admin configuration (Templates, Groups, Routing rules, Reminders) saves immediately** on submit — no draft state, no publish step. Every create/update/delete/activate/deactivate writes a best-effort row to the Audit Log; a failure to write that audit row never blocks or rolls back the actual change.
- **Scheduled broadcasts** are the one place with a genuine in-between state: queued rows sit in `notification_outbox` until their `scheduled_for` time, and can be cancelled up to the moment the drain cron converts them into a real, delivered message (see §4 and §6).

Offline — confirmed for compose, staff-side only. Composing or replying goes through the same offline queue described in the app's PWA architecture: when `ComposeForm` detects `navigator.onLine === false`, it blocks the normal form POST and instead calls `enqueueSubmission({ moduleKey: "communications", action: "submit", payload })` from `src/lib/offline/use-sync-queue.ts` — the client never writes to `offline_sync_queue` directly. The submit button relabels to **Save offline**, and after queuing you see a **"Saved on this device"** confirmation ("this message is queued and will send automatically once you're back online"). Once connectivity returns, the service worker replays the queued submission to `/api/offline-sync`, which for `moduleKey === "communications"` dispatches to `handleMessageReplay()` in `src/app/reports/communications/_lib/offline.ts`. That handler re-runs the exact same parse → validate → permission → persist pipeline as the online action (`persistMessage` in `_lib/submit.ts`), re-resolving `isAdmin` fresh from the replaying user's live session, and is idempotent via the queue's `local_id` claim — a duplicate replay (e.g. the service worker retrying after a lost response) is a no-op rather than a second send.

Everything else in this module is online-only. Acknowledging an alert (`acknowledgeAlert`), acknowledging a message (`acknowledgeMessage`), and the automatic mark-as-read on opening a message (`markMessageRead`) are plain server actions with no `enqueueSubmission` call anywhere in their components — unlike compose, there is no offline queueing path for them at all. ⚠ VERIFY the exact behavior a staffer sees tapping **Acknowledge** while offline (the code has no special offline handling for it, so it most likely surfaces as a generic failed-request error rather than a friendly "saved for later" message); the practical guidance is to acknowledge alerts and messages only once you have a live connection. The entire **admin console** (Broadcast, Templates, Groups, Routing, Reminders, Deliveries, Audit Log) is likewise plain server actions with no offline path — it's an admin workstation tool, not a floor-level PWA flow.

9. Troubleshooting & FAQ

"You don't have permission to view communications." Your account lacks **view** on the Communications module. Ask an admin to grant it (Admin → Permissions).

"You don't have permission to send messages." You have **view** but not **submit**. Ask an admin to grant submit access, or ask someone who has it to send on your behalf.

"Account not set up" / "Your account isn't fully set up yet." Your login isn't linked to an active employee record in this facility. Contact your administrator.

"There aren't any active groups to send to in this facility." No group is both active and flagged "Staff can message this group" (unless you're replying, which doesn't need a group). Ask an admin to create or flag one on the Groups tab.

"Can't reply to that message." The message either isn't in your inbox, or it was sent by the system (no human sender to reply to) — e.g. a scheduled broadcast that already went out.

Why can't I message a specific person directly? Ordinary staff sends are group-only, with one exception: replying, which is locked to the original sender. Direct arbitrary messaging isn't a feature

for non-admins — ask an admin to add the person to a shared group, or to broadcast on your behalf.

"Your account has admin console access but not the communications module's admin permission." You reach the Admin Center but haven't been granted the module-scoped `admin` action specifically for Communications. Ask a super admin or another facility admin to grant it under Admin → Permissions.

An alert I care about never generated a message or email. Check the **Routing** tab: alerts always show up in everyone's inbox regardless of rules, but only a matching, active routing rule fans one out as a notification. Use **Preview recipients** on the rule to confirm it resolves to the people you expect, and check the rule's severity/area filters aren't excluding this alert.

"Nothing left to cancel — it may have already been sent." The scheduled broadcast's send time already passed and the drain cron already converted it into a delivered message before you clicked Cancel.

"Template in use by N reminders; deactivate instead." / "Group is referenced by routing rules or reminders; deactivate instead." You can't delete a template or group that something else still points at. Deactivate it instead — it stays available to whatever references it but drops out of new pickers.

A failed email/notification won't go away. Go to **Deliveries** and click **Retry** — it resets the attempt counter and re-queues the row for the next scheduled send/drain run. If it keeps failing, the underlying error text (shown in the row) is your best clue — usually a bad email address or an email-provider error.

Can I attach a photo or file to a message? No. Compose has no file/photo upload; a routing rule can attach a **generated PDF of the source submission**, but that's system-produced, not something a sender uploads.

I submitted a message while offline — did it send? It's saved on your device ("Saved on this device") and sends automatically once you reconnect, going through the same checks as an online send. You'll find it in your **Sent** tab once it syncs. Acknowledging alerts/messages, by contrast, needs a live connection (see §8).

Source (footnote)

Staff flow: `src/app/reports/communications/page.tsx`, `actions.ts`, `types.ts`, `compose/page.tsx`, `compose/done/page.tsx`, and `_components/` (`acknowledge-alert-form.tsx`, `alerts-list.tsx`, `compose-form.tsx`, `format.ts`, `inbox-tabs.tsx`, `message-detail.tsx`, `messages-list.tsx`, `receipts-list.tsx`, `sent-list.tsx`). Pure/tested logic: `_lib/compose.ts` + `_lib/compose.test.ts`. Server-only pipeline: `_lib/submit.ts`, `_lib/offline.ts`, and the `moduleKey === "communications"` dispatch in `src/app/api/offline-sync/route.ts`.

Admin flow: `src/app/admin/communications/page.tsx`, `actions.ts`, `types.ts`, `_lib/validate.ts` (+ `validate.test.ts`), and `_components/` (`audit-tab.tsx`, `broadcast-tab.tsx`, `deliveries-tab.tsx`, `groups-tab.tsx`, `inbox-tab.tsx`, `reminders-tab.tsx`, `routing-tab.tsx`, `templates-tab.tsx`).

Schema / RLS / permissions: `supabase/migrations/000000000000009_communications_schema.sql` (base schema + RLS), `...045/...063` (routing rule columns: `target_department_id`, `timing`, `attach_pdf`, `requires_acknowledgement` — as referenced in `admin/communications/types.ts`), `...059_communication_groups_staff_can_message.sql`, `...170_communications_security_remediation.sql`, `src/lib/permissions/actions.ts` (module/action model), `src/lib/permissions/check.ts` (`currentUserCan`).

Adapted from: docs/training-guide/src/ch06-admin-modules.md, §6.9 "Communications Admin" and §6.10 "How a submission becomes an alert (the pipeline)" — reused for the pipeline explanation in §1 and the admin-tab overview in §4, cross-checked against the source files above.

Employee Scheduling

Brand: RinkReports. Primary green **#4DFF00**, navy **#002244**. **You only see data for your own facility — this is automatic.** You never switch facilities or see another rink's schedule; RinkReports keeps each facility's records separate for you.

1. What this module is for

Employee Scheduling is how a facility **plans who works which shift** and how staff **manage their own availability and requests**. It has two sides:

- **The admin scheduling grid** — managers build a week of shifts by dragging on a calendar, assign people to them, and then publish the schedule so staff can see it.
- **The staff scheduling app** — each employee sees their upcoming shifts, sets weekly availability, requests time off, picks up open shifts, and asks to swap shifts with a coworker.

Important — there is no payroll, timekeeping, or clock-in/out here. Scheduling is **assignment only**: it says who is scheduled to work when. It does **not** record hours actually worked, clock anyone in or out, produce timesheets, or calculate pay. The one pay-adjacent number, "**max weekly hours**," is purely an **overtime/scheduling warning threshold** — when an assignment would push someone over it, the grid shows a warning. It changes nothing about pay.

The module also runs **scheduling rule checks** as you assign people: required-certification gaps, double-booking, approved time-off overlap, minor hour caps, overtime thresholds, and required breaks. Most are advisory warnings; a missing required certification is a hard block (see §8).

2. Who can use it

Access is **permission-driven**, not a fixed rank ladder. The table below is the typical default mapping; because an admin can customize any person's permissions per module, two people with the same title can differ. ⚠ **VERIFY** any place a tier label feels too rigid for a specific facility.

Tier (doc vocabulary)	Real app role	Scheduling access (default)
super_admin	super_admin (platform-wide)	Full admin grid + all approvals across the system; sees everything an org/facility admin sees.
org_admin	(no exact equivalent — ⚠ VERIFY ; treat as super_admin)	No separate tier exists in RinkReports.
facility_manager	admin (facility administrator)	Full admin grid: build/assign/publish shifts, approve publish requests, decide time-off and swaps, configure settings and job areas. Needs the admin permission for this facility.
supervisor	manager (or a custom role)	Mid-tier. Whatever the admin grid permission grants them; commonly review/decision rights. ⚠ VERIFY per facility — gated by permissions, not title.

Tier (doc vocabulary)	Real app role	Scheduling access (default)
staff	staff (or a custom role such as driver)	Staff scheduling app only (my shifts, availability, time off, claim open shifts, request/accept swaps). Needs the scheduling view permission; submitting requests needs submit. No access to the admin grid.

- The **admin grid** (/admin/scheduling) requires the **admin** permission (or **super_admin**). Non-admins who try are shown a **Forbidden** message.
- The **staff app** (/reports/scheduling) requires an active employee account with the scheduling view permission. Without **view**, the page shows "You don't have access to scheduling yet." Submitting availability or time off additionally needs **submit**; if you only have **view**, you can see your schedule but not file requests.
- A **deactivated** account is denied everywhere.

3. How to get there

Staff side — /reports/scheduling ("Scheduling" in the staff menu / mobile bottom tabs):

- The landing screen shows a **Next shift** hero, a 7-day week strip, your **Upcoming** shifts, any **Open · Pick up** shifts you can claim, **Your claims · Awaiting approval**, and **Quick links** to My schedule, Time off, Availability, Shift swaps, and Notifications.

Admin side — /admin/scheduling (Admin Center → Module Admin → **Scheduling Admin**):

- A sticky sub-navigation runs across the top: **Overview · Shifts · Templates · Publish history · Publish requests · Time-Off · Swaps · Compliance · Job areas · Settings · Notifications**.
- **Overview** is the dashboard (KPI cards, pending swaps/time-off, open shifts, week-at-a-glance, module tiles).
- **Shifts** is the drag-and-drop grid where you actually build the schedule.

4. Setup & configuration (admins)

Before staff see a useful schedule, an admin sets up a few things. These all live under **Scheduling Admin**.

Job areas (positions) — /admin/scheduling/job-areas

"Job areas" are the **positions** people can be scheduled into (e.g. Skate Rental, Front Desk, Resurfacer).

- **Add** a job area by typing a name and pressing **Add area** (or **Bulk upload**).
- **Reorder** with the up/down arrows, **rename** inline, and **deactivate** (toggle) or **delete**.
 - Deactivating hides an area from new assignments without affecting people already assigned.
 - Deleting is only allowed when **no one is assigned**; otherwise deactivate instead.
- **Required certifications**: under each area you can add cert names (e.g. CPR). RinkReports will then **hard-block** assigning anyone who lacks a current matching certification to a shift in that area (a manager can override — see §8).

Assigning people to job areas (cross-reference)

Adding an employee, choosing their role, and seeding permissions is covered in the **Admin Control Center** chapter. The **scheduling-specific** parts of the employee form are:

- **Job areas** — multi-select, **up to 4** per employee (cross-training). One can be marked the **Primary job area**. These determine which positions the person can be scheduled/qualified for, and which areas they can pick when setting availability.
- **Max weekly hours** — an optional **per-employee cap**. Leave blank for "No individual cap." This is the overtime-warning threshold only (not payroll).

Scheduling settings — /admin/scheduling/settings

One settings form controls scheduling behavior facility-wide. There is also a **Seed defaults** button that fills in sensible starting values and a few default compliance rules.

Key settings (see §7 for the full list):

- **Week start day, Default shift minutes.**
- **Minor max weekly hours, Overtime weekly hours, Minimum break minutes / after hours** — feed the advisory warnings.
- **Swaps require manager approval, Swap request expiry (hours)** (default 72), **Open shifts: first-come, first-served.**
- **Notify employees when schedule is published, Notify on overtime warnings.**
- **Allow staff to submit weekly availability** (on/off).
- **Require employees to be assigned to a shift's job area.**
- **Block scheduling-grid saves that raise warnings** — when **on**, advisory warnings (hours cap, overlap, etc.) block the save; when **off**, they are advisory and a manager can confirm and save anyway. (Missing required certs always block regardless.)

Templates & compliance

- **Templates** (/admin/scheduling/templates) are reusable shift patterns you can apply to a week. You can also save a block straight from the grid (see §5).
- **Compliance** (/admin/scheduling/compliance) lists the rule set (minor hours, overtime, required break) that drives the warnings; rules can be enabled/disabled, edited, and reordered.

5. Screen-by-screen walkthrough

Admin grid — /admin/scheduling/shifts

The grid header reads: "Drag in a day column to create a shift; drag a block to move it, or its edges to resize. Click a shift to assign, duplicate, or delete."

- **Views:** Day / Week / Month toggle, with Previous/Next and **Today**. Month view is read-only — click a day to drop into the editable week.
- **Drag-create:** drag in a day column to paint a new shift; an **assign popover** opens.
- **Move / resize:** drag a block to move it; drag its top/bottom edge to change start/end. Changes save immediately (optimistically).
- **Assign popover** (create or edit): set **Start time / End time**, choose an **Employee** (or Open / unassigned), and a **Job area**. As you choose, RinkReports checks for conflicts live ("Checking for conflicts...") and shows any cert block or advisory warnings. You can also **Save as template** from here.

- **Click a shift** → a detail panel on the right lets you **assign**, **duplicate**, **edit**, or **delete** it.
- **Position (job-area) filter**: a filter chip row narrows the board to one job area so the whole view — grid, hours, crew tally — stays consistent.
- **Right rail**: Open shifts needing coverage, Swap requests, Time-off, and a Crew roster with each person's scheduled hours vs. their cap.
- **Export** writes the visible week to CSV. A KPI strip shows scheduled hours, shift count, open shifts, and pending swaps.
- **Request publish · (week)** button — files a publish request for the visible week (see below).

Publishing

- **Publish requests** (/admin/scheduling/publish/requests): the queue of pending requests. Each shows the range, who filed it, and notes. A different admin clicks **Approve & publish** (confirm dialog: "Approve and publish all draft shifts in this window?") or **Reject** with a required reason. **You cannot approve or reject your own request** — the buttons are hidden and the system blocks it, showing "You filed this request — a different admin must approve or reject it."
- **Publish history** (/admin/scheduling/publish): an append-only log of every publish event (when, range, shift count, who published).

Time-off & swaps (admin)

- **Time-Off** (/admin/scheduling/time-off): pending employee requests; **Approve** / **Deny** (with an optional note) — only pending requests can be decided. The employee is notified of the decision.
- **Swaps** (/admin/scheduling/swaps): pending swap requests; a manager can **assign a target**, **Approve** (re-runs all rule checks and applies the trade or a one-way coverage hand-off), **Deny**, or **Cancel**. An applied swap can't be denied or cancelled afterward.

Staff app — /reports/scheduling

- **My schedule** — your upcoming published shifts.
- **Availability** (/reports/scheduling/availability) — your weekly availability rows. Add an entry: **Day of week**, **Start/End time**, **Type** (Available / Preferred / Unavailable), an optional **Area/department you want to work** (only areas you're assigned to), optional **Effective from/to** dates, and notes. Edit or remove entries. If your facility has turned availability submission off, you'll see a message instead.
- **Time off** (/reports/scheduling/time-off) — **New request** with **Starts**, **Ends**, and an optional reason. Track status (pending/approved/denied) and cancel your own (see §8).
- **Open shifts** — on the landing screen, published shifts with no one assigned appear under **Open · Pick up** with a **Claim shift** button. Some are marked **Approval req.** A claim you make appears under **Your claims · Awaiting approval** until a manager approves it.
- **Shift swaps** (/reports/scheduling/swaps) — **New swap request**: pick **your shift to give up**, optionally a **coworker**, and optionally **their shift to take** (a trade) or leave it as just covering yours. Lists show **Outgoing** and **Incoming** requests with status; you can **Accept** an incoming pending swap directed at you, or **Cancel** your own pending/accepted one.
- **Notifications** (/reports/scheduling/notifications) — schedule alerts (publish, decisions, swap notices); mark one or all read.

6. Step-by-step: common tasks

Place and edit a shift (admin)

1. Go to **Scheduling Admin → Shifts**. Navigate to the right week.
2. **Drag** in a day column to create a shift (or click **Add shift**).
3. In the popover, set **Start/End time**, pick an **Employee** (or leave Open), and a **Job area**. Resolve any warnings, then **Save**.
4. To edit later, **drag** to move/resize, or **click** the shift and use the detail panel (assign / duplicate / edit / delete).

Publish a schedule

1. Build the week's draft shifts on the grid.
2. Click **Request publish · (week)**, add optional notes for the approver, and **File request**.
3. A **different** admin opens **Publish requests**, clicks **Approve & publish**, and confirms. Drafts in that window become **published**; any unassigned published shifts open for claims.

Request a publish & approve it (two-person flow)

1. Admin A files the publish request (above).
2. Admin B opens **Publish requests → Approve & publish** (or **Reject** with a reason). Admin A cannot approve their own — that's by design.

Submit availability (staff)

1. **Scheduling → Availability → New availability** (or per-day).
2. Choose day, times, **Type**, optional **area** and effective dates, then **Add availability**. (Queues offline if you're disconnected — see §8.)

Request time off (staff)

1. **Scheduling → Time off → New request**.
2. Set **Starts** and **Ends**, add an optional reason, **Submit request**. Track its status; cancel while pending/approved if plans change.

Claim an open shift (staff)

1. On the **Scheduling** landing screen, find a shift under **Open · Pick up**.
2. Tap **Claim shift**. If it needs approval, it moves to **Your claims · Awaiting approval** until a manager approves. (Online only.)

Request a swap (staff)

1. **Scheduling → Shift swaps → New swap request**.
2. Pick **your shift to give up**, optionally a **coworker** and **their shift**, add a note, **Send swap request**.
3. The coworker (if chosen) is notified and can **Accept**; a manager then approves it if your facility requires approval. (Online only.)

7. Field reference

Admin shift (assign popover / detail)

Field	What it is
Start time / End time	The shift's hours. End must be after start.

Field	What it is
Employee	Who is assigned, or Open / unassigned .
Job area	The position the shift is for (drives qualification & cert checks).
Break minutes	Unpaid break length, subtracted from scheduled hours in tallies.
Role label	Optional free-text label shown on the shift.
Notes	Optional shift notes.
Status	Draft, Published, or Cancelled (set by the publish / cancel flows, not typed directly).
Override reason	Optional note recorded when a manager overrides a certification block.

Scheduling settings

Setting	Effect
Week start day	First day of the scheduling week.
Default shift minutes	Default length for a newly created shift.
Minor max weekly hours	Cap used to warn when a minor is over-scheduled.
Overtime weekly hours	Threshold for the overtime warning.
Minimum break minutes / after hours	When a long shift should include a break (advisory).
Swaps require manager approval	If on, an accepted swap still needs a manager to approve/apply it.
Swap request expiry (hours)	Undecided swaps lapse to "expired" after this (default 72, capped at the shift start).
Open shifts: first-come, first-served	How open-shift claims are resolved.
Notify on publish / on overtime	Whether staff get notified on those events.
Allow staff to submit weekly availability	Turns the staff availability form on/off.
Require employees to be assigned to a shift's job area	Enforces job-area qualification.
Block scheduling-grid saves that raise warnings	On = advisory warnings block; off = advisory only (manager confirms).

Staff availability

Field	What it is
Day of week	Sunday–Saturday.
Start / End time	The window for that day.
Type	Available / Preferred / Unavailable.
Area / department you want to work	Optional; must be a job area you're assigned to.
Effective from / to	Optional date range the entry applies to.
Notes	Optional.

Staff time-off / swap

Field	What it is
Starts / Ends (time off)	The requested off period. End must be after start.
Reason (time off)	Optional.
Your shift to give up (swap)	The shift you want covered/traded.
Coworker (swap)	Optional target coworker.
Their shift to take (swap)	Optional — makes it a trade rather than one-way coverage.
Note (swap)	Optional message.

8. Locking, saving & offline

Draft → Published lifecycle

- New shifts are **drafts** — admins build and rearrange them freely; staff don't see drafts.
- A schedule is released through a **publish request** that a **second admin approves** (requester ≠ approver). Approving turns drafts in that window **published** and visible to staff; any **unassigned** published shift becomes an **open shift** staff can claim.

Publish-lock (what changes after publishing)

- Once a shift is **published it is locked** from free editing or deletion. Admins can still change a published shift, but only through a **controlled, audited path** that re-checks the scheduling rules and notifies affected staff (the editor shows "This shift is published — saving republishes it and notifies affected staff").
- **Deleting a published shift is a soft cancel** — it is marked **cancelled**, not erased, so there's a record. (Draft shifts are deleted outright.)

Approvals as locks

- **Time-off** and **swaps** move **pending** → **approved/denied** (swaps also pass through **accepted** and end at **approved & applied**). You can **cancel your own pending (or approved) time-off**, and **cancel your own pending/accepted swap**, but you **cannot reverse a manager's decision**, and an **applied swap** can't be undone here.

Certification & warning gates (when assigning)

- A **missing/expired required certification** for the shift's job area is a **hard block**. Only a facility manager can **override**, and every override is **logged** (with an optional reason).
- Other signals — overtime, minor hours, double-booking, approved-time-off overlap, missing break, job-area mismatch — are **advisory**. They warn and require a **Confirm & save**, unless the facility turned on "block saves that raise warnings," in which case they block too.

What queues offline vs. online-only

RinkReports is a PWA: some staff actions are **saved on your device and sync when you reconnect**. Scheduling's offline scope is deliberately narrow:

- **Queues offline (staff): Submit availability** and **Request time off**. When offline, the form saves locally and shows "Saved offline — will sync when you're back online," replaying once reconnected.

- **Online-only (by design):** the **entire admin grid** (drag-create / edit / move / delete / publish), **claiming an open shift**, **swap requests**, **accepting a swap**, and **cancelling a swap**. These depend on live shift state and on cert/hour-cap/publish-lock checks that can't be safely replayed later, so they require a connection. (Confirmed in code: swap request/accept/cancel and open-shift claim all run as direct online actions with no offline queue.)

9. Troubleshooting & FAQ

"Where do I see hours worked or run payroll?"

You don't — that isn't in RinkReports. Scheduling is **assignment only**: it plans who works when. There's no clock-in/out, timesheet, or pay calculation. "Max weekly hours" is only a warning threshold for over-scheduling.

"I can't approve my own publish request."

Correct — a publish must be approved by a **different** admin than the one who filed it. Ask a colleague with admin access to approve it.

"The grid won't let me assign this person."

If it's a **certification block**, the person lacks a required cert for that job area — only a facility manager can override (it's logged). If it's an **advisory warning** (overtime, overlap, etc.), click **Confirm & save** — unless your facility blocks warning saves, in which case fix the conflict first.

"I claimed/swapped a shift while offline and nothing happened."

Claiming and swaps are **online-only**. Reconnect and try again. (Availability and time-off do queue offline.)

"Staff say they can't see the schedule."

Check that the schedule for that week has been **published**, the staff member has the scheduling view permission, and their account is **active** and assigned to your facility. To let them file requests too, they also need **submit**.

"A published shift disappeared from history after delete."

It didn't — deleting a **published** shift is a **soft cancel** (status cancelled), so the record remains. Only **drafts** are removed outright.

"Why can't this employee be scheduled in this position?"

They may not be assigned to that **job area**, or the area may be **deactivated**. Add the area to their employee record (up to 4) or reactivate the area under **Job areas**.

Source (footnote)

```
src/app/reports/scheduling/ (page.tsx, actions.ts, swaps/, time-off/, availability/, notifications/,
my-schedule/, _components/{availability-form, time-off-form, swap-form, swap-action-button,
claim-open-shift-button}.tsx); src/app/admin/scheduling/ (page.tsx, shifts/page.tsx +
_components/{week-board, assign-popover, publish-button}.tsx, publish/page.tsx,
publish/requests/_components/requests-client.tsx, job-areas/_components/job-areas-client.tsx,
settings/_components/settings-form.tsx, _components/scheduling-nav.tsx, _lib/{grid-actions,
publish-request-actions, enforcement, governance-actions}.ts);
src/app/admin/employees/_components/employee-form.tsx (job areas, max weekly hours);
```

`src/app/api/offline-sync/route.ts` (scheduling replay = availability + time-off only);
`docs/training/00-MANIFEST.md`.

Rink Scheduling

Brand: RinkReports. Primary green **#4DFF00**, navy **#002244**. **You only see data for your own facility — this is automatic.** You never switch facilities or see another rink's bookings, customers or invoices; RinkReports keeps each facility's records separate for you.

1. What this module is for

Rink Scheduling is ICE-TIME booking and billing — which rink is booked when, who it's booked for, what it costs, and whether it has been paid. It covers the whole commercial life of a booking: the calendar itself, rate cards and pricing, season-long contracts with clubs, invoicing and accounts receivable, ice-resurfacing ("cut") scheduling, locker-room assignment, a public ice-time request form, and lobby TV displays.

This is a completely different module from Employee Scheduling, even though both modules are called "Scheduling" in the app and sit next to each other in the admin menu. Employee Scheduling (/admin/scheduling) plans who works which shift. Rink Scheduling (/admin/rink-scheduling) books ice time and bills for it — there are no employees, shifts, or timesheets anywhere in it. The nav config carries a comment for exactly this reason:

"Sits next to Scheduling Admin deliberately: that one is EMPLOYEE scheduling (shifts), this one is ICE scheduling (bookings and billing). Adjacent placement with distinct labels is what keeps an admin from opening the wrong console." — src/components/admin/nav-config.ts

If you came here looking for shift assignment, time-off, or swaps, see the **Employee Scheduling** chapter instead.

Rink Scheduling is also **revenue-critical**: every booking can carry a price (via rate cards, with a prime/non-prime split and per-booking-type overrides), and that price flows into invoices, accounts receivable aging, and season-contract billing. Getting the setup right in §4 below — rinks, operating hours, rate cards, booking types — is what makes every downstream screen (the calendar, invoices, insights) correct.

2. Who can use it

Access here is **permission-driven**, through the same per-user, per-module, per-action grants used everywhere else in RinkReports (user_permissions, action = view / submit / edit / admin, module = rink_scheduling). There is **no separate role ladder** for this module — a "front desk" person and a "biller" are simply accounts with different action grants. Grants are set under **Admin → Employees → Permissions**.

Action on rink_scheduling	What it unlocks
view	Read the calendar: the staff dashboard schedule (/reports/rink-scheduling), Front Desk (/reports/rink-scheduling/desk), and — combined with the global Admin Center gate — the admin Ice Schedule grid in read-only form.

Action on <code>rink_scheduling</code>	What it unlocks
submit	Everything view grants, plus creating a tentative booking on the admin Ice Schedule grid. Cannot confirm a booking, move/resize one, cancel one, or touch anything money-related.
edit	Full day-to-day scheduling and billing: confirm/move/resize/cancel bookings, manage recurring series and resurface ("cut") status, assign locker rooms, decide public requests and the waitlist, manage season contracts, and everything in Invoices & Insights. Also covers Facility Setup (rinks, locker rooms, operating hours, exceptions) and Displays in the admin config console.
admin	Everything edit grants, plus Rate Cards and Module Settings in the admin config console — and, separately, admin is what's required just to open <code>/admin/rink-scheduling</code> at all (see below).

Two extra things worth knowing:

- **Reaching the admin config console** (`/admin/rink-scheduling`) **itself requires the module's admin action**, on top of the global Admin Center (`admin/admin`) grant — the page calls `requireAdmin()` and `requireModuleAdmin("rink_scheduling")`. Someone with only `edit` on `rink_scheduling` cannot open that console page at all, even though `edit` is enough to actually save most of what's on it (Facility Setup, Displays). In practice, grant a facility's rink-scheduling manager **both edit and admin** so the console opens and every tab saves. ⚠ VERIFY with a real account if your facility has split these grants across two people.
- **The admin Ice Schedule grid** (`/admin/rink-scheduling/schedule`) **only needs the global Admin Center gate plus view/submit/edit** on the module — it does **not** require the module's `admin` action. So a scheduler can be given Admin Center access and just `edit` on `rink_scheduling` to run the calendar day-to-day, without ever seeing the config console.
- There is also a genuinely **unauthenticated, public tier**: lobby TV boards, the ICS calendar feed, and the public ice-time request form. Nobody signs in on those — a long random link (a "display token") is the credential. See §3 and §4 ("Displays").

A **deactivated** account is denied everywhere, as usual.

3. How to get there

Admin configuration — Admin Center → Module Admin → **Rink Scheduling Admin**, or directly:

- `/admin/rink-scheduling` — the config console (tabs below are `?tab=` query params on this same URL):
 - `?tab=setup` — **Facility Setup** (default landing tab)
 - `?tab=rates` — **Rate Cards**
 - `?tab=lists` — **Lists**
 - `?tab=displays` — **Displays**
 - `?tab=settings` — **Settings**
- `/admin/rink-scheduling/schedule` — **Ice Schedule**, the live, editable booking calendar (a separate route, not a config tab — it carries its own `view/date/rink` query params).

Staff / front-desk / customer-facing surfaces — under **Rink Schedule** in the staff menu, or directly:

- /reports/rink-scheduling — **Rink Schedule**, the read-only dashboard calendar.
- /reports/rink-scheduling/desk — **Front Desk**, a quick-lookup agenda for phone questions.
- /reports/rink-scheduling/requests — **Requests & Waitlist** (?tab=requests or ?tab=waitlist).
- /reports/rink-scheduling/contracts — **Season Contracts**.
- /reports/rink-scheduling/invoices — **Invoices** (?tab=invoices, ?tab=aging, or ?tab=new), plus /reports/rink-scheduling/invoices/[invoiceId] for one invoice and /reports/rink-scheduling/invoices/[invoiceId]/pdf for its PDF.
- /reports/rink-scheduling/insights — **Insights** (?month=YYYY-MM).
- /offline-rink-schedule — a data-free, service-worker-cacheable shell showing the last-synced calendar when you have no connection (see §8).

Public, tokened surfaces — no login, no account (each is a random link an admin generates in Displays):

- /display/[token] — a lobby TV board: locker rooms or the ice schedule, depending on the token's type.
- /api/display/[token] — the JSON endpoint the TV board polls (not meant to be opened by a person).
- /api/rink-ics/[token] — a subscribable ICS calendar feed for coaches' and leagues' own calendar apps.
- /request-ice/[token] — a public ice-time request form.

4. Setup & configuration (admins)

Everything here lives at /admin/rink-scheduling, one tab per section. A brand-new facility (no operating-hours row yet) is shown a **Seed defaults** card at the top of Facility Setup — it calls the same idempotent database seeder the facility-creation trigger runs, filling in starter rinks, hours, booking types, customer types, payment methods and a zero-rate default rate card, all with **ON CONFLICT DO NOTHING** so it can never clobber anything you've already edited.

Facility Setup (?tab=setup) — **needs** edit

The root of the module: everything downstream (the calendar's columns, locker-room pickers, the TV display, coverage checks) reads from these four tables and only these four.

- **Rinks** — the ice surfaces this facility books, each a column on the calendar. Fields: name, an uppercase **short code** (≤8 characters — shown on the TV display and dense grid headers), a display colour, and optional per-rink overrides for resurface (cut) duration and ice-make buffer minutes, which fall back to the facility defaults on the Settings tab when left blank. **Capped at 10 active rinks** — this is an application rule (not a database constraint), because it bounds how many columns the calendar can render; deactivate one to add another past the cap. Deleting is refused while a rink has bookings (deactivate instead).
- **Locker rooms** — rooms that can be assigned to a booking. Fields: name, short code, optional capacity, an optional default rink ("rink-side"), notes, and sort order.
- **Operating hours** — one weekly grid, saved as a single form (all seven days upsert together). Times are the **facility's own local clock**, never UTC. A closing time earlier than the opening time means the day runs past midnight (e.g. a 6:00–2:00 rink) — enter it exactly that way rather than as

two rows. A booking outside these hours is **not blocked**; it is flagged as a coverage gap (see §7, §9).

- **Holidays & special days (exceptions)** — a specific date's hours entirely replace that day's usual weekly row. Only one exception per date.

Rate Cards (?tab=rates) — needs admin

What ice costs, and when. See §7 for the exact pricing model (rate-engine.ts). Editing a rate card, a prime window, or an override **never** changes a booking or invoice that already exists — a booking snapshots its price at save time.

- **Rate cards**: name, an effective date range (effective_start, optional effective_end), an **is_default** flag, and a flat **prime** / **non-prime** hourly rate. Only one default card may cover any given date (an exclusion constraint enforces it); non-default cards may overlap freely — useful for a "member rate" card a scheduler picks explicitly for one customer.
- **Prime windows** on a card: day-of-week + start/end time (facility-local). Anything not inside a prime window on that card bills at the card's non-prime rate.
- **Per-type overrides** on a card: a specific booking type can bill at its own prime/non-prime rate on that card instead of the card's own rate — e.g. \$0/hr for an internal program, or a discounted youth-league rate.

Lists (?tab=lists) — needs edit

Three lookup lists that fill the pickers elsewhere in the module:

- **Booking types** — what a calendar slot is for (Ice Rental, Public Skate, Figure Skating, Maintenance Block, ...). Each has a colour (the block's colour on the grid), an **is_billable** flag (a non-billable type occupies ice but never reaches an invoice — a booking of that type is also allowed to have **no customer**), and an **is_resurface** flag (marks it as an ice-cut booking rather than a bookable rental). One type, **Maintenance Block**, is a built-in **system** type: it can be renamed and recoloured but never deleted, and never made billable — flipping that would orphan the customer-less bookings that rely on it staying non-billable.
- **Customer types** — how booking parties are classified (internal program, team, league, school, ...). Name + sort order only.
- **Payment methods** — how a recorded payment was made. "Card (recorded)" logs that a card payment happened elsewhere; **no card details are ever stored in RinkReports**.

Displays (?tab=displays) — needs edit

Public, tokened links — a lobby TV board, a subscribable calendar feed, or a public request form. Each is created with a **type**, a **label**, and (for the two board types only) a **look-ahead window** (1–48 hours) and a **refresh interval** (15–3,600 seconds, defaulting to the facility's module setting).

- **locker_rooms** → renders at /display/{token} — a live board of which room holds which team, now and next.
- **ice_schedule** → renders at /display/{token} — a live board of what's on each rink, now and next.
- **rink_ics** → serves at /api/rink-ics/{token} — a read-only ICS feed a coach or league can subscribe to in their own calendar app.
- **request_form** → renders at /request-ice/{token} — the public ice-time request form.

The plaintext link is shown exactly once, right after creation, in a "copy this address now" card. Only a SHA-256 hash of the token is ever stored — RinkReports genuinely cannot show it to you again. Lose it, and the fix is to **revoke** that link and issue a new one; **revoking** stops it working immediately

(kept in the list, greyed out, for the record), while **deleting** removes the row outright.

Settings (?tab=settings) — needs admin

One row per facility, in four groups. See §7 for the full field list. Highlights:

- **Calendar:** default ice-make (buffer) minutes, default resurface (cut) duration, the calendar's drag/snap increment, and a toggle for whether ice-make time is billed as part of the rental hour or reserved after it. A **coverage-check** toggle turns the "no staff / outside hours" flags on the calendar on or off facility-wide.
- **Billing:** default payment terms (days), the invoice number prefix, and an optional tax rate (blank = no tax line at all, which is a different statement from entering 0).
- **Billing emails:** whether a booking confirmation email fires automatically, whether overdue-invoice reminder emails go out, and the reminder cadence (minimum days between reminders on the same invoice).
- **Locker rooms & displays:** how many minutes before/after a booking a locker room is held (lead / vacate time), and the default TV-display refresh interval.

5. Screen-by-screen walkthrough

Admin — Ice Schedule (/admin/rink-scheduling/schedule)

The live, editable booking calendar — **this is the only place any scheduling write happens** (see the callout in §8). Header: "Book, move and cancel ice across every surface. Times are the rink's own local clock; staff see this same calendar read-only on their dashboard."

- **Views:** Day / Week / Month / **Agenda**, with Previous/Today/Next navigation and a rink filter (week and month views, when the facility has more than one active rink). Agenda doubles as the printable daily schedule — the toolbar and shell chrome are `print:hidden`, so printing shows just the list.
- **Drag-create:** drag in a rink/day column to open the booking sheet pre-filled with that window.
- **Move / resize:** drag a block to move it to another time or rink column; drag its edge to resize. Only whole, undragged blocks move — a booking clipped by the visible day's edge (crossing midnight) is rescheduled through the sheet instead, never by dragging its visible fragment. Drags apply **optimistically** and roll back if the server rejects them (most commonly, a booking-overlap conflict once the resurfacing buffer is counted).
- **Booking sheet** (create or edit): rink, customer, booking type, title, start/end time, status (tentative/confirmed), notes. A live rate quote and any booking-overlap conflicts are fetched as you edit; on an edit, a **Locker Rooms** panel lets you assign/release rooms for that booking, and resurface bookings expose **Cut status** controls (Scheduled/Completed/Skipped) plus a shortcut to the matching Ice Operations record.
- **Toolbar tools:** **Find a slot** (available to submit+ accounts — a phone-call answer: pick a duration and date range, see every open window across the rinks); **Plan cuts** and **New series** (both edit-tier): see §6. **Coverage gaps** and **Show cancelled** are view filters, not tools.
- A  badge on a block marks a coverage gap (see §7/§9); it does not block anything.

Staff dashboard — Rink Schedule (/reports/rink-scheduling)

The **same calendar component**, deliberately locked read-only: `canCreate/canEdit` are pinned `false` here regardless of the account's actual grants, so no edit affordance ever renders on this page. A link to **Front Desk** always shows; a **Manage schedule** → link to the admin grid appears only for edit-tier accounts. Requires view.

Front Desk (/reports/rink-scheduling/desk)

A phone-call lookup tool, **also view-tier and also read-only** — no booking creation happens here despite the name. It shows: two pinned widgets ("Next resurface" per rink, "Next Public Skate"), a day picker (today $\pm$ a short window), an optional filter by booking type, and that day's agenda as a simple list (time, colour swatch, label, rink). A resurface booking is called out with a wrench icon rather than treated as an ordinary rental.

Requests & Waitlist (/reports/rink-scheduling/requests) — edit-tier

Two tabs:

- **Requests** — new submissions from the public request form land in an inbox (requester name/email/phone, organization, requested date/window, rink if named, purpose, received-at). Approving one **creates a tentative booking** (see §6) and marks the request approved, linked back to the booking; declining requires a reason. The last 20 decided requests are shown underneath for reference.
- **Waitlist** — manually add an entry (a customer, or a plain contact name/phone, for a date with an optional rink and an optional time window — leaving the window blank means "any time that day"). When a booking is later cancelled, matching entries are surfaced (§6) so the desk can call while the opportunity is still warm.

Season Contracts (/reports/rink-scheduling/contracts) — edit-tier

Open (draft/active) contracts render as cards; finished (completed/cancelled) contracts list at the bottom. Each card shows the customer, season dates, rate (a negotiated flat rate, or "Card pricing" to defer to the rate cards), billing mode (auto-invoiced on a day of the month, or manual), how far billing has progressed, notes, any **bound series** (recurring bookings this contract already covers), and the invoices it has produced. A contract inside its final 60 days with no renewal on file is flagged **expiring soon**.

Invoices (/reports/rink-scheduling/invoices) — edit-tier

Three tabs behind one summary strip (Outstanding, Overdue invoices, Billed this month, Collected this month):

- **Invoices** — every invoice for the facility, filterable/sortable in the table.
- **Aging** — the same A/R aging buckets (Current, 1-30, 31-60, 61-90, 90+ days) shown per-customer, driven by `buildAgingReport` in `ar.ts`.
- **New invoice** — pick a customer and a date window, see exactly which of their billable, non-cancelled, not-already-invoiced bookings fall in it (pre-selected, deselectable), then generate a draft. See §6.

One invoice (/reports/rink-scheduling/invoices/[invoiceId]) shows its line items, payment history, and status-specific actions (Send, Email, Record payment, Reverse a payment, Void). `.../pdf` streams the same PDF document a "Send" or "Email" action attaches — one render function shared by the download route and the email path, so what a biller sees and what a customer receives are never two different documents.

Insights (/reports/rink-scheduling/insights) — edit-tier

A month-by-month operating dashboard: ice utilization per rink (prime/non-prime/unclassified minutes, as a percentage of posted operating hours), a 6-month revenue trail (invoiced vs. collected), top customers by invoiced amount, ice time by booking type, and the same A/R aging table as Invoices. Navigable by month; defaults to the facility's current month.

Public displays and forms (no login)

- `/display/[token]` — a fixed navy/lime board (deliberately not theme-aware — it's a lobby TV, not an app surface) polling `/api/display/[token]` on the token's configured refresh interval. The payload is an **allow-list**, not a redaction: it is built field-by-field from only what a skater needs (room or rink, a label, a time window) — there is no code path by which a customer's contact details, a rate, an invoice, or a note can reach it.
- `/api/rink-ics/[token]` — the same allow-list discipline, served as an ICS feed. Both this and the display API are rate-limited per-token (and, for unrecognized tokens, per the guessing attempt itself) since they are unauthenticated internet endpoints.
- `/request-ice/[token]` — the public request form: name, email, phone, organization, an optional specific rink (else "any rink"), a date within the next two years, a start/end time (at least 30 minutes), and an optional message. An unknown, revoked, or wrong-type token renders the same neutral "this link isn't active" screen as a legitimately-inactive one — never a message that would confirm a guess was close.

6. Step-by-step: common tasks

Set up a brand-new facility for ice scheduling (admin)

1. Go to **Rink Scheduling Admin** → **Facility Setup** and click **Seed defaults** if it's offered — this seeds starter rinks, hours, booking types, customer types, payment methods, and a zero-rate default rate card.
2. Add or rename **rinks** (each needs a short code) and **locker rooms**.
3. Set **operating hours** for the week, and add any **holiday exceptions**.
4. Switch to **Rate Cards** and set real prime/non-prime rates (and prime windows) on the default card, or add a new one.

Book a single ice slot (admin, on the Ice Schedule grid)

1. Go to **Ice Schedule**, navigate to the right day/week, and **drag** in a rink column (or use **Find a slot** first if you don't know what's open).
2. In the booking sheet, pick the **booking type**, a **customer** (required unless the type is non-billable), and confirm the **start/end time**. Check the live rate quote.
3. Set **status** to Confirmed (needs edit) or leave it Tentative (available at submit), then Save. A conflicting slot is refused with the details of what it collides with.

Find and book an open slot (front-desk phone call)

1. On the Ice Schedule grid, click **Find a slot**.
2. Choose a duration, a date range, and optionally a specific rink; the results list every open window.
3. Pick a result to open the ordinary booking sheet, pre-filled — it re-checks everything at save time like any other booking.

Create a recurring booking series

1. On the Ice Schedule grid, click **New series** (edit-tier).
2. Pick the rink, booking type, customer, day(s) of the week, a time window, a date range, and the recurrence interval (every N weeks).
3. Click **Preview** — every occurrence is expanded and checked for conflicts before anything is written; conflicted dates default to **Skip**, but you can choose to book them anyway or pick something else.

4. **Commit** — each occurrence is priced against whichever rate card was in effect on that occurrence's own date, so a series spanning a rate-card change bills correctly across the boundary.

Plan and apply today's ice cuts

1. On the Ice Schedule grid, click **Plan cuts** (edit-tier).
2. Pick a day (and optionally one rink) and **Preview** — RinkReports proposes a resurface booking in every gap between consecutive rentals on that rink-day that is long enough for the configured cut duration, skipping gaps that already have (or sit next to) a cut.
3. **Apply** — the plan is **recomputed from scratch at apply time**, never trusted from the preview; any gap a concurrent booking has since filled is skipped and reported rather than double-booked.

Handle a public ice-time request

1. Open **Requests & Waitlist → Requests**.
2. Review the requester's details, date/window, and message. Pick a rink (required if the requester left it as "any rink").
3. **Approve** — this creates a tentative booking and links the request to it; or **Decline** with a reason. A conflict on approval leaves the request undecided so you can pick another rink or handle it by hand.

Set up a season contract and bind a recurring series to it

1. Open **Season Contracts → Create contract**: pick the customer, name it, set the season's start/end dates, an optional negotiated rate (blank defers to rate cards), and whether it auto-invoices (and on what day of the month, 1–28).
2. On the contract card, **bind** an existing recurring series for that same customer to the contract — only unbound series for that customer are offered.
3. From then on, the season-contract cron bills the bound series' bookings each month in arrears once the invoice day passes, advancing the contract's billed-through cursor one month at a time.

Generate, send, and collect on an invoice

1. Open **Invoices → New invoice**, pick the customer and a date window, and **Find** — every billable, uninvoiced booking in that window for that customer is listed and pre-selected.
2. Deselect anything you don't want on this invoice, add optional notes, and **Generate** — this creates a **draft**.
3. Open the invoice and **Send** (marks it sent; emails the customer if the facility's confirmation-email settings allow it) or use **Email** separately to resend the PDF.
4. As payments come in, **Record payment** (amount, date, method, optional reference) — the invoice's status (sent → partially paid → paid) is always **derived** from the payment total, never set by hand. A mistaken payment is corrected with **Reverse**, never edited or deleted.

Publish a lobby display or public request link

1. Open **Rink Scheduling Admin → Displays → Add link**, choose the type (locker room board, ice schedule board, calendar feed, or request form), and a label.
2. **Copy the address immediately** — it's shown exactly once and cannot be retrieved again.
3. Open it on the lobby TV browser (for a board) or share it (for the feed or the request form). If it's ever lost or compromised, **revoke** it and create a new one.

7. Field reference

Rate card pricing model (rate-engine.ts) — how a booking's price is actually derived

1. **Pick the rate card** in effect for the booking's **facility-local start date**: a card whose `effective_start-effective_end` covers that date. A customer's own `default_rate_card_id` wins outright if it covers the date; otherwise the **default** card covering the date wins; if several defaults somehow qualify, the one with the latest `effective_start` wins; with no default at all, the latest-starting covering card wins. No covering card at all means the booking cannot be priced (a problem string is returned rather than silently pricing at \$0).
2. **Slice the booking into prime / non-prime segments**. Prime windows are day-of-week + time-of-day ranges in the facility's own zone (not UTC), so the engine walks every local calendar day the booking touches, projects each day's prime windows into UTC instants for that specific date (correct across DST transitions), and cuts the booking wherever a boundary falls inside it. A slot straddling a prime boundary becomes two (or more) priced segments — never one rate "rounded" to a majority.
3. **Rate per segment**: the card's own prime/non-prime hourly rate, unless a **per-type override** exists for this card + this booking type, in which case the override's rate is used instead.
4. **Total and rounding**: each segment's amount is computed at full precision; rounding to the cent happens **once, on the sum** — never per segment, which is what stops two half-cent segments drifting a cent from the honest total.
5. **What gets saved**: a booking that lands wholly in prime or wholly in non-prime snapshots that single hourly rate (`rate_snapshot_hourly`) and whether it was prime (`rate_snapshot_prime = true/false`). A booking that straddles both snapshots `rate_snapshot_hourly` as **null** and `rate_snapshot_prime` as **null** — there is no single honest rate to name — and the blended total lives only in `computed_amount`. Invoicing always bills the **quoted** `computed_amount`, never a `hours × snapshot-rate` recompute, specifically because that recompute is wrong for a straddled booking (it would bill `hours × null = $0`) and drifts by a cent even for a uniform-rate booking due to hour rounding.
6. **Nothing here ever restates history**. A later edit to a rate card, a window, or an override changes only future quotes — every booking and invoice keeps the rate it was saved with.
7. A **non-billable** booking type (e.g. Maintenance Block) is never priced at all — its quote is \$0 with no rate card involved, and it never reaches an invoice.

Rate card

Field	Notes
Name	Free text, e.g. "2026-27 Season".
Effective start / end	End is optional (open-ended).
Default	At most one default card may cover any given date (enforced by a database exclusion constraint). Non-default cards may overlap freely.
Prime rate / hr	Flat rate for time inside a prime window.
Non-prime rate / hr	Flat rate for everything else on the card.
Prime windows	Day of week + start/end time, facility-local. A window may not wrap past midnight — enter it as two windows (...-23:59 and 00:00-...) instead.
Per-type overrides	Booking type → its own prime/non-prime rate on this card only.

Rink

Field	Notes
Name	Free text.
Short code	≤8 characters, upper-cased — shown on the TV display and grid headers.
Colour	The block colour for this rink's bookings.
Resurface (min) override	Cut duration for this sheet; blank uses the facility default.
Ice-make (min) override	Buffer minutes for this sheet; blank uses the facility default.
Sort order	Column order on the calendar.
Active	Deactivated rinks don't count toward new bookings; capped at 10 active.

Locker room

Field	Notes
Name / Short code	Short code shows on the TV board.
Capacity	Optional, 1–500.
Rink-side	Optional default rink association.
Notes	Optional, admin-only (never shown on the public display).

Booking (calendar / booking sheet)

Field	Notes
Rink	The sheet the booking occupies.
Customer	Required unless the booking type is non-billable.
Booking type	Drives colour, billability, and whether it's a resurface booking.
Title	Optional free-text label.
Start / End	Facility-local wall clock in the sheet; stored as UTC.
Status	Tentative, Confirmed, or Cancelled (cancellation requires a reason and is a soft state, not a delete).
Notes	Optional.
Buffer (ice-make) minutes	Snapshot at save time from the rink/facility settings in effect then.
Rate snapshot (hourly, prime)	Set for a wholly prime or wholly non-prime booking; both null for a straddled one.
Computed amount	The quoted total actually billed; null (not \$0) when nothing could price it.

Module settings

Setting	Effect
Ice-make time (minutes)	Facility default buffer after a rental; overridable per rink.
Resurface duration (minutes)	Facility default cut length; overridable per rink.
Calendar slot (minutes)	The grid's drag/snap increment (5/10/15/20/30/60).
Ice-make included in rental	On: buffer time is billed as part of the booked hour and nothing is blocked afterward. Off (most rinks): buffer minutes are reserved after each booking. Either way, invoicing always bills the booked hours.
Flag bookings with no staff / outside hours	Turns the coverage-gap badges on/off.
Payment terms (days)	Default days-to-due on a new invoice; a customer's own terms override it.
Invoice prefix	≤12 chars, letters/numbers/hyphens.
Tax rate (%)	Blank = no tax line at all (different from 0%).
Send booking confirmation emails	Emails the customer's billing contact on booking creation.
Send overdue invoice reminders / cadence	A daily job nags customers past their due date, at most once per cadence period per invoice.
Locker lead / vacate minutes	Padding before/after a booking a locker room is considered held.
Display refresh (seconds)	Default poll interval offered when creating a new board-type display link.

Booking type

Field	Notes
Name / Colour	Colour is the block colour on the grid.
Billable	Off = occupies ice, never invoiced, and bookings of this type may have no customer.
Ice resurface	Marks bookings of this type as cuts, not rentals.
Built-in (system)	Only true for Maintenance Block : cannot be deleted or made billable.

Season contract

Field	Notes
Customer	Required.
Name	≤160 characters.
Season start / end	End must be after start.
Contract rate	Optional flat hourly rate; blank defers to rate cards.
Auto-invoice	Bills the bound series' bookings monthly in arrears once on.

Field	Notes
Auto-send	Sends (and emails) each auto-generated invoice immediately rather than leaving it a draft for review.
Invoice day of month	1–28 (so it exists in every month).
Status	Draft → Active → Completed, or Cancelled (with a reason).

Invoice / line item / payment

Field	Notes
Status	Draft, Sent, Partially paid, Paid, or Void — always derived from the payment total against the invoice total, never set directly. A void invoice stays void no matter what has been paid against it.
Issue date / Due date	Due date = issue date + payment terms.
Subtotal / Tax / Total	Tax computed once on the subtotal, snapshot at generation — a later settings change never restates an issued invoice.
Line item	Either tied to a booking (its description, hours, and rate are derived from that booking) or a manual line (booking_id is null) for something the calendar doesn't cover.
Payment	Always positive; correcting one means recording a reversal (a negative payment referencing the original), never editing or deleting the row. Overpayment is refused outright.

Waitlist entry

Field	Notes
Customer or contact name/phone	At least one is required.
Desired date	Required.
Rink	Optional — blank matches any rink.
Start / End minute	Optional — blank matches any time that day.
Notes	Optional.

8. Locking, saving & offline

There is no draft/publish lock the way Employee Scheduling has one

A booking has its own small lifecycle — **tentative** → **confirmed**, either of which can move to **cancelled** — but there is no separate "publish" step and no requester/approver split the way shift publishing works. submit-tier accounts can create tentative bookings; confirming, moving, resizing, or cancelling any booking needs edit. Cancelling is a **soft** state (status = cancelled, with a stamped reason, actor, and timestamp) — bookings are never hard-deleted from the calendar.

The database exclusion constraint is the real conflict authority

Every booking write is attempted first and only translated into a friendly message on rejection — RinkReports does not pre-check-then-hope. Two bookings on the same rink whose windows overlap **once the resurfacing buffer is counted** are refused by a database exclusion constraint (23P01), and the app looks up what it collided with purely to explain the rejection. Drag-move and drag-resize on the calendar apply **optimistically** and roll back to the server's answer if the write is refused.

Snapshotting: nothing rewrites history

A booking's price (`rate_snapshot_hourly` / `rate_snapshot_prime` / `computed_amount`) and its buffer minutes are captured **at save time** and never silently recomputed later — editing a rate card, a rink's overrides, or the module's buffer settings only changes what a future booking or edit will use. The same principle governs invoices: tax rate and line amounts are snapshotted at generation, so a later settings change never restates a document already issued.

Online-only, by design — and a genuine difference from Employee Scheduling

CLAUDE.md documents that Employee Scheduling deliberately queues two staff actions offline (submit availability, request time off) because those don't depend on live shift state. **Rink Scheduling has no offline write path anywhere, for any account tier** — not the admin calendar, not the staff dashboard, not Front Desk, not requests, invoices, or contracts. Every write in this module goes through a server action that re-validates against the database's exclusion constraints at the moment it runs, and — as `calendar-client.tsx` states directly —

"Viewing the calendar online is what populates the offline copy — there is no separate sync step. Writes are NOT cached: booking conflicts are enforced by a database exclusion constraint that re-checks at insert time, so a booking queued offline could be accepted against a world that has since moved on."

What **does** exist offline is a **read-only cached view**: every time you view the staff dashboard calendar online, it silently saves a copy of what you saw (bookings, rinks, colours) to that browser's local IndexedDB, keyed to your account. `/offline-rink-schedule` — a deliberately data-free, service-worker-cacheable page outside the normal `/reports` layout (so its shell carries no user data before the client component loads) — reads that cache and shows it with a plain banner: **"You're offline — bookings can be viewed but not created or edited,"** noting when the snapshot is more than a day stale. If you've never opened the calendar online on that device, it says so instead of showing nothing unexplained.

The public display boards (`/display/[token]`) and the ICS feed similarly have no offline story of their own — a lobby TV or a subscribed calendar app simply shows whatever it last successfully fetched, per its own polling interval.

9. Troubleshooting & FAQ

"I opened Rink Scheduling Admin but every save says I don't have permission."

The console page itself only requires the module's `admin` action (plus the global Admin Center grant) — but most of what you'll actually save there (rinks, locker rooms, hours, exceptions, display links) is checked against the `edit` action, and rate cards / module settings are checked against `admin`. If your account has only one of the two, ask a super admin to grant **both** `edit` **and** `admin` on Rink Scheduling.

"A staff member says they can't book or edit anything on their dashboard."

That's correct and intentional — the dashboard calendar at `/reports/rink-scheduling` is **read-only for everyone**, whatever permissions the account holds. All scheduling writes happen on the admin Ice Schedule grid (`/admin/rink-scheduling/schedule`), which itself needs Admin Center access. Front Desk is also read-only, for the same reason.

"My booking was refused with an overlap error, but the calendar looked empty there."

The overlap check counts the **resurfacing buffer** after the existing booking, not just its visible end time — a slot that looks free right after another booking may still fall inside its ice-make window. Check the facility's ice-make setting (or that rink's override) on the Settings/Facility Setup tabs.

"Why does this booking have no price, or show \$0.00 for the wrong reason?"

A blank/"—" price means no rate card covers that date at all — add one, or extend an existing card's date range. A **genuine** \$0.00 only happens for a non-billable booking type (e.g. Maintenance Block) or an explicit \$0 per-type override; those are two different situations and the UI distinguishes "unpriced" (null) from "free" (zero) throughout.

"I edited a rate card and now an old invoice looks wrong to me."

It shouldn't have changed — every booking snapshots the rate it was priced at when saved, and every invoice snapshots its own tax and line amounts at generation. Editing a rate card only affects bookings priced after the edit.

"A recurring series skipped some dates when I created it."

That's the conflict-safe design: previewing a series shows every occurrence and defaults any date that collides with an existing booking to **Skip**, so you can review before anything is written. Choose "book anyway" on a previewed occurrence if you want it created despite the collision (it will still be refused at commit time if the conflict is still there).

"Plan cuts skipped a gap I expected a cut in."

Either the gap is shorter than the configured cut duration, it already has a resurface booking in it (or immediately next to one — cutting twice around a cut is treated as noise), or it isn't between two rentals (RinkReports never proposes a cut before the first booking of the day or after the last one — schedule opening/closing floods yourself).

"I lost the address for a lobby display / request form."

There's no way to look it up again — only a hash is stored. **Revoke** the old link and create a new one; revoking takes effect immediately.

"Someone's calling about an ice-time request/waitlist entry and I can't find it."

New public requests land under **Requests & Waitlist → Requests**; once decided (approved or declined) they drop into the "Decided recently" list (last 20). Waitlist entries never expire on their own — resolve or remove them once they're satisfied or no longer wanted.

"An invoice's status looks wrong after I recorded a payment."

Status is always **computed** from the payment total against the invoice total — you cannot set it directly. If a payment amount was wrong, **reverse** it (a reversal is its own negative payment record) rather than trying to edit or delete the original; the status recalculates automatically.

"The calendar showed something different when I went offline."

`/offline-rink-schedule` shows whatever was last cached the last time you viewed the calendar **online** on that device — it can be up to a day (or more) stale, and the banner says so. Reconnect and reopen

the live calendar to refresh it; you cannot create or edit anything while offline, on any screen in this module.

Source (footnote)

Admin config: `src/app/admin/rink-scheduling/{page.tsx,actions.ts,types.ts}, _lib/{config.ts,config.test.ts}, _components/{facility-setup-tab,rate-cards-tab,lists-tab,displays-tab,settings-tab,seed-defaults-card}.tsx, schedule/page.tsx.`

Admin booking calendar surface: `src/app/reports/rink-scheduling/_components/{calendar-client.tsx,booking-sheet.tsx,series-sheet.tsx,plan-cuts-sheet.tsx,find-slot-sheet.tsx,locker-room-panel.tsx,use-grid-drag.ts,rink-schedule-readonly.tsx,offline-rink-schedule.tsx,not-available.tsx}, _lib/{grid-model.ts,month-model.ts,drag-model.ts,find-slot.ts,series.ts,rate-engine.ts,rate-engine.test.ts,insights.ts,desk-agenda.ts,load-calendar.ts,types.ts}` and their test files.

Staff/front-desk/customer-facing pages and actions: `src/app/reports/rink-scheduling/{page.tsx,desk/page.tsx,requests/page.tsx,contracts/page.tsx,insights/page.tsx,invoices/page.tsx,invoices/[invoiceId]/page.tsx,invoices/[invoiceId]/pdf/route.tsx}, _components under requests/, contracts/, invoices/ (request-inbox,waitlist-panel,contract-card,contract-form,invoice-table,invoice-detail,new-invoice-panel,aging-table), and actions.ts, contract-actions.ts, invoice-actions.ts, find-slot-actions.ts, locker-actions.ts, request-actions.ts, resurface-actions.ts, resurface-plan-actions.ts, series-actions.ts.`

Domain logic: `src/lib/rink-scheduling/{booking-request.ts,booking-email.ts,contract-invoices.ts,coverage.ts,coverage-sweep.ts,buffer.ts,resurface.ts,resurface-plan.ts,invoice-generation.ts,invoice-pdf.tsx,invoice-pdf-data.ts,invoice-email.ts,overdue.ts,overdue-reminders.tsx,season-contracts.ts,waitlist-match.ts,locker-display.ts,ice-schedule-display.ts,display-rate-limit.ts,public-tokens.ts,ar.ts,deliver-booking-email.ts,deliver-invoice-email.tsx}` and their test files.

Public tokened surfaces: `src/app/display/[token]/page.tsx, src/app/api/display/[token]/route.ts, src/app/api/rink-ics/[token]/route.ts, src/app/request-ice/[token]/page.tsx, src/app/offline-rink-schedule/{page.tsx,layout.tsx}.`

Permission model: `src/lib/permissions/actions.ts, src/lib/permissions/check.ts, src/lib/auth/require-module-admin.ts, src/components/admin/nav-config.ts.`